

C8000V IPsec Testbed

Test evidence — every command run on every device

Run started	07 September 2026, 15:51 UTC
Duration	18.8 minutes
Result	204 of 204 tests passed
Suites	22
Device commands recorded	1405
Topology	hub-and-spoke: R1 hub, R2 and R3 spokes, one Linux host per router
Encryption	IKEv2 + static VTI, ESP-AES-256 / SHA256, one tunnel per spoke
Routing	eBGP inside the tunnels; AS 65001 hub, AS 65002 / 65003 spokes

Suite results

Suite	Tests	Passed	Failed	Commands	Time
01 Platform	5	5	0	10	1s
02 Configuration	5	5	0	14	2s
03 Link	6	6	0	15	0s
04 Ipsec	10	10	0	43	43s
05 Capture State	12	12	0	79	16s
06 Wire Encryption	3	3	0	27	7s
07 Ipsec Negative	3	3	0	62	112s
08 Management Api	6	6	0	7	3s
09 Bgp	26	26	0	622	294s
10 Host To Host	9	9	0	43	76s
11 Throughput	3	3	0	10	81s
12 Nat	10	10	0	51	77s
13 Nat Pool	13	13	0	51	92s
14 Ntp	12	12	0	39	3s
15 Syslog	10	10	0	21	47s
16 Snmpv3	13	13	0	87	15s

17 Snmp Polling	14	14	0	76	22s
18 Mtu	8	8	0	18	21s
19 Oob	9	9	0	68	31s
20 Tacacs	12	12	0	34	22s
21 Vty Acl	9	9	0	22	1s
22 Banner	6	6	0	6	21s

Each test below lists the commands it issued, in order, with the device that ran them and the output it returned. Router commands are marked in teal, Linux host commands in amber. Long output is trimmed to the first 14 lines; complete captures are in results/.

01 Platform

1.01 PASS Both Routers Report Catalyst 8000V Software (0.2s, 2 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output matches **Cisco IOS[-]XE Software, Version 1d+1.1d+** (checked 2×)
- output contains **C8000V** (checked 2×)

How it was checked

R1 show version

```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 by mcpre
```

```
Cisco IOS-XE software, Copyright (c) 2005-2026 by cisco Systems, Inc.
All rights reserved. Certain components of Cisco IOS-XE software are
licensed under the GNU General Public License ("GPL") Version 2.0. The
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with ABSOLUTELY NO WARRANTY. You can redistribute and/or modify such
GPL code under the terms of GPL Version 2.0. For more details, see the
documentation or "License Notice" file accompanying the IOS-XE software,
... 52 more lines
```

R2 show version

```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 by mcpre
```

```
Cisco IOS-XE software, Copyright (c) 2005-2026 by cisco Systems, Inc.
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licensed under the GNU General Public License ("GPL") Version 2.0. The
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with ABSOLUTELY NO WARRANTY. You can redistribute and/or modify such
GPL code under the terms of GPL Version 2.0. For more details, see the
documentation or "License Notice" file accompanying the IOS-XE software,
... 52 more lines
```

Teardown (suite) Close All Routers

1.02 PASS Hostnames Match The Day0 Bootstrap Configs (0.2s, 2 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **\${expected}** (checked 2×)

How it was checked

R1 show running-config | include ^hostname

```
hostname c8000v-r1
```

R2 show running-config | include ^hostname

```
hostname c8000v-r2
```

Teardown (suite) Close All Routers

1.03 PASS Management Interfaces Are Up With DHCP Addresses (0.1s, 2 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output matches **GigabitEthernet1s+10\0.0\2\1d+1s+YES\+DHCP\+up\+up** (checked 2×)

How it was checked

R1 `show ip interface brief GigabitEthernet1`

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1	10.0.2.15	YES	DHCP	up	up

R2 `show ip interface brief GigabitEthernet1`

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1	10.0.2.15	YES	DHCP	up	up

Teardown (suite) Close All Routers

1.04 **PASS** Crypto Feature License Is Active (0.1s, 2 commands)

Without a feature license the crypto CLI does not exist at all, so every IPsec test downstream depends on this.

Setup (suite) Open All Routers

Acceptance criteria

- output matches **License Level:ls*S+** (checked 2×)

How it was checked

R1 `show version | include License Level`

```
License Level: network-premier
Addon License Level: dna-premier
```

R2 `show version | include License Level`

```
License Level: network-premier
Addon License Level: dna-premier
```

Teardown (suite) Close All Routers

1.05 **PASS** Management Plane Services Are Enabled (0.2s, 2 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **netconf-yang** (checked 2×)
- output contains **restconf** (checked 2×)

How it was checked

R1 `show running-config | include ^netconf-yang|^restconf`

```
netconf-yang
restconf
```

R2 `show running-config | include ^netconf-yang|^restconf`

```
netconf-yang
restconf
```

Teardown (suite) Close All Routers

02 Configuration

2.01 **PASS** Create A Loopback Interface (0.1s, 6 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain % **Invalid input**
- output does not contain % **Incomplete command**
- output does not contain % **Ambiguous command**

How it was checked

R1 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1 interface Loopback100

#{output} =

R1 description ROBOT-MANAGED

#{output} =

R1 ip address 192.0.2.1 255.255.255.0

#{output} =

R1 no shutdown

#{output} =

R1 end

#{output} =

Teardown (this test) Run Keyword If Test Failed (Log, #{TEST_NAME} failed -- see the console on port 5001)

2.02 **PASS** Loopback Appears In The Interface Table (0.0s, 1 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output matches **#{LOOPBACK}#{s}#{LOOPBACK_IP}#{s}+YES#{s}+manual#{s}+up#{s}+up**

How it was checked

R1 show ip interface brief Loopback100

Interface	IP-Address	OK?	Method	Status	Protocol
Loopback100	192.0.2.1	YES	manual	up	up

Teardown (this test) Run Keyword If Test Failed (Log, #{TEST_NAME} failed -- see the console on port 5001)

2.03 **PASS** Loopback Is Reachable From The Device Itself (2.1s, 2 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **Success rate is 100 percent**

How it was checked

R1 ping 192.0.2.1 repeat 2

Type escape sequence to abort.

Sending 2, 100-byte ICMP Echos to 192.0.2.1, timeout is 2 seconds:

..!

Success rate is 50 percent (1/2), round-trip min/avg/max = 2/2/2 ms

R1 ping 192.0.2.1 repeat 5

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.0.2.1, timeout is 2 seconds:

!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

Teardown (this test) Run Keyword If Test Failed (Log, #{TEST_NAME} failed -- see the console on port 5001)

2.04 **PASS** Loopback Is Present In The Running Config (0.0s, 1 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **description** ROBOT-MANAGED
- output contains **ip address** \${LOOPBACK_IP} \${LOOPBACK_MASK}

How it was checked

R1 show running-config interface Loopback100

Building configuration...

```
Current configuration : 92 bytes
!
interface Loopback100
  description ROBOT-MANAGED
  ip address 192.0.2.1 255.255.255.0
end
```

Teardown (this test) Run Keyword If Test Failed (Log, \${TEST_NAME} failed -- see the console on port 5001)

2.05 **PASS** Removing The Loopback Cleans It Up (0.1s, 4 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain \${LOOPBACK_IP}

How it was checked

R1 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1 no interface Loopback100

\${output} =

R1 end

\${output} =

R1 show ip interface brief Loopback100

^

% Invalid input detected at '^' marker.

Teardown (this test) Run Keyword If Test Failed (Log, \${TEST_NAME} failed -- see the console on port 5001)

03 Link

3.01 PASS Hub Link Interfaces Are Up Toward Both Spokes (0.1s, 2 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output matches `${R2_HUB_INTF}\s+${R2_LINK_HUB}\s+YES\s+IS\s+up\s+up`
- output matches `${R3_HUB_INTF}\s+${R3_LINK_HUB}\s+YES\s+IS\s+up\s+up`

How it was checked

R1 `show ip interface brief GigabitEthernet2`

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet2	10.10.10.1	YES	TFTP	up	up

R1 `show ip interface brief GigabitEthernet4`

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet4	10.10.20.1	YES	TFTP	up	up

Teardown (suite) Close All Connections

3.02 PASS Spoke Link Interfaces Are Up Toward The Hub (0.1s, 2 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output matches `GigabitEthernet2\s+${R2_LINK_LOCAL}\s+YES\s+IS\s+up\s+up`
- output matches `GigabitEthernet2\s+${R3_LINK_LOCAL}\s+YES\s+IS\s+up\s+up`

How it was checked

R2 `show ip interface brief GigabitEthernet2`

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet2	10.10.10.2	YES	TFTP	up	up

R3 `show ip interface brief GigabitEthernet2`

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet2	10.10.20.2	YES	TFTP	up	up

Teardown (suite) Close All Connections

3.03 PASS Hub And Spokes Can Reach Each Other Across The Links (0.1s, 4 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **Success rate is 100 percent** (checked 4x)

How it was checked

R1 `ping 10.10.10.2 repeat 5`

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
```

R1 `ping 10.10.20.2 repeat 5`

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

R2 `ping 10.10.10.1 repeat 5`

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

R3 `ping 10.10.20.1 repeat 5`

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

Teardown (suite) Close All Connections

3.04 **PASS** Spokes Have No Direct Link To Each Other (0.0s, 2 commands)

Confirms this really is hub-and-spoke: neither spoke has any route to the other's transport subnet, so all spoke-to-spoke traffic must traverse the hub.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **% Subnet not in table** (checked 2×)

How it was checked

R2 `show ip route 10.10.20.1`
% Subnet not in table

R3 `show ip route 10.10.10.1`
% Subnet not in table

Teardown (suite) Close All Connections

3.05 **PASS** Each Router Learns Its Peer ARP Entry On The Link (0.0s, 2 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **#{R2_HUB_INTF}**
- output contains **#{R3_HUB_INTF}**

How it was checked

R1 `show ip arp 10.10.10.2`

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	10.10.10.2	138	5254.00c8.0202	ARPA	GigabitEthernet2

R1 `show ip arp 10.10.20.2`

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	10.10.20.2	138	5254.00c8.0302	ARPA	GigabitEthernet4

Teardown (suite) Close All Connections

3.06 **PASS** Protected Endpoints Exist On Every Router (0.1s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output matches **Loopback0ls+#{lo}ls+YESls+IS+ls+upls+up** (checked 3×)

How it was checked

R1 `show ip interface brief Loopback0`

Interface	IP-Address	OK?	Method	Status	Protocol
Loopback0	172.16.1.1	YES	TFTP	up	up

R2 `show ip interface brief Loopback0`

Interface	IP-Address	OK?	Method	Status	Protocol
Loopback0	172.16.2.1	YES	TFTP	up	up

R3 `show ip interface brief Loopback0`

Interface	IP-Address	OK?	Method	Status	Protocol
Loopback0	172.16.3.1	YES	TFTP	up	up

Teardown (suite) Close All Connections

04 Ipsec

4.01 PASS Tunnel Interfaces Are Up On Hub And Spokes (0.2s, 4 commands)

Setup (suite) Establish Tunnels

Acceptance criteria

- output matches `${R2_HUB_TUNNEL}\s+${R2_TUNNEL_HUB}\s+YES\s+\S+\s+up\s+up`
- output matches `${R3_HUB_TUNNEL}\s+${R3_TUNNEL_HUB}\s+YES\s+\S+\s+up\s+up`
- output matches `Tunnel0\s+${R2_TUNNEL_LOCAL}\s+YES\s+\S+\s+up\s+up`
- output matches `Tunnel0\s+${R3_TUNNEL_LOCAL}\s+YES\s+\S+\s+up\s+up`

How it was checked

R1 show ip interface brief Tunnel0

Interface	IP-Address	OK?	Method	Status	Protocol
Tunnel0	192.168.100.1	YES	manual	up	up

R1 show ip interface brief Tunnel1

Interface	IP-Address	OK?	Method	Status	Protocol
Tunnel1	192.168.101.1	YES	manual	up	up

R2 show ip interface brief Tunnel0

Interface	IP-Address	OK?	Method	Status	Protocol
Tunnel0	192.168.100.2	YES	manual	up	up

R3 show ip interface brief Tunnel0

Interface	IP-Address	OK?	Method	Status	Protocol
Tunnel0	192.168.101.2	YES	manual	up	up

Teardown (suite) Close All Connections

4.02 PASS Every Tunnel Is Bound To An IPsec Profile (0.1s, 4 commands)

Setup (suite) Establish Tunnels

Acceptance criteria

- output contains `tunnel mode ipsec ipv4` (checked 4×)
- output contains `tunnel protection ipsec profile LAB-IPSEC` (checked 4×)

How it was checked

R1 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 281 bytes
!
interface Tunnel0
  description to c8000v-r2
  ip address 192.168.100.1 255.255.255.252
  ip nat outside
  bfd interval 500 min_rx 500 multiplier 3
  tunnel source GigabitEthernet2
  tunnel mode ipsec ipv4
  tunnel destination 10.10.10.2
  tunnel protection ipsec profile LAB-IPSEC
end
```

R1 show running-config interface Tunnel1

Building configuration...

```
Current configuration : 281 bytes
!
interface Tunnel1
  description to c8000v-r3
  ip address 192.168.101.1 255.255.255.252
  ip nat outside
  bfd interval 500 min_rx 500 multiplier 3
  tunnel source GigabitEthernet4
  tunnel mode ipsec ipv4
  tunnel destination 10.10.20.2
  tunnel protection ipsec profile LAB-IPSEC
end
```

R2 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 291 bytes
```

```

!
interface Tunnel0
description to the hub (c8000v-r1)
ip address 192.168.100.2 255.255.255.252
ip nat outside
bfd interval 500 min_rx 500 multiplier 3
tunnel source GigabitEthernet2
tunnel mode ipsec ipv4
tunnel destination 10.10.10.1
tunnel protection ipsec profile LAB-IPSEC
end

```

R3 show running-config interface Tunnel0

Building configuration...

```

Current configuration : 291 bytes
!
interface Tunnel0
description to the hub (c8000v-r1)
ip address 192.168.101.2 255.255.255.252
ip nat outside
bfd interval 500 min_rx 500 multiplier 3
tunnel source GigabitEthernet2
tunnel mode ipsec ipv4
tunnel destination 10.10.20.1
tunnel protection ipsec profile LAB-IPSEC
end

```

Teardown (suite) Close All Connections

4.03 PASS Hub Holds An IKEv2 Security Association With Each Spoke (0.0s, 1 commands)

Setup (suite) Establish Tunnels

Acceptance criteria

- output contains **#{R2_LINK_LOCAL}**
 - output contains **#{R3_LINK_LOCAL}**
 - **#{ready}** has exactly 2 entries
- hub should hold one READY SA per spoke

How it was checked

R1 show crypto ikev2 sa

IPv4 Crypto IKEv2 SA

Tunnel-id	Local	Remote	fvrf/ivrf	Status
1	10.10.10.1/500	10.10.10.2/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/97 sec				
CE id: 1068, Session-id: 43				
Local spi: 2D80B89D4CD9F796		Remote spi: B3723E550B92C45A		
Tunnel-id	Local	Remote	fvrf/ivrf	Status
2	10.10.20.1/500	10.10.20.2/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/979 sec				
CE id: 1066, Session-id: 42				
... 3 more lines				

Teardown (suite) Close All Connections

4.04 PASS Each Spoke Holds An IKEv2 Security Association With The Hub (0.1s, 2 commands)

Setup (suite) Establish Tunnels

Acceptance criteria

- output contains **READY** (checked 2x)
- output contains **#{R2_LINK_HUB}**
- output contains **#{R3_LINK_HUB}**

How it was checked

R2 show crypto ikev2 sa

IPv4 Crypto IKEv2 SA

Tunnel-id	Local	Remote	fvrf/ivrf	Status
-----------	-------	--------	-----------	--------

```

3      10.10.10.2/500      10.10.10.1/500      none/none      READY
    Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
    Life/Active Time: 86400/97 sec
    CE id: 1042, Session-id: 16
    Local spi: B3723E550B92C45A      Remote spi: 2D80B89D4CD9F796

```

```

Tunnel-id Local      Remote      fvrf/ivrf      Status
1      10.10.10.2/500      10.10.10.1/500      none/none      DELETE
    Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
    Life/Active Time: 86400/980 sec
    CE id: 1040, Session-id: 16
... 3 more lines

```

R3 show crypto ikev2 sa

IPv4 Crypto IKEv2 SA

```

Tunnel-id Local      Remote      fvrf/ivrf      Status
1      10.10.20.2/500      10.10.20.1/500      none/none      READY
    Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
    Life/Active Time: 86400/979 sec
    CE id: 1026, Session-id: 21
    Local spi: 45E55BDBFF61B189      Remote spi: F58E80592B7534C5

```

IPv6 Crypto IKEv2 SA

Teardown (suite) Close All Connections

4.05 PASS IPsec Security Associations Are Installed In Both Directions (0.3s, 3 commands)

Setup (suite) Establish Tunnels

Acceptance criteria

- output contains **inbound esp sas**: (checked 3×)
- output contains **outbound esp sas**: (checked 3×)
- output matches **spi: 0x[0-9A-Fa-f]+** (checked 3×)

How it was checked

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 100593, #pkts encrypt: 100593, #pkts digest: 100593
#pkts decaps: 87290, #pkts decrypt: 87290, #pkts verify: 87290
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines

```

R2 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 544, #pkts encrypt: 544, #pkts digest: 544
#pkts decaps: 558, #pkts decrypt: 558, #pkts verify: 558
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines

```

R3 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.20.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)

```

```

current_peer 10.10.20.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 104781, #pkts encrypt: 104781, #pkts digest: 104781
#pkts decaps: 184242, #pkts decrypt: 184242, #pkts verify: 184242
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines

```

Teardown (suite) Close All Connections

4.06 PASS Negotiated Transform Matches The Configured Policy (0.3s, 3 commands)

Guards against silently falling back to a weaker or null cipher.

Setup (suite) Establish Tunnels

Acceptance criteria

- output contains **esp-256-aes** (checked 3×)
- output contains **esp-sha256-hmac** (checked 3×)

How it was checked

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 100603, #pkts encrypt: 100603, #pkts digest: 100603
#pkts decaps: 87290, #pkts decrypt: 87290, #pkts verify: 87290
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines

```

R2 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 554, #pkts encrypt: 554, #pkts digest: 554
#pkts decaps: 558, #pkts decrypt: 558, #pkts verify: 558
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines

```

R3 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.20.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.20.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 104781, #pkts encrypt: 104781, #pkts digest: 104781
#pkts decaps: 184242, #pkts decrypt: 184242, #pkts verify: 184242
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines

```

Teardown (suite) Close All Connections

4.07 PASS Peer Loopbacks Are Routed Through The Tunnels (0.1s, 4 commands)

Setup (suite) Establish Tunnels

Acceptance criteria

- output contains **\${R2_HUB_TUNNEL}**
- output contains **\${R3_HUB_TUNNEL}**
- output contains **Tunnel0** (checked 2×)

How it was checked

R1 show ip route 172.16.2.1

```
Routing entry for 172.16.2.1/32
  Known via "static", distance 1, metric 0 (connected)
  Routing Descriptor Blocks:
    * directly connected, via Tunnel0
      Route metric is 0, traffic share count is 1
```

R1 show ip route 172.16.3.1

```
Routing entry for 172.16.3.1/32
  Known via "static", distance 1, metric 0 (connected)
  Routing Descriptor Blocks:
    * directly connected, via Tunnel1
      Route metric is 0, traffic share count is 1
```

R2 show ip route 172.16.1.1

```
Routing entry for 172.16.1.1/32
  Known via "static", distance 1, metric 0 (connected)
  Routing Descriptor Blocks:
    * directly connected, via Tunnel0
      Route metric is 0, traffic share count is 1
```

R3 show ip route 172.16.1.1

```
Routing entry for 172.16.1.1/32
  Known via "static", distance 1, metric 0 (connected)
  Routing Descriptor Blocks:
    * directly connected, via Tunnel0
      Route metric is 0, traffic share count is 1
```

Teardown (suite) Close All Connections

4.08 PASS Loopback Traffic To Each Spoke Is Encrypted And Decrypted (11.3s, 14 commands)

The core assertion, run once per tunnel: send known traffic and prove the ESP counters moved by that amount on both peers.

Setup (suite) Establish Tunnels

Acceptance criteria

- output contains **Success rate is 100 percent** (checked 2×)
- **\${r1_enc} >= \${expected}** holds (checked 2×)
- **\${r1_dec} >= \${expected}** holds (checked 2×)
- **\${r2_enc} >= \${expected}** holds (checked 2×)
- **\${r2_dec} >= \${expected}** holds (checked 2×)
- **\${a_enc} >= \${expected}** holds (checked 2×)
- **\${a_dec} >= \${expected}** holds (checked 2×)
- **\${b_enc} >= \${expected}** holds (checked 2×)
- **\${b_dec} >= \${expected}** holds (checked 2×)

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 100603, #pkts encrypt: 100603, #pkts digest: 100603
  #pkts decaps: 87300, #pkts decrypt: 87300, #pkts verify: 87300
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
```

```

... 81 more lines
R2 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

  protected vrf: (none)
  local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  current_peer 10.10.10.1 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 554, #pkts encrypt: 554, #pkts digest: 554
    #pkts decaps: 558, #pkts decrypt: 558, #pkts verify: 558
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 33 more lines

R1 ping 172.16.2.1 source Loopback0 repeat 20
Type escape sequence to abort.
Sending 20, 100-byte ICMP Echos to 172.16.2.1, timeout is 2 seconds:
Packet sent with a source address of 172.16.1.1
!!!!!!!!!!!!!!!!!!!!!!
Success rate is 100 percent (20/20), round-trip min/avg/max = 1/1/1 ms

```

```

R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  current_peer 10.10.10.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 100603, #pkts encrypt: 100603, #pkts digest: 100603
    #pkts decaps: 87300, #pkts decrypt: 87300, #pkts verify: 87300
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 81 more lines

```

```

R2 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

  protected vrf: (none)
  local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  current_peer 10.10.10.1 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 554, #pkts encrypt: 554, #pkts digest: 554
    #pkts decaps: 583, #pkts decrypt: 583, #pkts verify: 583
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 33 more lines

```

```

R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  current_peer 10.10.10.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 100654, #pkts encrypt: 100654, #pkts digest: 100654
    #pkts decaps: 87351, #pkts decrypt: 87351, #pkts verify: 87351
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 81 more lines

```

```

R2 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

  protected vrf: (none)
  local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)

```

```

current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 604, #pkts encrypt: 604, #pkts digest: 604
#pkts decaps: 615, #pkts decrypt: 615, #pkts verify: 615
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines

R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 100660, #pkts encrypt: 100660, #pkts digest: 100660
#pkts decaps: 87351, #pkts decrypt: 87351, #pkts verify: 87351
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines

R3 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.20.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.20.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 104818, #pkts encrypt: 104818, #pkts digest: 104818
#pkts decaps: 184278, #pkts decrypt: 184278, #pkts verify: 184278
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines

R1 ping 172.16.3.1 source Loopback0 repeat 20
Type escape sequence to abort.
Sending 20, 100-byte ICMP Echos to 172.16.3.1, timeout is 2 seconds:
Packet sent with a source address of 172.16.1.1
!!!!!!!!!!!!!!!!!!!!!!
Success rate is 100 percent (20/20), round-trip min/avg/max = 1/1/1 ms

```

```

R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 100660, #pkts encrypt: 100660, #pkts digest: 100660
#pkts decaps: 87357, #pkts decrypt: 87357, #pkts verify: 87357
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines

R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.20.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.20.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 104818, #pkts encrypt: 104818, #pkts digest: 104818
#pkts decaps: 184278, #pkts decrypt: 184278, #pkts verify: 184278
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines

```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 100690, #pkts encrypt: 100690, #pkts digest: 100690
#pkts decaps: 87386, #pkts decrypt: 87386, #pkts verify: 87386
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.20.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.20.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 104866, #pkts encrypt: 104866, #pkts digest: 104866
#pkts decaps: 184334, #pkts decrypt: 184334, #pkts verify: 184334
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines
```

Teardown (suite) Close All Connections

4.09 PASS Traffic Outside The Protected Selector Is Not Encrypted (30.8s, 5 commands)

Negative control, self-calibrating. A delta of zero is not assertable: BGP peers across the tunnels and BFD now runs on them at 500ms x 2 tunnels, so the counters tick continuously whatever the data plane is doing -- that background alone is far larger than the ping count, which is what used to be the threshold. So measure it: time the ping window, then measure an idle window of the same length, and require the ping to have added no more than the control plane did by itself. If link-local echoes were being encrypted the difference could not stay under the number of pings sent.

Setup (suite) Establish Tunnels

Acceptance criteria

- output contains **Success rate is 100 percent**
- **#{attributable} < #{LINK_PINGS}** holds

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 100690, #pkts encrypt: 100690, #pkts digest: 100690
#pkts decaps: 87386, #pkts decrypt: 87386, #pkts verify: 87386
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines
```

R1 ping 10.10.10.2 repeat 10

```
Type escape sequence to abort.
Sending 10, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!!!!!!!!
Success rate is 100 percent (10/10), round-trip min/avg/max = 1/1/1 ms
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1
```



```

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 100778, #pkts encrypt: 100778, #pkts digest: 100778
#pkts decaps: 87475, #pkts decrypt: 87475, #pkts verify: 87475
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 100783, #pkts encrypt: 100783, #pkts digest: 100783
#pkts decaps: 87475, #pkts decrypt: 87475, #pkts verify: 87475
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 100872, #pkts encrypt: 100872, #pkts digest: 100872
#pkts decaps: 87569, #pkts decrypt: 87569, #pkts verify: 87569
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines

```

Teardown (suite) Close All Connections

4.10 PASS Anti-Replay And Lifetime Protections Are Present (0.3s, 3 commands)

Setup (suite) Establish Tunnels

Acceptance criteria

- output matches **replay detection support:ls*Y** (checked 3×)
- output matches **sa timing: remaining key lifetime** (checked 3×)

How it was checked

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 100872, #pkts encrypt: 100872, #pkts digest: 100872
#pkts decaps: 87569, #pkts decrypt: 87569, #pkts verify: 87569
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines

```

R2 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2
protected vrf: (none)

```

```
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 822, #pkts encrypt: 822, #pkts digest: 822
#pkts decaps: 827, #pkts decrypt: 827, #pkts verify: 827
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines
```

R3 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.20.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.20.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 105049, #pkts encrypt: 105049, #pkts digest: 105049
#pkts decaps: 184519, #pkts decrypt: 184519, #pkts verify: 184519
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines
```

Teardown (suite) Close All Connections

05 Capture State

5.01 PASS Capture Running Config From Every Router (12.7s, 3 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria

- output contains **hostname** (checked 3×)
- output matches **(?m)^endl*\$** (checked 3×)

How it was checked

R1 show running-config

Building configuration...

```
Current configuration : 21875 bytes
!
! Last configuration change at 19:51:03 UTC Mon Sep 7 2026 by lab
! NVRAM config last updated at 19:36:03 UTC Mon Sep 7 2026 by lab
!
version 17.15
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
platform qfp utilization monitor load 80
platform sslvpn use-pd
platform console serial
... 641 more lines
```

R2 show running-config

Building configuration...

```
Current configuration : 19825 bytes
!
! Last configuration change at 19:50:03 UTC Mon Sep 7 2026 by lab
! NVRAM config last updated at 19:36:03 UTC Mon Sep 7 2026 by lab
!
version 17.15
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
platform qfp utilization monitor load 80
platform sslvpn use-pd
platform console serial
... 575 more lines
```

R3 show running-config

Building configuration...

```
Current configuration : 19825 bytes
!
! Last configuration change at 19:50:03 UTC Mon Sep 7 2026 by lab
! NVRAM config last updated at 19:36:03 UTC Mon Sep 7 2026 by lab
!
version 17.15
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
platform qfp utilization monitor load 80
platform sslvpn use-pd
platform console serial
... 575 more lines
```

Teardown (suite) Close Nms, Close All Connections

5.02 PASS Capture Routing Table From Every Router (0.2s, 3 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria

- output contains **Gateway of last resort** (checked 3×)
- output contains **Codes:** (checked 3×)

How it was checked

R1 show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
... 30 more lines

R2 show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
... 24 more lines

R3 show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
... 24 more lines

Teardown (suite) Close Nms, Close All Connections

5.03 PASS Capture BGP Table And Neighbours From Every Router (0.2s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

R1 show bgp ipv4 unicast

BGP table version is 16, local router ID is 172.16.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.1.1/32	0.0.0.0	0		32768	i
*>	10.20.2.1/32	192.168.100.2	0		0	65002 i
*>	10.20.3.1/32	192.168.101.2	0		0	65003 i
*>	10.30.1.0/24	0.0.0.0	0		32768	i
*>	10.30.2.0/24	192.168.100.2	0		0	65002 i

... 4 more lines

R1 show bgp ipv4 unicast summary

BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 16, main routing table version 16
9 network entries using 2232 bytes of memory

```

9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
3 BGP route-map cache entries using 216 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5936 total bytes of memory
BGP activity 281/272 prefixes, 294/285 paths, scan interval 60 secs
22 networks peaked at 17:59:32 Sep 7 2026 UTC (01:52:32.507 ago)

Neighbor      V          AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines

```

R2 show bgp ipv4 unicast

```

BGP table version is 22, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.1.1/32	192.168.100.1	0		0	65001 i
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.20.3.1/32	192.168.100.1			0	65001 65003 i
*>	10.30.1.0/24	192.168.100.1	0		0	65001 i
*>	10.30.2.0/24	0.0.0.0	0		32768	i

... 4 more lines

R2 show bgp ipv4 unicast summary

```

BGP router identifier 172.16.2.1, local AS number 65002
BGP table version is 22, main routing table version 22
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6168 total bytes of memory
BGP activity 332/323 prefixes, 337/328 paths, scan interval 60 secs
31 networks peaked at 18:00:16 Sep 7 2026 UTC (01:51:48.580 ago)

```

```

Neighbor      V          AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 1 more lines

```

R3 show bgp ipv4 unicast

```

BGP table version is 334, local router ID is 172.16.3.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.1.1/32	192.168.101.1	0		0	65001 i
*>	10.20.2.1/32	192.168.101.1			0	65001 65002 i
*>	10.20.3.1/32	0.0.0.0	0		32768	i
*>	10.30.1.0/24	192.168.101.1	0		0	65001 i
*>	10.30.2.0/24	192.168.101.1			0	65001 65002 i

... 4 more lines

R3 show bgp ipv4 unicast summary

```

BGP router identifier 172.16.3.1, local AS number 65003
BGP table version is 334, main routing table version 334
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
3 BGP route-map cache entries using 216 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5952 total bytes of memory
BGP activity 157/148 prefixes, 171/162 paths, scan interval 60 secs
9 networks peaked at 17:33:25 Sep 7 2026 UTC (02:18:39.656 ago)

```

```

Neighbor      V          AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 1 more lines

```

Teardown (suite) Close Nms, Close All Connections

5.04 PASS Capture Crypto State From Every Router (0.3s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 100965, #pkts encrypt: 100965, #pkts digest: 100965
  #pkts decaps: 87662, #pkts decrypt: 87662, #pkts verify: 87662
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 81 more lines
```

R1 show crypto ikev2 sa

IPv4 Crypto IKEv2 SA

Tunnel-id	Local	Remote	fvrf/ivrf	Status
1	10.10.10.1/500	10.10.10.2/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/156 sec				
CE id: 1068, Session-id: 43				
Local spi: 2D80B89D4CD9F796 Remote spi: B3723E550B92C45A				

Tunnel-id	Local	Remote	fvrf/ivrf	Status
2	10.10.20.1/500	10.10.20.2/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/1038 sec				
CE id: 1066, Session-id: 42				

... 3 more lines

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2
```

```
protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 915, #pkts encrypt: 915, #pkts digest: 915
  #pkts decaps: 925, #pkts decrypt: 925, #pkts verify: 925
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 33 more lines
```

R2 show crypto ikev2 sa

IPv4 Crypto IKEv2 SA

Tunnel-id	Local	Remote	fvrf/ivrf	Status
3	10.10.10.2/500	10.10.10.1/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/156 sec				
CE id: 1042, Session-id: 16				
Local spi: B3723E550B92C45A Remote spi: 2D80B89D4CD9F796				

IPv6 Crypto IKEv2 SA

R3 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.20.2
```

```
protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.20.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 105149, #pkts encrypt: 105149, #pkts digest: 105149
  #pkts decaps: 184612, #pkts decrypt: 184612, #pkts verify: 184612
  #pkts compressed: 0, #pkts decompressed: 0
```

```

#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines

```

R3 show crypto ikev2 sa

IPv4 Crypto IKEv2 SA

```

Tunnel-id Local Remote fvrf/ivrf Status
1 10.10.20.2/500 10.10.20.1/500 none/none READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
Life/Active Time: 86400/1039 sec
CE id: 1026, Session-id: 21
Local spi: 45E55BDBFF61B189 Remote spi: F58E80592B7534C5

```

IPv6 Crypto IKEv2 SA

Teardown (suite) Close Nms, Close All Connections

5.05 PASS Capture NAT State From Every Router (0.5s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

R1 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.1.128	192.168.10.88	---	---
---	10.30.1.129	192.168.10.90	---	---
---	10.30.1.113	192.168.10.91	---	---
---	10.30.1.114	192.168.10.87	---	---
---	10.30.1.116	192.168.10.89	---	---
---	10.30.1.119	192.168.10.71	---	---
---	10.30.1.123	192.168.10.81	---	---
---	10.30.1.120	192.168.10.75	---	---
---	10.30.1.126	192.168.10.72	---	---
---	10.30.1.125	192.168.10.85	---	---
---	10.30.1.101	192.168.10.92	---	---
---	10.30.1.102	192.168.10.83	---	---
---	10.30.1.111	192.168.10.73	---	---

... 28 more lines

R1 show ip nat statistics

```

Total active translations: 40 (10 static, 30 dynamic; 0 extended)
Outside interfaces:
  Tunnel0, Tunnel1
Inside interfaces:
  GigabitEthernet3
Hits: 457 Misses: 123
Reserved port setting disabled provisioned no
Dynamic overload mapping configured: 0
Expired translations: 126
Dynamic mappings:
-- Inside Source
[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 30
  pool SITE-POOL: id 1, netmask 255.255.255.0
    start 10.30.1.100 end 10.30.1.129
... 9 more lines

```

R2 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.2.128	192.168.20.86	---	---
---	10.30.2.129	192.168.20.75	---	---
---	10.30.2.106	192.168.20.80	---	---
---	10.30.2.105	192.168.20.82	---	---
---	10.30.2.111	192.168.20.89	---	---
---	10.30.2.108	192.168.20.79	---	---
---	10.30.2.101	192.168.20.76	---	---
---	10.30.2.102	192.168.20.90	---	---
---	10.30.2.126	192.168.20.74	---	---
---	10.30.2.125	192.168.20.67	---	---
---	10.30.2.123	192.168.20.77	---	---
---	10.30.2.120	192.168.20.66	---	---
---	10.30.2.116	192.168.20.91	---	---

... 28 more lines

R2 show ip nat statistics

```

Total active translations: 40 (10 static, 30 dynamic; 0 extended)
Outside interfaces:

```

```

Tunnel0
Inside interfaces:
  GigabitEthernet3
Hits: 337 Misses: 83
Reserved port setting disabled provisioned no
Dynamic overload mapping configured: 0
Expired translations: 83
Dynamic mappings:
-- Inside Source
[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 30
  pool SITE-POOL: id 1, netmask 255.255.255.0
    start 10.30.2.100 end 10.30.2.129
... 9 more lines

```

R3 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.3.104	192.168.30.85	---	---
---	10.30.3.107	192.168.30.92	---	---
---	10.30.3.109	192.168.30.91	---	---
---	10.30.3.110	192.168.30.93	---	---
---	10.30.3.103	192.168.30.90	---	---
---	10.30.3.100	192.168.30.84	---	---
---	10.30.3.124	192.168.30.82	---	---
---	10.30.3.127	192.168.30.88	---	---
---	10.30.3.121	192.168.30.86	---	---
---	10.30.3.122	192.168.30.83	---	---
---	10.30.3.118	192.168.30.89	---	---
---	10.30.3.117	192.168.30.79	---	---
---	10.30.3.115	192.168.30.76	---	---

... 28 more lines

R3 show ip nat statistics

```

Total active translations: 40 (10 static, 30 dynamic; 0 extended)
Outside interfaces:
  Tunnel0
Inside interfaces:
  GigabitEthernet3
Hits: 120 Misses: 40
Reserved port setting disabled provisioned no
Dynamic overload mapping configured: 0
Expired translations: 40
Dynamic mappings:
-- Inside Source
[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 30
  pool SITE-POOL: id 1, netmask 255.255.255.0
    start 10.30.3.100 end 10.30.3.129
... 9 more lines

```

Teardown (suite) Close Nms, Close All Connections

5.06 PASS Capture NTP State From Every Router (0.1s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

R1 show ntp status

```

Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0017 Hz, precision is 2**10
ntp uptime is 834000 (1/100 of seconds), resolution is 4000
reference time is EE4993D1.74FDF4F8 (19:49:37.457 UTC Mon Sep 7 2026)
clock offset is -6.2328 msec, root delay is 66.10 msec
root dispersion is 18.60 msec, peer dispersion is 4.18 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000006803 s/s
system poll interval is 256, last update was 148 sec ago.

```

R1 show ntp associations

```

address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10  96.60.247.29   3   148   256   377  0.908  -6.232  4.181
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured

```

R2 show ntp status

```

Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0018 Hz, precision is 2**10
ntp uptime is 834000 (1/100 of seconds), resolution is 4000
reference time is EE49941C.716873E8 (19:50:52.443 UTC Mon Sep 7 2026)
clock offset is -2.7727 msec, root delay is 66.11 msec
root dispersion is 7947.92 msec, peer dispersion is 3938.54 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007224 s/s

```


system poll interval is 128, last update was 73 sec ago.

R2 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10  96.60.247.29    3   32   128    1 0.918 -2.772 3938.5
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0017 Hz, precision is 2**10
ntp uptime is 834000 (1/100 of seconds), resolution is 4000
reference time is EE499001.7CAC0988 (19:33:21.487 UTC Mon Sep 7 2026)
clock offset is -4.6683 msec, root delay is 66.11 msec
root dispersion is 30.73 msec, peer dispersion is 3.69 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000006973 s/s
system poll interval is 256, last update was 1124 sec ago.
```

R3 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10  96.60.247.29    3   74   256   377 0.070 -4.668 3.691
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

Teardown (suite) Close Nms, Close All Connections

5.07 PASS Capture Logging State From Every Router (0.2s, 3 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

R1 show logging | include Trap logging|Logging to|Syslog logging

```
Syslog logging: enabled (0 messages dropped, 3 messages rate-limited, 0 flushes, 0 overruns, xml disabled,
filtering disabled)
Trap logging: level informational, 1734 message lines logged
Logging to 192.168.99.10 (MGMT) (udp port 514, audit disabled,
```

R2 show logging | include Trap logging|Logging to|Syslog logging

```
Syslog logging: enabled (0 messages dropped, 3 messages rate-limited, 0 flushes, 0 overruns, xml disabled,
filtering disabled)
Trap logging: level informational, 2150 message lines logged
Logging to 192.168.99.10 (MGMT) (udp port 514, audit disabled,
001231: .Sep 7 19:08:18.101 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514
(MGMT) started - reconnection
001234: .Sep 7 19:08:29.639 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514
(MGMT) started - reconnection
001655: .Sep 7 19:29:34.838 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514
(MGMT) started - reconnection
001658: .Sep 7 19:29:49.653 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514
(MGMT) started - reconnection
002098: .Sep 7 19:49:48.655 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514
(MGMT) started - reconnection
002101: .Sep 7 19:49:59.699 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514
(MGMT) started - reconnection
```

R3 show logging | include Trap logging|Logging to|Syslog logging

```
Syslog logging: enabled (0 messages dropped, 3 messages rate-limited, 0 flushes, 0 overruns, xml disabled,
filtering disabled)
Trap logging: level informational, 1221 message lines logged
Logging to 192.168.99.10 (MGMT) (udp port 514, audit disabled,
```

Teardown (suite) Close Nms, Close All Connections

5.08 PASS Capture SNMP State From Every Router (0.2s, 9 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

R1 show snmp user

```
User name: labmon
Engine ID: 800000090300525400C80101
storage-type: nonvolatile active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: LABGRP
```

R1 show snmp host

```
Notification host: 192.168.99.10      udp-port: 1162  VRFName: MGMT  type: trap
user: labmon      security model: v3 priv

Notification host: 192.168.99.10      udp-port: 162   VRFName: MGMT  type: trap
user: labmon      security model: v3 priv
```

R1 show snmp engineID

```
Local SNMP engineID: 800000090300525400C80101
Remote Engine ID      IP-addr      Port
```

R2 show snmp user

```
User name: labmon
Engine ID: 800000090300525400C80201
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: LABGRP
```

R2 show snmp host

```
Notification host: 192.168.99.10      udp-port: 1162  VRFName: MGMT  type: trap
user: labmon      security model: v3 priv

Notification host: 192.168.99.10      udp-port: 162   VRFName: MGMT  type: trap
user: labmon      security model: v3 priv
```

R2 show snmp engineID

```
Local SNMP engineID: 800000090300525400C80201
Remote Engine ID      IP-addr      Port
```

R3 show snmp user

```
User name: labmon
Engine ID: 800000090300525400C80301
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: LABGRP
```

R3 show snmp host

```
Notification host: 192.168.99.10      udp-port: 1162  VRFName: MGMT  type: trap
user: labmon      security model: v3 priv

Notification host: 192.168.99.10      udp-port: 162   VRFName: MGMT  type: trap
user: labmon      security model: v3 priv
```

R3 show snmp engineID

```
Local SNMP engineID: 800000090300525400C80301
Remote Engine ID      IP-addr      Port
```

Teardown (suite) Close Nms, Close All Connections

5.09 PASS Capture The NMS View Of Every Router (0.5s, 16 commands)

The same system group, but seen over SNMPv3 from the NMS rather than read off the console. Captured next to the CLI output so the two can be compared in the evidence PDF.

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

NMS ip -br addr; ip route

```
lo          UNKNOWN      127.0.0.1/8  ::1/128
ens3        UP           10.0.2.15/24 metric 100 fec0::5054:ff:febb:101/64 fe80::5054:ff:febb:101/64
ens4        UP           192.168.10.20/24 fe80::5054:ff:febb:102/64
ens5        UP           192.168.99.10/24 fe80::5054:ff:febb:103/64
default via 10.0.2.2 dev ens3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev ens3 proto kernel scope link src 10.0.2.15 metric 100
10.0.2.2 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.0.2.3 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.30.0.0/16 via 192.168.10.1 dev ens4
192.168.10.0/24 dev ens4 proto kernel scope link src 192.168.10.20
192.168.20.0/24 via 192.168.10.1 dev ens4
192.168.30.0/24 via 192.168.10.1 dev ens4
192.168.99.0/24 dev ens5 proto kernel scope link src 192.168.99.10
```

NMS ip -br addr; ip route

```
lo          UNKNOWN      127.0.0.1/8  ::1/128
ens3        UP           10.0.2.15/24 metric 100 fec0::5054:ff:febb:101/64 fe80::5054:ff:febb:101/64
ens4        UP           192.168.10.20/24 fe80::5054:ff:febb:102/64
ens5        UP           192.168.99.10/24 fe80::5054:ff:febb:103/64
default via 10.0.2.2 dev ens3 proto dhcp src 10.0.2.15 metric 100
```

```

10.0.2.0/24 dev ens3 proto kernel scope link src 10.0.2.15 metric 100
10.0.2.2 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.0.2.3 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.30.0.0/16 via 192.168.10.1 dev ens4
192.168.10.0/24 dev ens4 proto kernel scope link src 192.168.10.20
192.168.20.0/24 via 192.168.10.1 dev ens4
192.168.30.0/24 via 192.168.10.1 dev ens4
192.168.99.0/24 dev ens5 proto kernel scope link src 192.168.99.10

```

```

NMS snmpget --version 2>&1 | head -1
NET-SNMP version: 5.9.4.pre2

```

```

NMS snmpget --version 2>&1 | head -1
NET-SNMP version: 5.9.4.pre2

```

```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.10.1
1.3.6.1.2.1.1

```

```

.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (839598) 2:19:55.98
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00

```

```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.10.1
1.3.6.1.2.1.1

```

```

.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (839598) 2:19:55.98
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00

```

```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.10.1
1.3.6.1.2.1.2.2.1.2

```

```

.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "GigabitEthernet5"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.10 = STRING: "Tunnel1"
.1.3.6.1.2.1.2.2.1.2.11 = STRING: "Loopback1"

```

```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.10.1
1.3.6.1.2.1.2.2.1.2

```

```

.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "GigabitEthernet5"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.10 = STRING: "Tunnel1"
.1.3.6.1.2.1.2.2.1.2.11 = STRING: "Loopback1"

```

```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.20.1
1.3.6.1.2.1.1

```

```

.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (839048) 2:19:50.48
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

```

```

.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.20.1
1.3.6.1.2.1.1
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (839048) 2:19:50.48
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.20.1
1.3.6.1.2.1.2.2.1.2
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.20.1
1.3.6.1.2.1.2.2.1.2
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.30.1
1.3.6.1.2.1.1
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (839020) 2:19:50.20
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.30.1
1.3.6.1.2.1.1
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (839020) 2:19:50.20
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.30.1
1.3.6.1.2.1.2.2.1.2
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"

```

```
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.30.1
1.3.6.1.2.1.2.2.1.2
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
```

Teardown (suite) Close Nms, Close All Connections

5.10 PASS Capture Collector State From The NMS (0.3s, 12 commands)

What the collectors actually hold: the tail of each router's syslog file and of snmptrapd's decrypted log. This is the management-plane evidence that used to come off h1.

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

NMS systemctl is-active rsyslog snmptrapd lab-trapcap

```
active
active
active
```

NMS systemctl is-active rsyslog snmptrapd lab-trapcap

```
active
active
active
```

NMS sudo ls -l /var/log/lab

```
total 4208
-rw-r--r-- 1 syslog syslog 226349 Sep  7 19:51 192.168.99.1.log
-rw-r--r-- 1 syslog syslog 268289 Sep  7 19:51 192.168.99.2.log
-rw-r--r-- 1 syslog syslog 160684 Sep  7 19:51 192.168.99.3.log
-rw-r--r-- 1 root  root  1133863 Sep  7 19:52 snmptrapd.log
-rw-r--r-- 1 root  root  2489750 Sep  7 19:52 traps-raw.log
```

NMS sudo ls -l /var/log/lab

```
total 4208
-rw-r--r-- 1 syslog syslog 226349 Sep  7 19:51 192.168.99.1.log
-rw-r--r-- 1 syslog syslog 268289 Sep  7 19:51 192.168.99.2.log
-rw-r--r-- 1 syslog syslog 160684 Sep  7 19:51 192.168.99.3.log
-rw-r--r-- 1 root  root  1133863 Sep  7 19:52 snmptrapd.log
-rw-r--r-- 1 root  root  2489750 Sep  7 19:52 traps-raw.log
```

NMS sudo tail -40 /var/log/lab/192.168.10.1.log

```
tail: cannot open '/var/log/lab/192.168.10.1.log' for reading: No such file or directory
```

NMS sudo tail -40 /var/log/lab/192.168.10.1.log

```
tail: cannot open '/var/log/lab/192.168.10.1.log' for reading: No such file or directory
```

NMS sudo tail -40 /var/log/lab/192.168.20.1.log

```
tail: cannot open '/var/log/lab/192.168.20.1.log' for reading: No such file or directory
```

NMS sudo tail -40 /var/log/lab/192.168.20.1.log

```
tail: cannot open '/var/log/lab/192.168.20.1.log' for reading: No such file or directory
```

NMS sudo tail -40 /var/log/lab/192.168.30.1.log

```
tail: cannot open '/var/log/lab/192.168.30.1.log' for reading: No such file or directory
```

NMS sudo tail -40 /var/log/lab/192.168.30.1.log

```
tail: cannot open '/var/log/lab/192.168.30.1.log' for reading: No such file or directory
```

NMS sudo tail -40 /var/log/lab/snmptrapd.log

```
2026-09-07 19:51:04 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:2:18:53.61 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.286 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.286 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.286 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.286 "lab"
2026-09-07 19:51:04 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:2:18:53.63 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.287 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.287 2
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.287 3 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.287 "lab"
2026-09-07 19:51:04 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:2:18:53.67 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.2
iso.3.6.1.4.1.9.9.43.1.1.1.0 833366 iso.3.6.1.4.1.9.9.43.1.1.6.1.6.7 5
iso.3.6.1.4.1.9.9.43.1.1.6.1.8.7 "lab"
```

```

2026-09-07 19:51:05 <UNKNOWN> [UDP: [192.168.99.2]:52037->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:2:18:48.96 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.0.1
iso.3.6.1.4.1.9.2.9.3.1.1.434.1 6 iso.3.6.1.2.1.6.13.1.1.10.0.2.15.22.10.0.2.2.55266 8
iso.3.6.1.4.1.9.2.6.1.1.5.10.0.2.15.22.10.0.2.2.55266 72
iso.3.6.1.4.1.9.2.6.1.1.1.10.0.2.15.22.10.0.2.2.55266 2208
iso.3.6.1.4.1.9.2.6.1.1.2.10.0.2.15.22.10.0.2.2.55266 14371 iso.3.6.1.4.1.9.2.9.2.1.18.434 "lab"
2026-09-07 19:51:05 <UNKNOWN> [UDP: [192.168.99.3]:52396->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:2:18:48.52 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.0.1
iso.3.6.1.4.1.9.2.9.3.1.1.434.1 6 iso.3.6.1.2.1.6.13.1.1.10.0.2.15.22.10.0.2.2.51432 8
iso.3.6.1.4.1.9.2.6.1.1.5.10.0.2.15.22.10.0.2.2.51432 61
iso.3.6.1.4.1.9.2.6.1.1.1.10.0.2.15.22.10.0.2.2.51432 2208
iso.3.6.1.4.1.9.2.6.1.1.2.10.0.2.15.22.10.0.2.2.51432 14371 iso.3.6.1.4.1.9.2.9.2.1.18.434 "lab"
2026-09-07 19:51:05 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:2:18:53.72 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.288 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.288 2
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.288 3 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.288 "lab"
2026-09-07 19:51:05 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:2:18:53.76 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.2
iso.3.6.1.4.1.9.9.43.1.1.1.0 833372 iso.3.6.1.4.1.9.9.43.1.1.6.1.6.8 5
iso.3.6.1.4.1.9.9.43.1.1.6.1.8.8 "lab"
... 26 more lines

```

NMS sudo tail -40 /var/log/lab/snmptrapd.log

```

2026-09-07 19:51:04 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:2:18:53.61 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.286 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.286 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.286 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.286 "lab"
2026-09-07 19:51:04 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:2:18:53.63 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.287 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.287 2
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.287 3 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.287 "lab"
2026-09-07 19:51:04 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:2:18:53.67 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.2
iso.3.6.1.4.1.9.9.43.1.1.1.0 833366 iso.3.6.1.4.1.9.9.43.1.1.6.1.6.7 5
iso.3.6.1.4.1.9.9.43.1.1.6.1.8.7 "lab"
2026-09-07 19:51:05 <UNKNOWN> [UDP: [192.168.99.2]:52037->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:2:18:48.96 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.0.1
iso.3.6.1.4.1.9.2.9.3.1.1.434.1 6 iso.3.6.1.2.1.6.13.1.1.10.0.2.15.22.10.0.2.2.55266 8
iso.3.6.1.4.1.9.2.6.1.1.5.10.0.2.15.22.10.0.2.2.55266 72
iso.3.6.1.4.1.9.2.6.1.1.1.10.0.2.15.22.10.0.2.2.55266 2208
iso.3.6.1.4.1.9.2.6.1.1.2.10.0.2.15.22.10.0.2.2.55266 14371 iso.3.6.1.4.1.9.2.9.2.1.18.434 "lab"
2026-09-07 19:51:05 <UNKNOWN> [UDP: [192.168.99.3]:52396->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:2:18:48.52 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.0.1
iso.3.6.1.4.1.9.2.9.3.1.1.434.1 6 iso.3.6.1.2.1.6.13.1.1.10.0.2.15.22.10.0.2.2.51432 8
iso.3.6.1.4.1.9.2.6.1.1.5.10.0.2.15.22.10.0.2.2.51432 61
iso.3.6.1.4.1.9.2.6.1.1.1.10.0.2.15.22.10.0.2.2.51432 2208
iso.3.6.1.4.1.9.2.6.1.1.2.10.0.2.15.22.10.0.2.2.51432 14371 iso.3.6.1.4.1.9.2.9.2.1.18.434 "lab"
2026-09-07 19:51:05 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:2:18:53.72 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.288 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.288 2
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.288 3 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.288 "lab"
2026-09-07 19:51:05 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:2:18:53.76 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.2
iso.3.6.1.4.1.9.9.43.1.1.1.0 833372 iso.3.6.1.4.1.9.9.43.1.1.6.1.6.8 5
iso.3.6.1.4.1.9.9.43.1.1.6.1.8.8 "lab"
... 26 more lines

```

Teardown (suite) Close Nms, Close All Connections

5.11 PASS Capture Config And Routing Table From Every Host (0.1s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria

- output contains **eth1** (checked 3×)
- output contains **default via** (checked 3×)

How it was checked

H1 ip addr

```

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:01:01 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0

```

```

        valid_lft 78074sec preferred_lft 67274sec
inet6 fec0::5054:ff:feaa:101/64 scope site dynamic noprefixroute flags 100
        valid_lft 86355sec preferred_lft 14355sec
inet6 fe80::5054:ff:feaa:101/64 scope link
        valid_lft forever preferred_lft forever
... 84 more lines

```

H1 ip route

```

default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.10.1 dev eth1
192.168.20.10 via 192.168.10.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.229.55 metric 1003
192.168.10.0/24 dev eth1 scope link src 192.168.10.10
192.168.20.0/24 via 192.168.10.1 dev eth1
192.168.20.10 via 192.168.10.1 dev eth1
192.168.30.0/24 via 192.168.10.1 dev eth1
192.168.20.10 via 192.168.10.1 dev eth1

```

H2 ip addr

```

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:02:01 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 78075sec preferred_lft 67275sec
    inet6 fec0::5054:ff:feaa:201/64 scope site dynamic noprefixroute flags 100
        valid_lft 86383sec preferred_lft 14383sec
    inet6 fe80::5054:ff:feaa:201/64 scope link
        valid_lft forever preferred_lft forever
... 84 more lines

```

H2 ip route

```

default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.20.1 dev eth1
192.168.30.10 via 192.168.20.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.199.41 metric 1003
192.168.10.0/24 via 192.168.20.1 dev eth1
192.168.30.10 via 192.168.20.1 dev eth1
192.168.20.0/24 dev eth1 scope link src 192.168.20.10
192.168.30.0/24 via 192.168.20.1 dev eth1
192.168.30.10 via 192.168.20.1 dev eth1

```

H3 ip addr

```

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:03:01 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 78075sec preferred_lft 67275sec
    inet6 fec0::5054:ff:feaa:301/64 scope site dynamic noprefixroute flags 100
        valid_lft 86379sec preferred_lft 14379sec
    inet6 fe80::5054:ff:feaa:301/64 scope link
        valid_lft forever preferred_lft forever
... 84 more lines

```

H3 ip route

```

default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.30.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.207.242 metric 1003
192.168.10.0/24 via 192.168.30.1 dev eth1
192.168.20.0/24 via 192.168.30.1 dev eth1
192.168.30.0/24 dev eth1 scope link src 192.168.30.10

```

Teardown (suite) Close Nms, Close All Connections

5.12 PASS Every Router Routing Table Holds All Three LANs (0.2s, 3 commands)

A content check on what was just captured: in a working hub-and-spoke every router reaches every LAN, the spokes reaching each other's through the hub.

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria

- output contains **#{R1_LAN_NET}** (checked 3×)
- output contains **#{R2_LAN_NET}** (checked 3×)
- output contains **#{R3_LAN_NET}** (checked 3×)

How it was checked

R1 show ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected
```

```
Gateway of last resort is 10.0.2.2 to network 0.0.0.0
... 30 more lines
```

R2 show ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected
```

```
Gateway of last resort is 10.0.2.2 to network 0.0.0.0
... 24 more lines
```

R3 show ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected
```

```
Gateway of last resort is 10.0.2.2 to network 0.0.0.0
... 24 more lines
```

Teardown (suite) Close Nms, Close All Connections

06 Wire Encryption

6.01 PASS Protected Traffic Appears On The Link As ESP (2.3s, 9 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **ESP**
- output matches `${R2_LINK_HUB}\s+>\s+${R2_LINK_LOCAL}`

How it was checked

R1 no monitor capture ROBOTCAP

Capture does not exist

R1 monitor capture ROBOTCAP interface GigabitEthernet2 both
\${output} =

R1 monitor capture ROBOTCAP match ipv4 any any
\${output} =

R1 monitor capture ROBOTCAP buffer size 5
\${output} =

R1 monitor capture ROBOTCAP start
Started capture point : ROBOTCAP

R1 ping 172.16.2.1 source Loopback0 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.2.1, timeout is 2 seconds:
Packet sent with a source address of 172.16.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

R1 monitor capture ROBOTCAP stop
Stopped capture point : ROBOTCAP

R1 show monitor capture ROBOTCAP buffer brief

#	size	timestamp	source	destination	dscp	protocol
0	186	0.000000	10.10.10.1	-> 10.10.10.2	0 BE	ESP
1	186	0.000992	10.10.10.2	-> 10.10.10.1	0 BE	ESP
2	186	0.000992	10.10.10.1	-> 10.10.10.2	0 BE	ESP
3	186	0.001999	10.10.10.2	-> 10.10.10.1	0 BE	ESP
4	186	0.003006	10.10.10.1	-> 10.10.10.2	0 BE	ESP
5	186	0.003998	10.10.10.2	-> 10.10.10.1	0 BE	ESP
6	186	0.003998	10.10.10.1	-> 10.10.10.2	0 BE	ESP
7	186	0.003998	10.10.10.2	-> 10.10.10.1	0 BE	ESP
8	186	0.005005	10.10.10.1	-> 10.10.10.2	0 BE	ESP
9	186	0.005005	10.10.10.2	-> 10.10.10.1	0 BE	ESP
10	122	0.027999	10.10.10.2	-> 10.10.10.1	48 CS6	ESP

... 21 more lines

R1 no monitor capture ROBOTCAP
\${output} =

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, Close All Routers

6.02 PASS Protected Endpoint Addresses Never Appear In Cleartext On The Link (2.3s, 9 commands)

The strongest statement this lab can make about encryption: traffic between the loopbacks crosses the wire without either loopback address being visible, and with no cleartext ICMP.

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain `${R1_LOOPBACK}`
- output does not contain `${R2_LOOPBACK}`
- output does not contain **ICMP**

How it was checked

R1 no monitor capture ROBOTCAP
Capture does not exist

R1 monitor capture ROBOTCAP interface GigabitEthernet2 both

```

${output} =
R1 monitor capture ROBOTCAP match ipv4 any any
${output} =
R1 monitor capture ROBOTCAP buffer size 5
${output} =
R1 monitor capture ROBOTCAP start
Started capture point : ROBOTCAP
R1 ping 172.16.2.1 source Loopback0 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.2.1, timeout is 2 seconds:
Packet sent with a source address of 172.16.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R1 monitor capture ROBOTCAP stop
Stopped capture point : ROBOTCAP
R1 show monitor capture ROBOTCAP buffer brief
-----
#    size    timestamp      source          destination      dscp    protocol
-----
0    186      0.000000    10.10.10.1      -> 10.10.10.2      0 BE    ESP
1    186      0.001059    10.10.10.2      -> 10.10.10.1      0 BE    ESP
2    186      0.001059    10.10.10.1      -> 10.10.10.2      0 BE    ESP
3    186      0.002050    10.10.10.2      -> 10.10.10.1      0 BE    ESP
4    186      0.002050    10.10.10.1      -> 10.10.10.2      0 BE    ESP
5    186      0.003057    10.10.10.2      -> 10.10.10.1      0 BE    ESP
6    186      0.003057    10.10.10.1      -> 10.10.10.2      0 BE    ESP
7    186      0.004049    10.10.10.2      -> 10.10.10.1      0 BE    ESP
8    186      0.004049    10.10.10.1      -> 10.10.10.2      0 BE    ESP
9    186      0.005056    10.10.10.2      -> 10.10.10.1      0 BE    ESP
10   122      0.120040    10.10.10.1      -> 10.10.10.2      48 CS6   ESP
... 22 more lines
R1 no monitor capture ROBOTCAP
${output} =

```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, Close All Routers

6.03 PASS Unprotected Link Traffic Is Visible As Cleartext ICMP (2.3s, 9 commands)

Positive control for the capture itself. Without this, the absence asserted above could just mean the capture was broken.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **ICMP**
- output matches `${R2_LINK_HUB}\s+->\s+${R2_LINK_LOCAL}\s+\S+\s+\S+\s+\S+\s+ICMP`

How it was checked

```

R1 no monitor capture ROBOTCAP
Capture does not exist
R1 monitor capture ROBOTCAP interface GigabitEthernet2 both
${output} =
R1 monitor capture ROBOTCAP match ipv4 any any
${output} =
R1 monitor capture ROBOTCAP buffer size 5
${output} =
R1 monitor capture ROBOTCAP start
Started capture point : ROBOTCAP
R1 ping 10.10.10.2 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R1 monitor capture ROBOTCAP stop
Stopped capture point : ROBOTCAP
R1 show monitor capture ROBOTCAP buffer brief
-----
#    size    timestamp      source          destination      dscp    protocol
-----
0    114      0.000000    10.10.10.1      -> 10.10.10.2      0 BE    ICMP
1    114      0.001007    10.10.10.2      -> 10.10.10.1      0 BE    ICMP
2    114      0.002014    10.10.10.1      -> 10.10.10.2      0 BE    ICMP

```

```
3 114 0.002014 10.10.10.2 -> 10.10.10.1 0 BE ICMP
4 114 0.003006 10.10.10.1 -> 10.10.10.2 0 BE ICMP
5 114 0.003006 10.10.10.2 -> 10.10.10.1 0 BE ICMP
6 114 0.003006 10.10.10.1 -> 10.10.10.2 0 BE ICMP
7 114 0.004013 10.10.10.2 -> 10.10.10.1 0 BE ICMP
8 114 0.004013 10.10.10.1 -> 10.10.10.2 0 BE ICMP
9 114 0.005005 10.10.10.2 -> 10.10.10.1 0 BE ICMP
10 138 0.105997 10.10.10.1 -> 10.10.10.2 48 CS6 ESP
... 25 more lines
```

R1 no monitor capture ROBOTCAP

`${output} =`

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, Close All Routers

07 Ipsec Negative

7.01 **PASS** IKEv2 Refuses To Establish With A Mismatched Pre-Shared Key (13.3s, 13 commands)

Proves authentication is actually enforced. Without this, a tunnel that accepted any peer would still pass suite 04.

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain **Success rate is 100 percent**
- output does not contain **READY**

How it was checked

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 crypto ikev2 keyring LAB-KR

#{output} =

R2 peer R1

#{output} =

R2 pre-shared-key WrongKeyOnPurpose999

#{output} =

R2 end

#{output} =

R1 clear crypto sa

#{output} =

R1 clear crypto ikev2 sa

#{output} =

R2 clear crypto sa

#{output} =

R2 clear crypto ikev2 sa

#{output} =

R3 clear crypto sa

#{output} =

R3 clear crypto ikev2 sa

#{output} =

R1 ping 172.16.2.1 source Loopback0 repeat 5

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 172.16.2.1, timeout is 2 seconds:

Packet sent with a source address of 172.16.1.1

.....

Success rate is 0 percent (0/5)

R2 show crypto ikev2 sa

#{output} =

Teardown (suite) Restore Baseline Crypto, Close All Routers

7.02 **PASS** Tunnel Recovers Once The Correct Key Is Restored (3.3s, 12 commands)

Also guarantees the previous test cannot leave the lab broken.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **Success rate is 100 percent**

How it was checked

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 crypto ikev2 keyring LAB-KR

#{output} =

R2 peer R1

#{output} =

R2 pre-shared-key LabPreSharedKey123

#{output} =

R2 end

#{output} =

```

R1 clear crypto sa
  ${output} =
R1 clear crypto ikev2 sa
  ${output} =
R2 clear crypto sa
  ${output} =
R2 clear crypto ikev2 sa
  ${output} =
R3 clear crypto sa
  ${output} =
R3 clear crypto ikev2 sa
  ${output} =
R1 ping 172.16.2.1 source Loopback0 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.2.1, timeout is 2 seconds:
Packet sent with a source address of 172.16.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

```

Teardown (suite) Restore Baseline Crypto, Close All Routers

7.03 PASS IPsec Rekeys And Traffic Still Flows Afterwards (95.2s, 37 commands)

Drops the SA lifetime to the 120s platform minimum, waits for the outbound SPI to rotate, then re-checks connectivity. Rekey is where real tunnels break.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **Success rate is 100 percent** (checked 2×)
 - **\${m}** is not empty (checked 11×)
 - **\${new}** differs from **\${old}** (checked 10×)
- SPI **\${new}** has not rotated yet

How it was checked

```

R1 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1 crypto ipsec security-association lifetime seconds 120
Warning! Lifetime value of 120 sec is lower than the recommended optimum value of 900 sec
R1 end
  ${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 crypto ipsec security-association lifetime seconds 120
Warning! Lifetime value of 120 sec is lower than the recommended optimum value of 900 sec
R2 end
  ${output} =
R3 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3 crypto ipsec security-association lifetime seconds 120
Warning! Lifetime value of 120 sec is lower than the recommended optimum value of 900 sec
R3 end
  ${output} =
R1 clear crypto sa
  ${output} =
R1 clear crypto ikev2 sa
  ${output} =
R2 clear crypto sa
  ${output} =
R2 clear crypto ikev2 sa
  ${output} =
R3 clear crypto sa
  ${output} =
R3 clear crypto ikev2 sa
  ${output} =
R1 ping 172.16.2.1 source Loopback0 repeat 5

```

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.2.1, timeout is 2 seconds:
Packet sent with a source address of 172.16.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 1, #pkts encrypt: 1, #pkts digest: 1
  #pkts decaps: 1, #pkts decrypt: 1, #pkts verify: 1
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 81 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 7, #pkts encrypt: 7, #pkts digest: 7
  #pkts decaps: 1, #pkts decrypt: 1, #pkts verify: 1
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 81 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 22, #pkts encrypt: 22, #pkts digest: 22
  #pkts decaps: 16, #pkts decrypt: 16, #pkts verify: 16
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 81 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 81, #pkts encrypt: 81, #pkts digest: 81
  #pkts decaps: 80, #pkts decrypt: 80, #pkts verify: 80
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 81 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 140, #pkts encrypt: 140, #pkts digest: 140
```

```

#pkts decaps: 139, #pkts decrypt: 139, #pkts verify: 139
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 200, #pkts encrypt: 200, #pkts digest: 200
#pkts decaps: 198, #pkts decrypt: 198, #pkts verify: 198
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 258, #pkts encrypt: 258, #pkts digest: 258
#pkts decaps: 256, #pkts decrypt: 256, #pkts verify: 256
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 317, #pkts encrypt: 317, #pkts digest: 317
#pkts decaps: 315, #pkts decrypt: 315, #pkts verify: 315
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 383, #pkts encrypt: 383, #pkts digest: 383
#pkts decaps: 375, #pkts decrypt: 375, #pkts verify: 375
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}

```

```

#pkts encaps: 441, #pkts encrypt: 441, #pkts digest: 441
#pkts decaps: 432, #pkts decrypt: 432, #pkts verify: 432
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 464, #pkts encrypt: 464, #pkts digest: 464
#pkts decaps: 457, #pkts decrypt: 457, #pkts verify: 457
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines

```

R1 ping 172.16.2.1 source Loopback0 repeat 5

```

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.2.1, timeout is 2 seconds:
Packet sent with a source address of 172.16.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

```

R1 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1 no crypto ipsec security-association lifetime seconds 120

```

${output} =

```

R1 end

```

${output} =

```

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 no crypto ipsec security-association lifetime seconds 120

```

${output} =

```

R2 end

```

${output} =

```

R3 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R3 no crypto ipsec security-association lifetime seconds 120

```

${output} =

```

R3 end

```

${output} =

```

Teardown (this test) Restore Default SA Lifetime

08 Management Api

8.01 PASS RESTCONF Reports The Same Hostname As The CLI (0.4s, 3 commands)

Setup (suite) Open All Routers, Create RESTCONF Sessions

Acceptance criteria

- output contains `${name}` (checked 3×)

How it was checked

```
R1 show running-config | include ^hostname
hostname c8000v-r1
```

```
R2 show running-config | include ^hostname
hostname c8000v-r2
```

```
R3 show running-config | include ^hostname
hostname c8000v-r3
```

Teardown (suite) Run Keyword And Ignore Error, Delete RESTCONF Loopback, Close All Routers

8.02 PASS RESTCONF Interface State Agrees With The CLI (0.1s, 1 commands)

Setup (suite) Open All Routers, Create RESTCONF Sessions

Acceptance criteria

- `${status}` equals up
- output matches `GigabitEthernet2` is `${R2_LINK_HUB}` is YES is `S` is up is up

How it was checked

```
R1 show ip interface brief GigabitEthernet2
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet2	10.10.10.1	YES	TFTP	up	up

Teardown (suite) Run Keyword And Ignore Error, Delete RESTCONF Loopback, Close All Routers

8.03 PASS Config Pushed Over RESTCONF Appears In The CLI (0.3s, 1 commands)

Write path plus cross-check: RESTCONF and the CLI must be two views of one datastore, not two datastores.

Setup (suite) Open All Routers, Create RESTCONF Sessions

Acceptance criteria

- `${resp.status_code}` in (201, 204) holds
- output contains `${RC_LOOPBACK_IP}`

How it was checked

```
R1 show running-config interface Loopback200
Building configuration...

Current configuration : 68 bytes
!
interface Loopback200
 ip address 198.51.100.1 255.255.255.0
end
```

Teardown (suite) Run Keyword And Ignore Error, Delete RESTCONF Loopback, Close All Routers

8.04 PASS Config Deleted Over RESTCONF Disappears From The CLI (0.8s, 1 commands)

Runs after the PUT test, which creates the interface; a 404 here would mean the create silently failed.

Setup (suite) Open All Routers, Create RESTCONF Sessions

Acceptance criteria

- `${resp.status_code}` equals 204
- RESTCONF DELETE returned `${resp.status_code}`; expected 204 for the loopback created by the previous test
- output does not contain `${RC_LOOPBACK_IP}`

How it was checked

```
R1 show ip interface brief Loopback200
```

```
^  
% Invalid input detected at '^' marker.
```

Teardown (suite) Run Keyword And Ignore Error, Delete RESTCONF Loopback, Close All Routers

8.05 **PASS** NETCONF Reports The Same Hostname As The CLI (0.8s, 1 commands)

Setup (suite) Open All Routers, Create RESTCONF Sessions

Acceptance criteria

- `${name}` equals `${R1_NAME}`
- output contains `${name}`

How it was checked

```
R1 show running-config | include ^hostname  
hostname c8000v-r1
```

Teardown (suite) Run Keyword And Ignore Error, Delete RESTCONF Loopback, Close All Routers

8.06 **PASS** NETCONF Advertises The Expected Base Capabilities (0.5s, 0 commands)

IOS-XE does not list per-module URIs such as Cisco-IOS-XE-native in the hello; modules are exposed via yang-library. That the native model is usable is proven by the hostname test above, which queries the native namespace directly.

Setup (suite) Open All Routers, Create RESTCONF Sessions

Acceptance criteria

- `${hit}` is not empty (checked 3×)

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Run Keyword And Ignore Error, Delete RESTCONF Loopback, Close All Routers

09 Bgp

9.01 PASS BGP Sessions Are Established With The Expected Peer AS (0.1s, 4 commands)

The hub holds one session per spoke; each spoke holds exactly one.

Setup (suite) Establish BGP

Acceptance criteria

- output matches `${peer}\s+4\s+${peer_asn}(\s+ld+){5}\s+!S+\s+ld+` (checked 4x)

How it was checked

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 52, main routing table version 52
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6152 total bytes of memory
BGP activity 299/290 prefixes, 312/303 paths, scan interval 60 secs
22 networks peaked at 17:59:32 Sep 7 2026 UTC (01:54:46.721 ago)
```

```
Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 52, main routing table version 52
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6152 total bytes of memory
BGP activity 299/290 prefixes, 312/303 paths, scan interval 60 secs
22 networks peaked at 17:59:32 Sep 7 2026 UTC (01:54:46.757 ago)
```

```
Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R2 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.2.1, local AS number 65002
BGP table version is 49, main routing table version 49
6 network entries using 1488 bytes of memory
6 path entries using 816 bytes of memory
7/4 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5016 total bytes of memory
BGP activity 344/338 prefixes, 349/343 paths, scan interval 60 secs
31 networks peaked at 18:00:16 Sep 7 2026 UTC (01:54:02.787 ago)
```

```
Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 1 more lines
```

R3 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.3.1, local AS number 65003
BGP table version is 364, main routing table version 364
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6168 total bytes of memory
BGP activity 172/163 prefixes, 186/177 paths, scan interval 60 secs
9 networks peaked at 17:33:25 Sep 7 2026 UTC (02:20:53.825 ago)
```

```
Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 1 more lines
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.02 **PASS** Each Spoke Learns The Other Spoke Prefix Through The Hub (25.2s, 7 commands)

The defining property of this topology: spokes never peer with each other, so R3's prefix can only reach R2 via the hub, and its AS path must show both hops.

Setup (suite) Establish BGP

Acceptance criteria

- output contains `${via}` (checked 7×)
- output matches `${R1_ASN}\s+${far_asn}` (checked 7×)

How it was checked

```
R2 show bgp ipv4 unicast 10.20.3.1
% Network not in table

R2 show bgp ipv4 unicast 10.20.3.1
% Network not in table

R2 show bgp ipv4 unicast 10.20.3.1
% Network not in table

R2 show bgp ipv4 unicast 10.20.3.1
% Network not in table

R2 show bgp ipv4 unicast 10.20.3.1
% Network not in table

R2 show bgp ipv4 unicast 10.20.3.1
BGP routing table entry for 10.20.3.1/32, version 50
Paths: (1 available, best #1, table default)
  Flag: 0x8100
  Not advertised to any peer
  Refresh Epoch 1
  65001 65003
    192.168.100.1 from 192.168.100.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65003:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 7 2026 19:54:43 UTC

R3 show bgp ipv4 unicast 10.20.2.1
BGP routing table entry for 10.20.2.1/32, version 362
Paths: (1 available, best #1, table default)
  Flag: 0x8100
  Not advertised to any peer
  Refresh Epoch 1
  65001 65002
    192.168.101.1 from 192.168.101.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65002:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 7 2026 19:54:15 UTC
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.03 **PASS** Every Peer Is Configured With MD5 Authentication (0.3s, 3 commands)

The negative test below proves a wrong password is rejected; this proves no peer was left without one in the first place.

Setup (suite) Establish BGP

Acceptance criteria

- `${remote}` is not empty (checked 3×)
 - `${remote}` matches `${secured}` exactly (checked 3×)
- `msg=${r}: neighbours ${remote} but only ${secured} carry a password`

How it was checked

```
R1 show running-config | include ^ neighbor .* remote-as|^ neighbor .* password
neighbor 192.168.100.2 remote-as 65002
neighbor 192.168.100.2 password LabBgpSecret
neighbor 192.168.101.2 remote-as 65003
neighbor 192.168.101.2 password LabBgpSecret

R2 show running-config | include ^ neighbor .* remote-as|^ neighbor .* password
```

```
neighbor 192.168.100.1 remote-as 65001
neighbor 192.168.100.1 password LabBgpSecret
```

R3 `show running-config | include ^ neighbor .* remote-as|^ neighbor .* password`

```
neighbor 192.168.101.1 remote-as 65001
neighbor 192.168.101.1 password LabBgpSecret
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.04 PASS A BFD Session Is Up For Every BGP Peer (0.1s, 3 commands)

One BFD session per peer, on the tunnel the peer address lives on -- the hub therefore holds two.

Setup (suite) Establish BGP

Acceptance criteria

- output contains `${R2_TUNNEL_LOCAL}`
- output contains `${R3_TUNNEL_LOCAL}`
- output does not contain **Down** (checked 3×)
- output contains `${{r}_TUNNEL_HUB}` (checked 2×)

How it was checked

R1 `show bfd neighbors`

```
IPv4 Sessions
NeighAddr          LD/RD          RH/RS    State    Int
192.168.100.2      4098/4097      Up       Up       Tu0
192.168.101.2      4097/4097      Up       Up       Tu1
```

R2 `show bfd neighbors`

```
IPv4 Sessions
NeighAddr          LD/RD          RH/RS    State    Int
192.168.100.1      4097/4098      Up       Up       Tu0
```

R3 `show bfd neighbors`

```
IPv4 Sessions
NeighAddr          LD/RD          RH/RS    State    Int
192.168.101.1      4097/4097      Up       Up       Tu0
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.05 PASS BGP Uses BFD For Fall-over On Every Peer (0.4s, 6 commands)

A BFD session that no protocol consumes would detect failures and change nothing, so this checks BGP is the registered client rather than just that BFD is running.

Setup (suite) Establish BGP

Acceptance criteria

- `${remote}` matches `${bfd}` exactly (checked 3×)
- `msg=${r}: neighbours ${remote} but only ${bfd} use BFD fall-over`
- output does not contain **Down** (checked 3×)

How it was checked

R1 `show running-config | include ^ neighbor .* remote-as|fall-over bfd`

```
neighbor 192.168.100.2 remote-as 65002
neighbor 192.168.100.2 fall-over bfd
neighbor 192.168.101.2 remote-as 65003
neighbor 192.168.101.2 fall-over bfd
```

R1 `show bfd neighbors client bgp`

```
IPv4 Sessions
NeighAddr          LD/RD          RH/RS    State    Int
192.168.100.2      4098/4097      Up       Up       Tu0
192.168.101.2      4097/4097      Up       Up       Tu1
```

R2 `show running-config | include ^ neighbor .* remote-as|fall-over bfd`

```
neighbor 192.168.100.1 remote-as 65001
neighbor 192.168.100.1 fall-over bfd
```

R2 `show bfd neighbors client bgp`

```
IPv4 Sessions
NeighAddr          LD/RD          RH/RS    State    Int
192.168.100.1      4097/4098      Up       Up       Tu0
```

R3 `show running-config | include ^ neighbor .* remote-as|fall-over bfd`

```
neighbor 192.168.101.1 remote-as 65001
```

```
neighbor 192.168.101.1 fall-over bfd
```

R3 show bfd neighbors client bgp

IPv4 Sessions	LD/RD	RH/RS	State	Int
NeighAddr			Up	Tu0
192.168.101.1	4097/4097	Up	Up	Tu0

Teardown (suite) Restore BGP Baseline, Close All Routers

9.06 PASS BFD Tears The Session Down Long Before The Hold Time (12.6s, 26 commands)

The test that makes BFD more than configuration. An ACL on the hub's tunnel drops only BFD control packets (UDP 3784 and 3785) and leaves BGP's TCP session untouched, so ordinary BGP liveness is unaffected. Without fall-over the session would survive until the 30s hold time expires; with it, BFD declares the peer down in about a second and BGP follows.

Setup (suite) Establish BGP

Acceptance criteria

- output matches `${peer}\s+4\s+${peer_asn}(\s+ld+){5}\s+S\s+ld+` (checked 4x)
- output does not match `^ld+$` (checked 3x)
- `${elapsed} < ${HOLD_TIME}` holds
- output contains **Known via "bgp"** (checked 2x)

How it was checked

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 52, main routing table version 52
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5720 total bytes of memory
BGP activity 299/290 prefixes, 312/303 paths, scan interval 60 secs
22 networks peaked at 17:59:32 Sep 7 2026 UTC (01:55:12.889 ago)
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
...	2	more	lines						

R1 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1 ip access-list extended BLOCK-BFD

`${output}` =

R1 deny udp any any eq 3784

`${output}` =

R1 deny udp any any eq 3785

`${output}` =

R1 permit ip any any

`${output}` =

R1 end

`${output}` =

R1 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1 interface Tunnel0

`${output}` =

R1 ip access-group BLOCK-BFD in

`${output}` =

R1 end

`${output}` =

R1 show bgp ipv4 unicast summary | include 192.168.100.2

192.168.100.2	4	65002	10	13	52	0	0	00:00:31	3
---------------	---	-------	----	----	----	---	---	----------	---

R1 show bgp ipv4 unicast summary | include 192.168.100.2

192.168.100.2	4	65002	10	13	52	0	0	00:00:32	3
---------------	---	-------	----	----	----	---	---	----------	---

R1 show bgp ipv4 unicast summary | include 192.168.100.2

192.168.100.2	4	65002	0	0	1	0	0	00:00:00 Idle	
---------------	---	-------	---	---	---	---	---	---------------	--

R1 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1 interface Tunnel0

```

${output} =
R1 no ip access-group BLOCK-BFD in
${output} =
R1 end
${output} =
R1 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1 no ip access-list extended BLOCK-BFD
${output} =
R1 end
${output} =
R1 show bgp ipv4 unicast summary
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 55, main routing table version 55
6 network entries using 1488 bytes of memory
6 path entries using 816 bytes of memory
4/4 BGP path/bestpath attribute entries using 1184 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
3 BGP community entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3584 total bytes of memory
BGP activity 299/290 prefixes, 312/306 paths, scan interval 60 secs
22 networks peaked at 17:59:32 Sep 7 2026 UTC (01:55:15.350 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines

R1 show bgp ipv4 unicast summary
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 55, main routing table version 55
6 network entries using 1488 bytes of memory
6 path entries using 816 bytes of memory
4/4 BGP path/bestpath attribute entries using 1184 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
3 BGP community entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3584 total bytes of memory
BGP activity 299/290 prefixes, 312/306 paths, scan interval 60 secs
22 networks peaked at 17:59:32 Sep 7 2026 UTC (01:55:20.386 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines

R1 show bgp ipv4 unicast summary
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 58, main routing table version 58
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
3 BGP route-map cache entries using 216 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5936 total bytes of memory
BGP activity 299/290 prefixes, 315/306 paths, scan interval 60 secs
22 networks peaked at 17:59:32 Sep 7 2026 UTC (01:55:25.420 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines

R1 show ip route 10.20.2.1
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 192.168.100.2 00:00:03 ago
  Routing Descriptor Blocks:
  * 192.168.100.2, from 192.168.100.2, 00:00:03 ago
    opaque_ptr 0x74C7DDCF3428
    Route metric is 0, traffic share count is 1
    AS Hops 1
    Route tag 65002
    MPLS label: none

R3 show ip route 10.20.2.1
Routing entry for 10.20.2.1/32
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 192.168.101.1 00:00:42 ago

```

```

Routing Descriptor Blocks:
* 192.168.101.1, from 192.168.101.1, 00:00:42 ago
  opaque_ptr 0x73EA5EFE0A68
  Route metric is 0, traffic share count is 1
  AS Hops 2
  Route tag 65001
  MPLS label: none

```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.07 **PASS** Every Peer Has An Outbound Route-map Over Per-class Prefix-lists (0.6s, 13 commands)

One prefix-list per prefix class, and a route-map clause per class so each can carry its own community. The clauses are the whitelist: the route-map's implicit deny is what stops anything unnamed being advertised.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **RM-OUT- $\{peer\}$** (checked 4×)
- output contains **PL-OWN- $\{cls\}$** (checked 9×)

How it was checked

```
R2 show running-config | include route-map RM-OUT-R2 out
neighbor 192.168.100.2 route-map RM-OUT-R2 out
```

```
R3 show running-config | include route-map RM-OUT-R3 out
neighbor 192.168.101.2 route-map RM-OUT-R3 out
```

```
R1 show ip prefix-list PL-OWN-L0
ip prefix-list PL-OWN-L0: 1 entries
seq 5 permit 10.20.1.1/32
```

```
R1 show ip prefix-list PL-OWN-NAT
ip prefix-list PL-OWN-NAT: 1 entries
seq 5 permit 10.30.1.0/24
```

```
R1 show ip prefix-list PL-OWN-LAN
ip prefix-list PL-OWN-LAN: 1 entries
seq 5 permit 192.168.10.0/24
```

```
R1 show running-config | include route-map RM-OUT-R1 out
neighbor 192.168.100.1 route-map RM-OUT-R1 out
```

```
R2 show ip prefix-list PL-OWN-L0
ip prefix-list PL-OWN-L0: 1 entries
seq 5 permit 10.20.2.1/32
```

```
R2 show ip prefix-list PL-OWN-NAT
ip prefix-list PL-OWN-NAT: 1 entries
seq 5 permit 10.30.2.0/24
```

```
R2 show ip prefix-list PL-OWN-LAN
ip prefix-list PL-OWN-LAN: 1 entries
seq 5 permit 192.168.20.0/24
```

```
R1 show running-config | include route-map RM-OUT-R1 out
neighbor 192.168.101.1 route-map RM-OUT-R1 out
```

```
R3 show ip prefix-list PL-OWN-L0
ip prefix-list PL-OWN-L0: 1 entries
seq 5 permit 10.20.3.1/32
```

```
R3 show ip prefix-list PL-OWN-NAT
ip prefix-list PL-OWN-NAT: 1 entries
seq 5 permit 10.30.3.0/24
```

```
R3 show ip prefix-list PL-OWN-LAN
ip prefix-list PL-OWN-LAN: 1 entries
seq 5 permit 192.168.30.0/24
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.08 **PASS** Every Peer Has An Inbound Policy Matching Community And AS Path (0.6s, 11 commands)

*Both matches sit in one permit clause, so they are ANDed and the clause's implicit deny is the drop: a prefix with an unknown community *or* an unexpected AS path falls through.*

Setup (suite) Establish BGP

Acceptance criteria

- output contains **RM-IN- $\{peer\}$** (checked 4×)
- output contains **community (community-list filter): CL-VALID** (checked 4×)
- output matches **as-path (as-path filter): $\{id\}^+$** (checked 4×)
- **$\{entries\}$** has exactly **9** entries (checked 3×)

How it was checked

```
R2 show running-config | include route-map RM-IN-R2 in
neighbor 192.168.100.2 route-map RM-IN-R2 in
```

```
R2 show route-map RM-IN-R2
route-map RM-IN-R2, permit, sequence 10
Match clauses:
  as-path (as-path filter): 10
  community (community-list filter): CL-VALID
Set clauses:
Policy routing matches: 0 packets, 0 bytes
```

```
R3 show running-config | include route-map RM-IN-R3 in
neighbor 192.168.101.2 route-map RM-IN-R3 in
```

```
R3 show route-map RM-IN-R3
route-map RM-IN-R3, permit, sequence 10
Match clauses:
  as-path (as-path filter): 20
  community (community-list filter): CL-VALID
Set clauses:
Policy routing matches: 0 packets, 0 bytes
```

```
R1 show ip community-list CL-VALID
Named Community standard list CL-VALID
permit 65001:1
permit 65001:2
permit 65001:3
permit 65002:1
permit 65002:2
permit 65002:3
permit 65003:1
permit 65003:2
permit 65003:3
```

```
R1 show running-config | include route-map RM-IN-R1 in
neighbor 192.168.100.1 route-map RM-IN-R1 in
```

```
R1 show route-map RM-IN-R1
route-map RM-IN-R1, permit, sequence 10
Match clauses:
  as-path (as-path filter): 10
  community (community-list filter): CL-VALID
Set clauses:
Policy routing matches: 0 packets, 0 bytes
```

```
R2 show ip community-list CL-VALID
Named Community standard list CL-VALID
permit 65001:1
permit 65001:2
permit 65001:3
permit 65002:1
permit 65002:2
permit 65002:3
permit 65003:1
permit 65003:2
permit 65003:3
```

```
R1 show running-config | include route-map RM-IN-R1 in
neighbor 192.168.101.1 route-map RM-IN-R1 in
```

```
R1 show route-map RM-IN-R1
route-map RM-IN-R1, permit, sequence 10
Match clauses:
  as-path (as-path filter): 10
  community (community-list filter): CL-VALID
Set clauses:
Policy routing matches: 0 packets, 0 bytes
```

```
R3 show ip community-list CL-VALID
Named Community standard list CL-VALID
permit 65001:1
permit 65001:2
permit 65001:3
permit 65002:1
```

```
permit 65002:2
permit 65002:3
permit 65003:1
permit 65003:2
permit 65003:3
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.09 PASS Every Received Prefix Carries The Community Of Its Origin (0.6s, 18 commands)

A prefix is tagged by the router that owns it, so the community identifies both the origin AS and the kind of prefix. Checked on the receiver, which is the only place it proves the tag survived the wire.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **\${expected}** (checked 18×)

How it was checked

R1 show bgp ipv4 unicast 10.20.2.1/32

```
BGP routing table entry for 10.20.2.1/32, version 58
Paths: (1 available, best #1, table default)
  Flag: 0x8100
  Advertised to update-groups: (Pending Update Generation)
    72
  Refresh Epoch 1
  65002
    192.168.100.2 from 192.168.100.2 (172.16.2.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65002:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 7 2026 19:54:54 UTC
```

R1 show bgp ipv4 unicast 10.20.3.1/32

```
BGP routing table entry for 10.20.3.1/32, version 50
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
    73
  Refresh Epoch 1
  65003
    192.168.101.2 from 192.168.101.2 (172.16.3.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65003:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 7 2026 19:54:15 UTC
```

R1 show bgp ipv4 unicast 10.30.2.0/24

```
BGP routing table entry for 10.30.2.0/24, version 57
Paths: (1 available, best #1, table default)
  Flag: 0x8100
  Advertised to update-groups: (Pending Update Generation)
    72
  Refresh Epoch 1
  65002
    192.168.100.2 from 192.168.100.2 (172.16.2.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65002:2
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 7 2026 19:54:54 UTC
```

R1 show bgp ipv4 unicast 10.30.3.0/24

```
BGP routing table entry for 10.30.3.0/24, version 52
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
    73
  Refresh Epoch 1
  65003
    192.168.101.2 from 192.168.101.2 (172.16.3.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65003:2
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 7 2026 19:54:15 UTC
```

R1 show bgp ipv4 unicast 192.168.20.0/24

```
BGP routing table entry for 192.168.20.0/24, version 56
Paths: (1 available, best #1, table default)
  Flag: 0x8100
  Advertised to update-groups: (Pending Update Generation)
    72
  Refresh Epoch 1
```

```
65002
 192.168.100.2 from 192.168.100.2 (172.16.2.1)
   Origin IGP, metric 0, localpref 100, valid, external, best
   Community: 65002:3
   rx pathid: 0, tx pathid: 0x0
   Updated on Sep 7 2026 19:54:54 UTC
```

R1 show bgp ipv4 unicast 192.168.30.0/24

```
BGP routing table entry for 192.168.30.0/24, version 51
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
    73
  Refresh Epoch 1
65003
 192.168.101.2 from 192.168.101.2 (172.16.3.1)
   Origin IGP, metric 0, localpref 100, valid, external, best
   Community: 65003:3
   rx pathid: 0, tx pathid: 0x0
   Updated on Sep 7 2026 19:54:15 UTC
```

R2 show bgp ipv4 unicast 10.20.1.1/32

```
BGP routing table entry for 10.20.1.1/32, version 64
Paths: (1 available, best #1, table default)
  Flag: 0x8100
  Not advertised to any peer
  Refresh Epoch 1
65001
 192.168.100.1 from 192.168.100.1 (172.16.1.1)
   Origin IGP, metric 0, localpref 100, valid, external, best
   Community: 65001:1
   rx pathid: 0, tx pathid: 0x0
   Updated on Sep 7 2026 19:54:54 UTC
```

R2 show bgp ipv4 unicast 10.20.3.1/32

```
BGP routing table entry for 10.20.3.1/32, version 61
Paths: (1 available, best #1, table default)
  Flag: 0x8100
  Not advertised to any peer
  Refresh Epoch 1
65001 65003
 192.168.100.1 from 192.168.100.1 (172.16.1.1)
   Origin IGP, localpref 100, valid, external, best
   Community: 65003:1
   rx pathid: 0, tx pathid: 0x0
   Updated on Sep 7 2026 19:54:54 UTC
```

R2 show bgp ipv4 unicast 10.30.1.0/24

```
BGP routing table entry for 10.30.1.0/24, version 63
Paths: (1 available, best #1, table default)
  Flag: 0x8100
  Not advertised to any peer
  Refresh Epoch 1
65001
 192.168.100.1 from 192.168.100.1 (172.16.1.1)
   Origin IGP, metric 0, localpref 100, valid, external, best
   Community: 65001:2
   rx pathid: 0, tx pathid: 0x0
   Updated on Sep 7 2026 19:54:54 UTC
```

R2 show bgp ipv4 unicast 10.30.3.0/24

```
BGP routing table entry for 10.30.3.0/24, version 59
Paths: (1 available, best #1, table default)
  Flag: 0x8100
  Not advertised to any peer
  Refresh Epoch 1
65001 65003
 192.168.100.1 from 192.168.100.1 (172.16.1.1)
   Origin IGP, localpref 100, valid, external, best
   Community: 65003:2
   rx pathid: 0, tx pathid: 0x0
   Updated on Sep 7 2026 19:54:54 UTC
```

R2 show bgp ipv4 unicast 192.168.10.0/24

```
BGP routing table entry for 192.168.10.0/24, version 62
Paths: (1 available, best #1, table default)
  Flag: 0x8100
  Not advertised to any peer
  Refresh Epoch 1
65001
 192.168.100.1 from 192.168.100.1 (172.16.1.1)
   Origin IGP, metric 0, localpref 100, valid, external, best
   Community: 65001:3
   rx pathid: 0, tx pathid: 0x0
   Updated on Sep 7 2026 19:54:54 UTC
```

R2 show bgp ipv4 unicast 192.168.30.0/24

BGP routing table entry for 192.168.30.0/24, version 60
Paths: (1 available, best #1, table default)
Flag: 0x8100
Not advertised to any peer
Refresh Epoch 1
65001 65003
192.168.100.1 from 192.168.100.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65003:3
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 19:54:54 UTC

R3 show bgp ipv4 unicast 10.20.1.1/32

BGP routing table entry for 10.20.1.1/32, version 361
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001
192.168.101.1 from 192.168.101.1 (172.16.1.1)
Origin IGP, metric 0, localpref 100, valid, external, best
Community: 65001:1
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 19:54:15 UTC

R3 show bgp ipv4 unicast 10.20.2.1/32

BGP routing table entry for 10.20.2.1/32, version 362
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001 65002
192.168.101.1 from 192.168.101.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65002:1
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 19:54:15 UTC

R3 show bgp ipv4 unicast 10.30.1.0/24

BGP routing table entry for 10.30.1.0/24, version 360
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001
192.168.101.1 from 192.168.101.1 (172.16.1.1)
Origin IGP, metric 0, localpref 100, valid, external, best
Community: 65001:2
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 19:54:15 UTC

R3 show bgp ipv4 unicast 10.30.2.0/24

BGP routing table entry for 10.30.2.0/24, version 363
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001 65002
192.168.101.1 from 192.168.101.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65002:2
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 19:54:15 UTC

R3 show bgp ipv4 unicast 192.168.10.0/24

BGP routing table entry for 192.168.10.0/24, version 359
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001
192.168.101.1 from 192.168.101.1 (172.16.1.1)
Origin IGP, metric 0, localpref 100, valid, external, best
Community: 65001:3
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 19:54:15 UTC

R3 show bgp ipv4 unicast 192.168.20.0/24

BGP routing table entry for 192.168.20.0/24, version 364
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001 65002
192.168.101.1 from 192.168.101.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65002:3
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 19:54:15 UTC

Teardown (suite) Restore BGP Baseline, Close All Routers

9.10 **PASS** Re-advertised Prefixes Keep Their Origin Community Through The Hub (0.1s, 4 commands)

A spoke-to-spoke prefix crosses the hub, which re-advertises it. If the hub re-tagged it the community would say 65001 and the origin would be lost, so this asserts the far spoke's own community survives the hop.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **\${expected}** (checked 4x)

How it was checked

R2 show bgp ipv4 unicast 10.20.3.1/32

```
BGP routing table entry for 10.20.3.1/32, version 61
Paths: (1 available, best #1, table default)
  Flag: 0x8100
  Not advertised to any peer
  Refresh Epoch 1
  65001 65003
    192.168.100.1 from 192.168.100.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65003:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 7 2026 19:54:54 UTC
```

R2 show bgp ipv4 unicast 192.168.30.0/24

```
BGP routing table entry for 192.168.30.0/24, version 60
Paths: (1 available, best #1, table default)
  Flag: 0x8100
  Not advertised to any peer
  Refresh Epoch 1
  65001 65003
    192.168.100.1 from 192.168.100.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65003:3
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 7 2026 19:54:54 UTC
```

R3 show bgp ipv4 unicast 10.20.2.1/32

```
BGP routing table entry for 10.20.2.1/32, version 362
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65002
    192.168.101.1 from 192.168.101.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65002:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 7 2026 19:54:15 UTC
```

R3 show bgp ipv4 unicast 192.168.20.0/24

```
BGP routing table entry for 192.168.20.0/24, version 364
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65002
    192.168.101.1 from 192.168.101.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65002:3
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 7 2026 19:54:15 UTC
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.11 **PASS** A Prefix With An Unknown Community Is Rejected Inbound (26.0s, 40 commands)

*The prefix is explicitly permitted *outbound* by a temporary clause, so the outbound whitelist cannot be what stops it -- only the inbound community match can. The test asserts R2 really did advertise it before asserting R1 refused it, otherwise it would pass whenever the advertisement failed.*

Setup (suite) Establish BGP

Acceptance criteria

- output contains **#{BAD_COMM_ADDR}** (checked 6x)
R2 is not yet advertising the test prefix
- output contains **#{BAD_COMM_ADDR}**
R2 never advertised the prefix, so this proves nothing about the inbound policy
- output contains **Network not in table**
the hub accepted a prefix carrying community #{UNKNOWN_COMM}

How it was checked

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Loopback98

#{output} =

R2 ip address 10.88.88.88 255.255.255.255

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 ip prefix-list PL-TEST seq 5 permit 10.88.88.88/32

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 route-map RM-OUT-R1 permit 5

#{output} =

R2 match ip address prefix-list PL-TEST

#{output} =

R2 set community 65500:9

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002

#{output} =

R2 address-family ipv4 unicast

#{output} =

R2 network 10.88.88.88 mask 255.255.255.255

#{output} =

R2 end

#{output} =

R2 clear ip bgp * soft out

#{output} =

R2 show bgp ipv4 unicast neighbors 192.168.100.1 advertised-routes

BGP table version is 64, local router ID is 172.16.2.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.2.1/32	0.0.0.0	0		32768	i
*> 10.30.2.0/24	0.0.0.0	0		32768	i
*> 192.168.20.0	0.0.0.0	0		32768	i

Total number of prefixes 3

R2 show bgp ipv4 unicast neighbors 192.168.100.1 advertised-routes

BGP table version is 65, local router ID is 172.16.2.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
---------	----------	--------	--------	--------	------

```
*> 10.20.2.1/32      0.0.0.0      0      32768 i
*> 10.30.2.0/24      0.0.0.0      0      32768 i
*> 192.168.20.0      0.0.0.0      0      32768 i
```

Total number of prefixes 3

R2 show bgp ipv4 unicast neighbors 192.168.100.1 advertised-routes

BGP table version is 65, local router ID is 172.16.2.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.30.2.0/24	0.0.0.0	0		32768	i
*>	192.168.20.0	0.0.0.0	0		32768	i

Total number of prefixes 3

R2 show bgp ipv4 unicast neighbors 192.168.100.1 advertised-routes

BGP table version is 65, local router ID is 172.16.2.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.30.2.0/24	0.0.0.0	0		32768	i
*>	192.168.20.0	0.0.0.0	0		32768	i

Total number of prefixes 3

R2 show bgp ipv4 unicast neighbors 192.168.100.1 advertised-routes

BGP table version is 65, local router ID is 172.16.2.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.30.2.0/24	0.0.0.0	0		32768	i
*>	192.168.20.0	0.0.0.0	0		32768	i

Total number of prefixes 3

R2 show bgp ipv4 unicast neighbors 192.168.100.1 advertised-routes

BGP table version is 65, local router ID is 172.16.2.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.30.2.0/24	0.0.0.0	0		32768	i
*>	10.88.88.88/32	0.0.0.0	0		32768	i
*>	192.168.20.0	0.0.0.0	0		32768	i

... 1 more lines

R2 show bgp ipv4 unicast neighbors 192.168.100.1 advertised-routes

BGP table version is 65, local router ID is 172.16.2.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.30.2.0/24	0.0.0.0	0		32768	i
*>	10.88.88.88/32	0.0.0.0	0		32768	i

```

*> 192.168.20.0      0.0.0.0      0      32768 i

... 1 more lines
R1 show bgp ipv4 unicast 10.88.88.88
% Network not in table
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 no network 10.88.88.88 mask 255.255.255.255
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 no route-map RM-OUT-R1 permit 5
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip prefix-list PL-TEST
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 no interface Loopback98
${output} =
R2 end
${output} =

```

Teardown (this test) Remove Test Prefix From R2

9.12 PASS Every Peer Has A Maximum-prefix Limit With Headroom (0.5s, 6 commands)

Filtering controls what a peer may send; this controls how much. The limit must sit above normal load or it would trip in steady state -- a hub holds 3 prefixes per spoke, a spoke 6 -- and far below anything that could exhaust memory.

Setup (suite) Establish BGP

Acceptance criteria

- **\${remote}** matches **\${limited}** exactly (checked 3×)
msg=\${r}: neighbours \${remote} but only \${limited} have a maximum-prefix
- **\${n}** < **\${MAX_PREFIX}** holds (checked 3×)

How it was checked

```

R1 show running-config | include ^ neighbor .* remote-as|maximum-prefix
neighbor 192.168.100.2 remote-as 65002
neighbor 192.168.101.2 remote-as 65003
neighbor 192.168.100.2 maximum-prefix 15 80
neighbor 192.168.101.2 maximum-prefix 15 80

R1 show bgp ipv4 unicast summary | begin Neighbor
Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
192.168.100.2  4      65002    15    13     58    0    0 00:00:31      3
192.168.101.2  4      65003    14    19     58    0    0 00:01:10      3

R2 show running-config | include ^ neighbor .* remote-as|maximum-prefix
neighbor 192.168.100.1 remote-as 65001
neighbor 192.168.100.1 maximum-prefix 15 80

R2 show bgp ipv4 unicast summary | begin Neighbor
Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
192.168.100.1  4      65001    13    15     65    0    0 00:00:31      6

R3 show running-config | include ^ neighbor .* remote-as|maximum-prefix
neighbor 192.168.101.1 remote-as 65001
neighbor 192.168.101.1 maximum-prefix 15 80

```


R3 show bgp ipv4 unicast summary | begin Neighbor

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
192.168.101.1	4	65001	19	14	364	0	0	00:01:10	6

Teardown (suite) Restore BGP Baseline, Close All Routers

9.13 PASS A Peer That Floods Prefixes Is Torn Down Rather Than Absorbed (37.1s, 344 commands)

R2 originates 20 extra prefixes, tagged with a valid community and carrying a valid AS path, and explicitly permitted outbound -- so neither the whitelist, the community match nor the AS-path filter can be what stops them. Only the prefix count is left. The hub must drop the session instead of accepting an unbounded table.

Setup (suite) Establish BGP

Acceptance criteria

- output does not match `^ld+$` (checked 7×)
- output contains **PfxCt**
the session went down, but not because of the prefix count: `$(state)`
- output contains `$(R2_TUNNEL_LOCAL)`
the hub logged no maximum-prefix violation for `$(R2_TUNNEL_LOCAL)`

How it was checked

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 ip prefix-list PL-FLOOD seq 5 permit 10.99.0.0/16 le 32

\$(output) =

R2 end

\$(output) =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 route-map RM-OUT-R1 permit 7

\$(output) =

R2 match ip address prefix-list PL-FLOOD

\$(output) =

R2 set community 65002:1

\$(output) =

R2 end

\$(output) =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.0.0 255.255.255.0 Null0

\$(output) =

R2 end

\$(output) =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.1.0 255.255.255.0 Null0

\$(output) =

R2 end

\$(output) =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.2.0 255.255.255.0 Null0

\$(output) =

R2 end

\$(output) =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.3.0 255.255.255.0 Null0

\$(output) =

R2 end

\$(output) =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

```

R2 ip route 10.99.4.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.5.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.6.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.7.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.8.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.9.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.10.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.11.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.12.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.13.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.14.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal

```

```

Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.15.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.16.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.17.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.18.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.19.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 network 10.99.0.0 mask 255.255.255.0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 network 10.99.1.0 mask 255.255.255.0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 network 10.99.2.0 mask 255.255.255.0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =

```

```

R2 network 10.99.3.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.4.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.5.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.6.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.7.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.8.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.9.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal

```

```

Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.10.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.11.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.12.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.13.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.14.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.15.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =

```

```

R2 network 10.99.16.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.17.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.18.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.19.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R1 show bgp ipv4 unicast summary | include 192.168.100.2
  192.168.100.2  4      65002    15      13      58      0      0 00:00:35      3
R1 show bgp ipv4 unicast summary | include 192.168.100.2
  192.168.100.2  4      65002    16      14      58      0      0 00:00:40      3
R1 show bgp ipv4 unicast summary | include 192.168.100.2
  192.168.100.2  4      65002    16      15      58      0      0 00:00:45      3
R1 show bgp ipv4 unicast summary | include 192.168.100.2
  192.168.100.2  4      65002    17      15      58      0      0 00:00:50      3
R1 show bgp ipv4 unicast summary | include 192.168.100.2
  192.168.100.2  4      65002    17      16      58      0      0 00:00:55      3
R1 show bgp ipv4 unicast summary | include 192.168.100.2
  192.168.100.2  4      65002    18      16      58      0      0 00:01:00      3
R1 show bgp ipv4 unicast summary | include 192.168.100.2
  192.168.100.2  4      65002      0      0      1      0      0 00:00:04 Idle (PfxCt)
R1 show bgp ipv4 unicast summary | include 192.168.100.2
  192.168.100.2  4      65002      0      0      1      0      0 00:00:04 Idle (PfxCt)
R1 show logging | include MAXPFEXCEED
  001148: Sep  7 19:16:34.723 UTC: %BGP-3-MAXPFEXCEED: Number of prefixes received from 192.168.100.2 1/1
  (afi/safi): 16 exceeds limit 15
  001512: Sep  7 19:36:37.503 UTC: %BGP-3-MAXPFEXCEED: Number of prefixes received from 192.168.100.2 1/1
  (afi/safi): 16 exceeds limit 15
  001835: Sep  7 19:55:54.809 UTC: %BGP-3-MAXPFEXCEED: Number of prefixes received from 192.168.100.2 1/1
  (afi/safi): 16 exceeds limit 15
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.0.0 mask 255.255.255.0
  ${output} =
R2 end

```

```

${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.0.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 no network 10.99.1.0 mask 255.255.255.0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.1.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 no network 10.99.2.0 mask 255.255.255.0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.2.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 no network 10.99.3.0 mask 255.255.255.0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.3.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 no network 10.99.4.0 mask 255.255.255.0
${output} =
R2 end
${output} =

```

```

R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.4.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.5.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.5.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.6.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.6.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.7.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.7.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.8.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal

```



```

Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.8.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 no network 10.99.9.0 mask 255.255.255.0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.9.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 no network 10.99.10.0 mask 255.255.255.0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.10.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 no network 10.99.11.0 mask 255.255.255.0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.11.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 no network 10.99.12.0 mask 255.255.255.0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

```

```

R2 no ip route 10.99.12.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.13.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.13.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.14.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.14.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.15.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.15.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.16.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.16.0 255.255.255.0 Null0

```

```

    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 no network 10.99.17.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.17.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 no network 10.99.18.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.18.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 no network 10.99.19.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.19.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no route-map RM-OUT-R1 permit 7
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip prefix-list PL-FLOOD
    ${output} =
R2 end
    ${output} =
R1 clear ip bgp 192.168.100.2
    ${output} =

```

Teardown (this test) Stop Flooding From R2

9.14 PASS The Session Recovers Once The Flood Stops (25.2s, 8 commands)

A maximum-prefix teardown is deliberately sticky -- no "restart" is configured -- so recovery takes an operator clearing it. This asserts the lab returns to a converged state rather than leaving the next suite a broken topology.

Setup (suite) Establish BGP

Acceptance criteria

- output matches `${peer}\s+4\s+${peer_asn}(\s+1d+){5}\s+1S+\s+1d+` (checked 3×)
- output contains **Known via "bgp"** (checked 5×)

How it was checked

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 85, main routing table version 85
6 network entries using 1488 bytes of memory
6 path entries using 816 bytes of memory
8/4 BGP path/bestpath attribute entries using 2368 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
7 BGP community entries using 168 bytes of memory
4 BGP route-map cache entries using 288 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5176 total bytes of memory
BGP activity 312/306 prefixes, 328/322 paths, scan interval 60 secs
22 networks peaked at 17:59:32 Sep 7 2026 UTC (01:56:31.088 ago)
```

```
Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 85, main routing table version 85
6 network entries using 1488 bytes of memory
6 path entries using 816 bytes of memory
8/4 BGP path/bestpath attribute entries using 2368 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
7 BGP community entries using 168 bytes of memory
4 BGP route-map cache entries using 288 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5176 total bytes of memory
BGP activity 312/306 prefixes, 328/322 paths, scan interval 60 secs
22 networks peaked at 17:59:32 Sep 7 2026 UTC (01:56:36.126 ago)
```

```
Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 88, main routing table version 88
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
8/7 BGP path/bestpath attribute entries using 2368 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
7 BGP community entries using 168 bytes of memory
4 BGP route-map cache entries using 288 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6328 total bytes of memory
BGP activity 315/306 prefixes, 331/322 paths, scan interval 60 secs
22 networks peaked at 17:59:32 Sep 7 2026 UTC (01:56:41.163 ago)
```

```
Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 192.168.100.2 00:00:05 ago
  Routing Descriptor Blocks:
    * 192.168.100.2, from 192.168.100.2, 00:00:05 ago
      opaque_ptr 0x74C7DDCF3428
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none
```

R3 show ip route 10.20.2.1

```

% Subnet not in table
R3 show ip route 10.20.2.1
% Subnet not in table
R3 show ip route 10.20.2.1
% Subnet not in table
R3 show ip route 10.20.2.1
Routing entry for 10.20.2.1/32
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 192.168.101.1 00:00:03 ago
  Routing Descriptor Blocks:
    * 192.168.101.1, from 192.168.101.1, 00:00:03 ago
      opaque_ptr 0x73EA5EFE0A68
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none

```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.15 PASS A Prefix With An Invalid AS Path Is Rejected Inbound (15.9s, 39 commands)

Same prefix, this time with a valid community but an AS path this topology cannot produce. Only the AS-path match differs, so a rejection here isolates that filter.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **#{BAD_COMM_ADDR}** (checked 4x)
R2 is not yet advertising the test prefix
- output contains **#{BAD_COMM_ADDR}**
R2 never advertised the prefix, so this proves nothing about the inbound policy
- output contains **Network not in table**
the hub accepted a prefix whose AS path contained **#{BOGUS_AS}**

How it was checked

```

R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 interface Loopback98
  #{output} =
R2 ip address 10.88.88.88 255.255.255.255
  #{output} =
R2 end
  #{output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip prefix-list PL-TEST seq 5 permit 10.88.88.88/32
  #{output} =
R2 end
  #{output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 route-map RM-OUT-R1 permit 5
  #{output} =
R2 match ip address prefix-list PL-TEST
  #{output} =
R2 set community 65002:1
  #{output} =
R2 set as-path prepend 65099
  #{output} =
R2 end
  #{output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  #{output} =
R2 address-family ipv4 unicast

```

```

${output} =
R2 network 10.88.88.88 mask 255.255.255.255
${output} =
R2 end
${output} =
R2 clear ip bgp * soft out
${output} =
R2 show bgp ipv4 unicast neighbors 192.168.100.1 advertised-routes
BGP table version is 118, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

      Network          Next Hop              Metric LocPrf Weight Path
*>  10.20.2.1/32        0.0.0.0                  0           32768 i
*>  10.30.2.0/24        0.0.0.0                  0           32768 i
*>  192.168.20.0        0.0.0.0                  0           32768 i

Total number of prefixes 3
R2 show bgp ipv4 unicast neighbors 192.168.100.1 advertised-routes
BGP table version is 119, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

      Network          Next Hop              Metric LocPrf Weight Path
*>  10.20.2.1/32        0.0.0.0                  0           32768 i
*>  10.30.2.0/24        0.0.0.0                  0           32768 i
*>  192.168.20.0        0.0.0.0                  0           32768 i

Total number of prefixes 3
R2 show bgp ipv4 unicast neighbors 192.168.100.1 advertised-routes
BGP table version is 119, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

      Network          Next Hop              Metric LocPrf Weight Path
*>  10.20.2.1/32        0.0.0.0                  0           32768 i
*>  10.30.2.0/24        0.0.0.0                  0           32768 i
*>  192.168.20.0        0.0.0.0                  0           32768 i

Total number of prefixes 3
R2 show bgp ipv4 unicast neighbors 192.168.100.1 advertised-routes
BGP table version is 119, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

      Network          Next Hop              Metric LocPrf Weight Path
*>  10.20.2.1/32        0.0.0.0                  0           32768 i
*>  10.30.2.0/24        0.0.0.0                  0           32768 i
*>  10.88.88.88/32      0.0.0.0                  0           32768 i
*>  192.168.20.0        0.0.0.0                  0           32768 i

... 1 more lines
R2 show bgp ipv4 unicast neighbors 192.168.100.1 advertised-routes
BGP table version is 119, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found
      Network          Next Hop              Metric LocPrf Weight Path

```

```
*> 10.20.2.1/32      0.0.0.0      0      32768 i
*> 10.30.2.0/24      0.0.0.0      0      32768 i
*> 10.88.88.88/32    0.0.0.0      0      32768 i
*> 192.168.20.0      0.0.0.0      0      32768 i
```

... 1 more lines

R1 show bgp ipv4 unicast 10.88.88.88

% Network not in table

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002

#{output} =

R2 address-family ipv4 unicast

#{output} =

R2 no network 10.88.88.88 mask 255.255.255.255

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 no route-map RM-OUT-R1 permit 5

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 no ip prefix-list PL-TEST

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 no interface Loopback98

#{output} =

R2 end

#{output} =

Teardown (this test) Remove Test Prefix From R2

9.16 PASS Each Peer Is Advertised Exactly Its Whitelisted Prefixes (0.3s, 4 commands)

Not "at least" -- exactly. A whitelist that permits more than intended still passes a containment check, so the advertised set is compared for equality against what the peer should get: a spoke sends only its own prefixes, the hub sends its own plus the far spoke's.

Setup (suite) Establish BGP

Acceptance criteria

- **#{actual}** matches **#{expected}** exactly (checked 4x)
msg=#{from} advertises #{actual} to #{to}, expected #{expected}

How it was checked

R2 show bgp ipv4 unicast neighbors 192.168.100.2 advertised-routes

BGP table version is 88, local router ID is 172.16.1.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.1.1/32	0.0.0.0	0		32768	i
*>	10.20.3.1/32	192.168.101.2	0		0 65003	i
*>	10.30.1.0/24	0.0.0.0	0		32768	i
*>	10.30.3.0/24	192.168.101.2	0		0 65003	i
*>	192.168.10.0	0.0.0.0	0		32768	i

... 3 more lines

R3 show bgp ipv4 unicast neighbors 192.168.101.2 advertised-routes

```

BGP table version is 88, local router ID is 172.16.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.1.1/32	0.0.0.0	0		32768	i
*> 10.20.2.1/32	192.168.100.2	0		0 65002	i
*> 10.30.1.0/24	0.0.0.0	0		32768	i
*> 10.30.2.0/24	192.168.100.2	0		0 65002	i
*> 192.168.10.0	0.0.0.0	0		32768	i

... 3 more lines

R1 show bgp ipv4 unicast neighbors 192.168.100.1 advertised-routes

```

BGP table version is 120, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.2.1/32	0.0.0.0	0		32768	i
*> 10.30.2.0/24	0.0.0.0	0		32768	i
*> 192.168.20.0	0.0.0.0	0		32768	i

Total number of prefixes 3

R1 show bgp ipv4 unicast neighbors 192.168.101.1 advertised-routes

```

BGP table version is 370, local router ID is 172.16.3.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.3.1/32	0.0.0.0	0		32768	i
*> 10.30.3.0/24	0.0.0.0	0		32768	i
*> 192.168.30.0	0.0.0.0	0		32768	i

Total number of prefixes 3

Teardown (suite) Restore BGP Baseline, Close All Routers

9.17 PASS A Prefix Outside The Whitelist Is Not Advertised (10.5s, 20 commands)

The point of a whitelist. A spoke originates a prefix nobody authorised; it appears in that spoke's own BGP table, so the origination worked, and must still never reach the hub.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **\${OFF_LIST_ADDR}**
R2 never originated the prefix, so this test proves nothing
- output does not contain **\${OFF_LIST_ADDR}/32**
R2 advertised an unlisted prefix to the hub
- output contains **Network not in table**
an unlisted prefix reached the hub

How it was checked

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Loopback99

\${output} =

R2 ip address 10.77.77.77 255.255.255.255

\${output} =

R2 end

\${output} =

R2 configure terminal


```

Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 network 10.77.77.77 mask 255.255.255.255
${output} =
R2 end
${output} =
R2 show bgp ipv4 unicast 10.77.77.77
BGP routing table entry for 10.77.77.77/32, version 121
Paths: (1 available, best #1, table default)
Flag: 0x8100
Not advertised to any peer
Refresh Epoch 1
Local
0.0.0.0 from 0.0.0.0 (172.16.2.1)
Origin IGP, metric 0, localpref 100, weight 32768, valid, sourced, local, best
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 19:56:46 UTC

R2 show bgp ipv4 unicast neighbors 192.168.100.1 advertised-routes
BGP table version is 121, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

      Network          Next Hop           Metric LocPrf Weight Path
*>  10.20.2.1/32        0.0.0.0              0         32768 i
*>  10.30.2.0/24        0.0.0.0              0         32768 i
*>  192.168.20.0        0.0.0.0              0         32768 i

Total number of prefixes 3

R1 show bgp ipv4 unicast 10.77.77.77
% Network not in table

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 no network 10.77.77.77 mask 255.255.255.255
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 no interface Loopback99
${output} =
R2 end
${output} =

```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.18 PASS Each Router Receives The Peer Prefix (0.1s, 2 commands)

Setup (suite) Establish BGP

Acceptance criteria

- output matches `!*>ls+${R2_BGP_PREFIX}/32ls+${R2_TUNNEL_LOCAL}`
- output contains `${R2_ASN}`
- output matches `!*>ls+${R1_BGP_PREFIX}/32ls+${R2_TUNNEL_HUB}`
- output contains `${R1_ASN}`

How it was checked

```

R1 show bgp ipv4 unicast
BGP table version is 88, local router ID is 172.16.1.1

```

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
 r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
 x best-external, a additional-path, c RIB-compressed,
 t secondary path, L long-lived-stale,
 Origin codes: i - IGP, e - EGP, ? - incomplete
 RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.1.1/32	0.0.0.0	0		32768	i
*> 10.20.2.1/32	192.168.100.2	0		0	65002 i
*> 10.20.3.1/32	192.168.101.2	0		0	65003 i
*> 10.30.1.0/24	0.0.0.0	0		32768	i
*> 10.30.2.0/24	192.168.100.2	0		0	65002 i

... 4 more lines

R2 show bgp ipv4 unicast

BGP table version is 122, local router ID is 172.16.2.1
 Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
 r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
 x best-external, a additional-path, c RIB-compressed,
 t secondary path, L long-lived-stale,
 Origin codes: i - IGP, e - EGP, ? - incomplete
 RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.1.1/32	192.168.100.1	0		0	65001 i
*> 10.20.2.1/32	0.0.0.0	0		32768	i
*> 10.20.3.1/32	192.168.100.1			0	65001 65003 i
*> 10.30.1.0/24	192.168.100.1	0		0	65001 i
*> 10.30.2.0/24	0.0.0.0	0		32768	i

... 4 more lines

Teardown (suite) Restore BGP Baseline, Close All Routers

9.19 PASS Peer Prefix Is Learned Via BGP And Not A Static Route (0.0s, 1 commands)

Distinguishes a genuinely BGP-installed route from one that merely happens to exist. eBGP is distance 20; a static is 1.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **Known via "bgp \${R1_ASN}"**
- output contains **distance 20**
- output contains **type external**

How it was checked

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 192.168.100.2 00:00:47 ago
  Routing Descriptor Blocks:
    * 192.168.100.2, from 192.168.100.2, 00:00:47 ago
      opaque_ptr 0x74C7DDCF3428
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.20 PASS BGP Route Resolves Out The Encrypted Tunnel (0.0s, 1 commands)

The next hop is a tunnel address, so the forwarding path must recurse to Tunnel0 rather than the cleartext link.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **\${R2_TUNNEL_LOCAL}**
- output contains **Tunnel0**

How it was checked

R1 show ip cef 10.20.2.1

```
10.20.2.1/32
```

nexthop 192.168.100.2 Tunnel0

Teardown (suite) Restore BGP Baseline, Close All Routers

9.21 **PASS** Traffic Over The BGP-Learned Route Reaches The Peer (0.0s, 1 commands)

End to end: this only succeeds because BGP installed the route.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **Success rate is 100 percent**

How it was checked

R1 ping 10.20.2.1 source Loopback1 repeat 5

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:

Packet sent with a source address of 10.20.1.1

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

Teardown (suite) Restore BGP Baseline, Close All Routers

9.22 **PASS** BGP-Routed Traffic Is Encrypted (5.7s, 7 commands)

Setup (suite) Establish BGP

Acceptance criteria

- output contains **Success rate is 100 percent**
- **\${r1_enc} >= \${expected}** holds (checked 2x)
- **\${r1_dec} >= \${expected}** holds (checked 2x)
- **\${r2_enc} >= \${expected}** holds (checked 2x)
- **\${r2_dec} >= \${expected}** holds (checked 2x)

How it was checked

R1 show crypto ipsec sa

interface: Tunnel0

Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)

local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)

remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)

current_peer 10.10.10.2 port 500

PERMIT, flags={origin_is_acl,}

#pkts encaps: 866, #pkts encrypt: 866, #pkts digest: 866

#pkts decaps: 868, #pkts decrypt: 868, #pkts verify: 868

#pkts compressed: 0, #pkts decompressed: 0

#pkts not compressed: 0, #pkts compr. failed: 0

#pkts not decompressed: 0, #pkts decompress failed: 0

#send errors 0, #recv errors 0

... 81 more lines

R2 show crypto ipsec sa

interface: Tunnel0

Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

protected vrf: (none)

local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)

remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)

current_peer 10.10.10.1 port 500

PERMIT, flags={origin_is_acl,}

#pkts encaps: 2259, #pkts encrypt: 2259, #pkts digest: 2259

#pkts decaps: 2268, #pkts decrypt: 2268, #pkts verify: 2268

#pkts compressed: 0, #pkts decompressed: 0

#pkts not compressed: 0, #pkts compr. failed: 0

#pkts not decompressed: 0, #pkts decompress failed: 0

#send errors 0, #recv errors 0

... 33 more lines

R1 ping 10.20.2.1 source Loopback1 repeat 20

Type escape sequence to abort.

Sending 20, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:

Packet sent with a source address of 10.20.1.1

!!!!!!!!!!!!!!!!!!!!

Success rate is 100 percent (20/20), round-trip min/avg/max = 1/1/2 ms

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 866, #pkts encrypt: 866, #pkts digest: 866
  #pkts decaps: 868, #pkts decrypt: 868, #pkts verify: 868
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 81 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 2259, #pkts encrypt: 2259, #pkts digest: 2259
  #pkts decaps: 2299, #pkts decrypt: 2299, #pkts verify: 2299
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 33 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 927, #pkts encrypt: 927, #pkts digest: 927
  #pkts decaps: 930, #pkts decrypt: 930, #pkts verify: 930
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 81 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 2309, #pkts encrypt: 2309, #pkts digest: 2309
  #pkts decaps: 2329, #pkts decrypt: 2329, #pkts verify: 2329
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 33 more lines
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.23 PASS BGP Control Plane Traffic Is Encrypted On The Wire (15.5s, 8 commands)

The reason for peering over the tunnel. Captures a keepalive window and asserts the session is invisible: no TCP, no peering addresses, no advertised prefixes -- only ESP.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **ESP**
- output does not contain **TCP**
- output does not contain **#{R2_TUNNEL_HUB}**
- output does not contain **#{R2_TUNNEL_LOCAL}**
- output does not contain **#{R1_BGP_PREFIX}**
- output does not contain **#{R2_BGP_PREFIX}**

How it was checked

R1 no monitor capture BGPCAP

Capture does not exist

R1 monitor capture BGPCAP interface GigabitEthernet2 both

#{output} =

R1 monitor capture BGPCAP match ipv4 any any

#{output} =

R1 monitor capture BGPCAP buffer size 5

#{output} =

R1 monitor capture BGPCAP start

Started capture point : BGPCAP

R1 monitor capture BGPCAP stop

Stopped capture point : BGPCAP

R1 show monitor capture BGPCAP buffer brief

```
-----
#   size   timestamp      source          destination    dscp   protocol
-----
0  122     0.0000000  10.10.10.2      -> 10.10.10.1    48 CS6  ESP
1  122     0.0000000  10.10.10.1      -> 10.10.10.2    48 CS6  ESP
2  138     0.105997    10.10.10.1      -> 10.10.10.2    48 CS6  ESP
3  122     0.176993    10.10.10.1      -> 10.10.10.2    48 CS6  ESP
4  122     0.176993    10.10.10.2      -> 10.10.10.1    48 CS6  ESP
5  138     0.329984    10.10.10.2      -> 10.10.10.1    48 CS6  ESP
6  122     0.394984    10.10.10.2      -> 10.10.10.1    48 CS6  ESP
7  122     0.394984    10.10.10.1      -> 10.10.10.2    48 CS6  ESP
8  122     0.666973    10.10.10.1      -> 10.10.10.2    48 CS6  ESP
9  122     0.666973    10.10.10.2      -> 10.10.10.1    48 CS6  ESP
10 122     0.835955    10.10.10.2      -> 10.10.10.1    48 CS6  ESP
... 157 more lines
```

R1 no monitor capture BGPCAP

#{output} =

Teardown (suite) Restore BGP Baseline, Close All Routers

9.24 PASS Withdrawing A Prefix Removes It From The Peer (55.4s, 22 commands)

Shuts the advertised interface on R2 and requires R1 to lose both the BGP path and the route, then restores it.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **% Subnet not in table** (checked 6x)
- output contains **Known via "bgp"** (checked 8x)

How it was checked

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Loopback1

#{output} =

R2 shutdown

#{output} =

R2 end

#{output} =

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 192.168.100.2 00:01:08 ago
  Routing Descriptor Blocks:
    * 192.168.100.2, from 192.168.100.2, 00:01:08 ago
      opaque_ptr 0x74C7DDCF3428
      Route metric is 0, traffic share count is 1
```

```
AS Hops 1
Route tag 65002
MPLS label: none
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
Known via "bgp 65001", distance 20, metric 0
Tag 65002, type external
Last update from 192.168.100.2 00:01:13 ago
Routing Descriptor Blocks:
* 192.168.100.2, from 192.168.100.2, 00:01:13 ago
  opaque_ptr 0x74C7DDCF3428
  Route metric is 0, traffic share count is 1
  AS Hops 1
  Route tag 65002
  MPLS label: none
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
Known via "bgp 65001", distance 20, metric 0
Tag 65002, type external
Last update from 192.168.100.2 00:01:18 ago
Routing Descriptor Blocks:
* 192.168.100.2, from 192.168.100.2, 00:01:18 ago
  opaque_ptr 0x74C7DDCF3428
  Route metric is 0, traffic share count is 1
  AS Hops 1
  Route tag 65002
  MPLS label: none
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
Known via "bgp 65001", distance 20, metric 0
Tag 65002, type external
Last update from 192.168.100.2 00:01:23 ago
Routing Descriptor Blocks:
* 192.168.100.2, from 192.168.100.2, 00:01:23 ago
  opaque_ptr 0x74C7DDCF3428
  Route metric is 0, traffic share count is 1
  AS Hops 1
  Route tag 65002
  MPLS label: none
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
Known via "bgp 65001", distance 20, metric 0
Tag 65002, type external
Last update from 192.168.100.2 00:01:28 ago
Routing Descriptor Blocks:
* 192.168.100.2, from 192.168.100.2, 00:01:28 ago
  opaque_ptr 0x74C7DDCF3428
  Route metric is 0, traffic share count is 1
  AS Hops 1
  Route tag 65002
  MPLS label: none
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R2 configure terminal

```
Enter configuration commands, one per line. End with CNTL/Z.
```

R2 interface Loopback1

```
${output} =
```

R2 no shutdown

```
${output} =
```

R2 end

```
${output} =
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 192.168.100.2 00:00:00 ago
  Routing Descriptor Blocks:
    * 192.168.100.2, from 192.168.100.2, 00:00:00 ago
      opaque_ptr 0x74C7DDCF3428
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none
```

R2 show ip route 10.20.1.1

```
Routing entry for 10.20.1.1/32
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 192.168.100.1 00:02:03 ago
  Routing Descriptor Blocks:
    * 192.168.100.1, from 192.168.100.1, 00:02:03 ago
      opaque_ptr 0x716DDEC6FB58
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65001
      MPLS label: none
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.25 PASS Session Recovers After A Hard Clear (10.2s, 5 commands)

Setup (suite) Establish BGP

Acceptance criteria

- output matches `${peer}\s+4\s+${peer_asn}(\s+1d+){5}\s+1S+1s+1d+` (checked 3x)
- output contains **Known via "bgp"**

How it was checked

R1 clear ip bgp *

`${output}` =

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
2/0 BGP path/bestpath attribute entries using 592 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
1 BGP community entries using 24 bytes of memory
1 BGP route-map cache entries using 72 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1864 total bytes of memory
BGP activity 319/316 prefixes, 335/332 paths, scan interval 60 secs
22 networks peaked at 17:59:32 Sep 7 2026 UTC (01:58:39.744 ago)

Neighbor      V          AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
2/0 BGP path/bestpath attribute entries using 592 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
1 BGP community entries using 24 bytes of memory
1 BGP route-map cache entries using 72 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1864 total bytes of memory
BGP activity 319/316 prefixes, 335/332 paths, scan interval 60 secs
22 networks peaked at 17:59:32 Sep 7 2026 UTC (01:58:44.782 ago)

Neighbor      V          AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 10, main routing table version 10
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
```

```

7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6152 total bytes of memory
BGP activity 325/316 prefixes, 341/332 paths, scan interval 60 secs
22 networks peaked at 17:59:32 Sep 7 2026 UTC (01:58:49.818 ago)

Neighbor      V      AS MsgRcvd MsgSent   TblVer  InQ  OutQ  Up/Down  State/PfxRcd
... 2 more lines

```

R1 show ip route 10.20.2.1

```

Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 192.168.100.2 00:00:01 ago
  Routing Descriptor Blocks:
    * 192.168.100.2, from 192.168.100.2, 00:00:01 ago
      opaque_ptr 0x74C7DDCF3428
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none

```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.26 PASS Mismatched MD5 Password Prevents The Session Coming Up (50.4s, 19 commands)

Negative control on BGP authentication, mirroring the IPsec wrong-PSK test: without it, a session that accepted any peer would still pass everything above.

Setup (suite) Establish BGP

Acceptance criteria

- output does not match `${peer}\s+ld+\{6\}\s+\s+\s+ld+\s*$`
- output matches `${peer}\s+4\s+${peer_asn}\s+ld+\{5\}\s+\s+\s+ld+` (checked 2x)

How it was checked

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002

`${output}` =

R2 neighbor 192.168.100.1 password WrongBgpSecret999

`${output}` =

R2 end

`${output}` =

R1 clear ip bgp *

`${output}` =

R2 clear ip bgp *

`${output}` =

R1 show bgp ipv4 unicast summary

```

BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
6 network entries using 1488 bytes of memory
6 path entries using 816 bytes of memory
7/0 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5000 total bytes of memory
BGP activity 331/325 prefixes, 347/341 paths, scan interval 60 secs
22 networks peaked at 17:59:32 Sep 7 2026 UTC (01:59:34.990 ago)

```

```

Neighbor      V      AS MsgRcvd MsgSent   TblVer  InQ  OutQ  Up/Down  State/PfxRcd
... 2 more lines

```

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002

`${output}` =

R2 neighbor 192.168.100.1 password LabBgpSecret

`${output}` =


```

R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 interface Loopback1
  ${output} =
R2 no shutdown
  ${output} =
R2 end
  ${output} =
R1 clear ip bgp *
  ${output} =
R2 clear ip bgp *
  ${output} =
R1 show bgp ipv4 unicast summary
  BGP router identifier 172.16.1.1, local AS number 65001
  BGP table version is 1, main routing table version 1
  3 network entries using 744 bytes of memory
  3 path entries using 408 bytes of memory
  7/0 BGP path/bestpath attribute entries using 2072 bytes of memory
  2 BGP AS-PATH entries using 48 bytes of memory
  6 BGP community entries using 144 bytes of memory
  6 BGP route-map cache entries using 432 bytes of memory
  0 BGP filter-list cache entries using 0 bytes of memory
  BGP using 3848 total bytes of memory
  BGP activity 334/331 prefixes, 350/347 paths, scan interval 60 secs
  22 networks peaked at 17:59:32 Sep 7 2026 UTC (01:59:35.192 ago)

  Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
  ... 2 more lines
R1 show bgp ipv4 unicast summary
  BGP router identifier 172.16.1.1, local AS number 65001
  BGP table version is 1, main routing table version 1
  3 network entries using 744 bytes of memory
  3 path entries using 408 bytes of memory
  7/0 BGP path/bestpath attribute entries using 2072 bytes of memory
  2 BGP AS-PATH entries using 48 bytes of memory
  6 BGP community entries using 144 bytes of memory
  6 BGP route-map cache entries using 432 bytes of memory
  0 BGP filter-list cache entries using 0 bytes of memory
  BGP using 3848 total bytes of memory
  BGP activity 334/331 prefixes, 350/347 paths, scan interval 60 secs
  22 networks peaked at 17:59:32 Sep 7 2026 UTC (01:59:40.231 ago)

  Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
  ... 2 more lines

```

Teardown (this test) Restore BGP Baseline

10 Host To Host

10.01 **PASS** Every Host Holds Its LAN Address (0.1s, 3 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains `#{HOST_IP}[#{h}]` (checked 3×)

How it was checked

H1 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:01:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.229.55/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
... 70 more lines
```

H2 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:02:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.199.41/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
... 70 more lines
```

H3 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:03:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.30.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.207.242/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
... 70 more lines
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.02 **PASS** Every Host Routes The Other Two LANs Via Its Own Router (0.0s, 3 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output matches `#{net}/24 via #{HOST_GW}[#{h}]` (checked 9×)

How it was checked

H1 ip route

```
default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.10.1 dev eth1
```

```
169.254.0.0/16 dev eth1 scope link src 169.254.229.55 metric 1003
192.168.10.0/24 dev eth1 scope link src 192.168.10.10
192.168.20.0/24 via 192.168.10.1 dev eth1
192.168.30.0/24 via 192.168.10.1 dev eth1
```

H2 ip route

```
default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.20.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.199.41 metric 1003
192.168.10.0/24 via 192.168.20.1 dev eth1
192.168.20.0/24 dev eth1 scope link src 192.168.20.10
192.168.30.0/24 via 192.168.20.1 dev eth1
```

H3 ip route

```
default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.30.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.207.242 metric 1003
192.168.10.0/24 via 192.168.30.1 dev eth1
192.168.20.0/24 via 192.168.30.1 dev eth1
192.168.30.0/24 dev eth1 scope link src 192.168.30.10
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.03 PASS Every Host Can Reach Its Default Gateway (12.3s, 3 commands)

Isolates the host-to-router leg so a failure further along is not misread as a broken LAN.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **0% packet loss** (checked 3×)
ping from \${alias} to \${target} lost packets: \${out}

How it was checked

H1 ping -c 5 192.168.10.1

```
PING 192.168.10.1 (192.168.10.1) 56(84) bytes of data.
64 bytes from 192.168.10.1: icmp_seq=1 ttl=255 time=0.420 ms
64 bytes from 192.168.10.1: icmp_seq=2 ttl=255 time=0.404 ms
64 bytes from 192.168.10.1: icmp_seq=3 ttl=255 time=0.422 ms
64 bytes from 192.168.10.1: icmp_seq=4 ttl=255 time=0.504 ms
64 bytes from 192.168.10.1: icmp_seq=5 ttl=255 time=0.378 ms

--- 192.168.10.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4097ms
rtt min/avg/max/mdev = 0.378/0.425/0.504/0.042 ms
```

H2 ping -c 5 192.168.20.1

```
PING 192.168.20.1 (192.168.20.1) 56(84) bytes of data.
64 bytes from 192.168.20.1: icmp_seq=1 ttl=255 time=0.455 ms
64 bytes from 192.168.20.1: icmp_seq=2 ttl=255 time=0.438 ms
64 bytes from 192.168.20.1: icmp_seq=3 ttl=255 time=0.404 ms
64 bytes from 192.168.20.1: icmp_seq=4 ttl=255 time=0.347 ms
64 bytes from 192.168.20.1: icmp_seq=5 ttl=255 time=0.377 ms

--- 192.168.20.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4073ms
rtt min/avg/max/mdev = 0.347/0.404/0.455/0.039 ms
```

H3 ping -c 5 192.168.30.1

```
PING 192.168.30.1 (192.168.30.1) 56(84) bytes of data.
64 bytes from 192.168.30.1: icmp_seq=1 ttl=255 time=0.433 ms
64 bytes from 192.168.30.1: icmp_seq=2 ttl=255 time=0.398 ms
64 bytes from 192.168.30.1: icmp_seq=3 ttl=255 time=0.527 ms
64 bytes from 192.168.30.1: icmp_seq=4 ttl=255 time=0.392 ms
64 bytes from 192.168.30.1: icmp_seq=5 ttl=255 time=0.413 ms

--- 192.168.30.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4083ms
rtt min/avg/max/mdev = 0.392/0.432/0.527/0.049 ms
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.04 PASS Every Host Can Reach Every Other Host (24.1s, 6 commands)

The full mesh, including h2 <-> h3 which never touches a direct link and can only work if the hub re-advertises between spokes.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **0% packet loss** (checked 6×)
ping from \${alias} to \${target} lost packets: \${out}

How it was checked

H2 ping -c 5 192.168.20.10

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.  
64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=1.25 ms  
64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.26 ms  
64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=29.0 ms  
64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.28 ms  
64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.23 ms  
  
--- 192.168.20.10 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4005ms  
rtt min/avg/max/mdev = 1.226/6.799/28.980/11.090 ms
```

H3 ping -c 5 192.168.30.10

```
PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.  
64 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.46 ms  
64 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.38 ms  
64 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=1.07 ms  
64 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.25 ms  
64 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.23 ms  
  
--- 192.168.30.10 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4006ms  
rtt min/avg/max/mdev = 1.072/1.279/1.458/0.133 ms
```

H1 ping -c 5 192.168.10.10

```
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.  
64 bytes from 192.168.10.10: icmp_seq=1 ttl=62 time=1.62 ms  
64 bytes from 192.168.10.10: icmp_seq=2 ttl=62 time=1.17 ms  
64 bytes from 192.168.10.10: icmp_seq=3 ttl=62 time=1.24 ms  
64 bytes from 192.168.10.10: icmp_seq=4 ttl=62 time=1.32 ms  
64 bytes from 192.168.10.10: icmp_seq=5 ttl=62 time=1.32 ms  
  
--- 192.168.10.10 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4006ms  
rtt min/avg/max/mdev = 1.172/1.334/1.621/0.153 ms
```

H3 ping -c 5 192.168.30.10

```
PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.  
64 bytes from 192.168.30.10: icmp_seq=1 ttl=61 time=1.61 ms  
64 bytes from 192.168.30.10: icmp_seq=2 ttl=61 time=1.64 ms  
64 bytes from 192.168.30.10: icmp_seq=3 ttl=61 time=1.68 ms  
64 bytes from 192.168.30.10: icmp_seq=4 ttl=61 time=1.87 ms  
64 bytes from 192.168.30.10: icmp_seq=5 ttl=61 time=1.63 ms  
  
--- 192.168.30.10 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4007ms  
rtt min/avg/max/mdev = 1.611/1.687/1.874/0.095 ms
```

H1 ping -c 5 192.168.10.10

```
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.  
64 bytes from 192.168.10.10: icmp_seq=1 ttl=62 time=1.46 ms  
64 bytes from 192.168.10.10: icmp_seq=2 ttl=62 time=1.21 ms  
64 bytes from 192.168.10.10: icmp_seq=3 ttl=62 time=1.23 ms  
64 bytes from 192.168.10.10: icmp_seq=4 ttl=62 time=1.33 ms  
64 bytes from 192.168.10.10: icmp_seq=5 ttl=62 time=1.25 ms  
  
--- 192.168.10.10 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4006ms  
rtt min/avg/max/mdev = 1.213/1.296/1.460/0.090 ms
```

H2 ping -c 5 192.168.20.10

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.  
64 bytes from 192.168.20.10: icmp_seq=1 ttl=61 time=1.60 ms  
64 bytes from 192.168.20.10: icmp_seq=2 ttl=61 time=1.66 ms  
64 bytes from 192.168.20.10: icmp_seq=3 ttl=61 time=1.74 ms  
64 bytes from 192.168.20.10: icmp_seq=4 ttl=61 time=1.77 ms  
64 bytes from 192.168.20.10: icmp_seq=5 ttl=61 time=41.4 ms  
  
--- 192.168.20.10 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4007ms  
rtt min/avg/max/mdev = 1.603/9.626/41.352/15.863 ms
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.05 PASS Spoke To Spoke Traffic Is Routed Through The Hub (0.1s, 3 commands)

R2 must reach R3's LAN via the hub's tunnel address, not by any direct path -- there is no spoke-to-spoke link.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **Known via "bgp \${R2_ASN}"**
- output contains **\${R2_TUNNEL_HUB}**
- output contains **Tunnel0** (checked 2×)

How it was checked

R2 show ip route 192.168.30.0

```
Routing entry for 192.168.30.0/24
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 192.168.100.1 00:01:29 ago
  Routing Descriptor Blocks:
    * 192.168.100.1, from 192.168.100.1, 00:01:29 ago
      opaque_ptr 0x716DDEC6FB58
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

R2 show ip cef 192.168.30.10

```
192.168.30.0/24
  nexthop 192.168.100.1 Tunnel0
```

R3 show ip cef 192.168.20.10

```
192.168.20.0/24
  nexthop 192.168.101.1 Tunnel0
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.06 PASS Hub Reaches Both Spoke LANs On Separate Tunnels (0.0s, 2 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **\${R2_HUB_TUNNEL}**
- output contains **\${R3_HUB_TUNNEL}**

How it was checked

R1 show ip cef 192.168.20.10

```
192.168.20.0/24
  nexthop 192.168.100.2 Tunnel0
```

R1 show ip cef 192.168.30.10

```
192.168.30.0/24
  nexthop 192.168.101.2 Tunnel1
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.07 PASS Host Traffic Is Encrypted In Transit (19.5s, 5 commands)

No host runs crypto: the routers encrypt for them, so the ESP counters must move by the traffic the hosts generated.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **0% packet loss**
ping from \${alias} to \${target} lost packets: \${out}
- **\${r1_enc} >= \${expected}** holds
- **\${r1_dec} >= \${expected}** holds
- **\${r2_enc} >= \${expected}** holds
- **\${r2_dec} >= \${expected}** holds

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
```

```
Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1
```

```
protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 2035, #pkts encrypt: 2035, #pkts digest: 2035
#pkts decaps: 2025, #pkts decrypt: 2025, #pkts verify: 2025
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 3409, #pkts encrypt: 3409, #pkts digest: 3409
#pkts decaps: 3444, #pkts decrypt: 3444, #pkts verify: 3444
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines
```

H1 ping -c 20 192.168.20.10

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.
64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=1.28 ms
64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.26 ms
64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=14.8 ms
64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.21 ms
64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.29 ms
64 bytes from 192.168.20.10: icmp_seq=6 ttl=62 time=1.18 ms
64 bytes from 192.168.20.10: icmp_seq=7 ttl=62 time=1.31 ms
64 bytes from 192.168.20.10: icmp_seq=8 ttl=62 time=35.9 ms
64 bytes from 192.168.20.10: icmp_seq=9 ttl=62 time=38.8 ms
64 bytes from 192.168.20.10: icmp_seq=10 ttl=62 time=1.15 ms
64 bytes from 192.168.20.10: icmp_seq=11 ttl=62 time=1.32 ms
64 bytes from 192.168.20.10: icmp_seq=12 ttl=62 time=1.24 ms
64 bytes from 192.168.20.10: icmp_seq=13 ttl=62 time=1.25 ms
... 11 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 2172, #pkts encrypt: 2172, #pkts digest: 2172
#pkts decaps: 2161, #pkts decrypt: 2161, #pkts verify: 2161
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 3541, #pkts encrypt: 3541, #pkts digest: 3541
#pkts decaps: 3574, #pkts decrypt: 3574, #pkts verify: 3574
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.08 PASS Host Addresses Never Appear In Cleartext On The Spoke Link (10.0s, 9 commands)

Capture the R1<->R2 link while the hosts talk: only ESP between the link addresses should be visible.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **0% packet loss**
ping from \${alias} to \${target} lost packets: \${out}
- output contains **ESP**
- output does not contain **\${H1_IP}**
- output does not contain **\${H2_IP}**
- output does not contain **ICMP**

How it was checked

R1 no monitor capture HOSTCAP

Capture does not exist

R1 monitor capture HOSTCAP interface GigabitEthernet2 both

\${output} =

R1 monitor capture HOSTCAP match ipv4 any any

\${output} =

R1 monitor capture HOSTCAP buffer size 5

\${output} =

R1 monitor capture HOSTCAP start

Started capture point : HOSTCAP

H1 ping -c 10 192.168.20.10

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.  
64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=1.66 ms  
64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.38 ms  
64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=1.32 ms  
64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.30 ms  
64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.21 ms  
64 bytes from 192.168.20.10: icmp_seq=6 ttl=62 time=18.5 ms  
64 bytes from 192.168.20.10: icmp_seq=7 ttl=62 time=1.25 ms  
64 bytes from 192.168.20.10: icmp_seq=8 ttl=62 time=1.18 ms  
64 bytes from 192.168.20.10: icmp_seq=9 ttl=62 time=1.21 ms  
64 bytes from 192.168.20.10: icmp_seq=10 ttl=62 time=1.27 ms  
  
--- 192.168.20.10 ping statistics ---  
10 packets transmitted, 10 received, 0% packet loss, time 9014ms  
... 1 more lines
```

R1 monitor capture HOSTCAP stop

Stopped capture point : HOSTCAP

R1 show monitor capture HOSTCAP buffer brief

```
-----  
#    size    timestamp    source            destination      dscp    protocol  
-----  
0    170     0.000000    10.10.10.1        -> 10.10.10.2      0 BE    ESP  
1    170     0.001007    10.10.10.2        -> 10.10.10.1      0 BE    ESP  
2    122     0.122995    10.10.10.1        -> 10.10.10.2      48 CS6   ESP  
3    122     0.122995    10.10.10.2        -> 10.10.10.1      48 CS6   ESP  
4    122     0.226993    10.10.10.2        -> 10.10.10.1      48 CS6   ESP  
5    122     0.226993    10.10.10.1        -> 10.10.10.2      48 CS6   ESP  
6    138     0.546029    10.10.10.2        -> 10.10.10.1      48 CS6   ESP  
7    122     0.571022    10.10.10.1        -> 10.10.10.2      48 CS6   ESP  
8    122     0.572029    10.10.10.2        -> 10.10.10.1      48 CS6   ESP  
9    122     0.621022    10.10.10.2        -> 10.10.10.1      48 CS6   ESP  
10   122     0.621022    10.10.10.1        -> 10.10.10.2      48 CS6   ESP  
... 117 more lines
```

R1 no monitor capture HOSTCAP

\${output} =

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.09 PASS Spoke To Spoke Traffic Is Encrypted On Both Legs (10.0s, 9 commands)

h2 -> h3 crosses two tunnels. Capture the second leg (the hub's link to R3) and show the spoke hosts are invisible there too.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **0% packet loss**
ping from \${alias} to \${target} lost packets: \${out}
- output contains **ESP**
- output does not contain **\${H2_IP}**
- output does not contain **\${H3_IP}**
- output does not contain **ICMP**

How it was checked

R1 no monitor capture HOSTCAP

Capture does not exist

R1 monitor capture HOSTCAP interface GigabitEthernet4 both

\${output} =

R1 monitor capture HOSTCAP match ipv4 any any

\${output} =

R1 monitor capture HOSTCAP buffer size 5

\${output} =

R1 monitor capture HOSTCAP start

Started capture point : HOSTCAP

H2 ping -c 10 192.168.30.10

```
PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.  
64 bytes from 192.168.30.10: icmp_seq=1 ttl=61 time=2.13 ms  
64 bytes from 192.168.30.10: icmp_seq=2 ttl=61 time=1.81 ms  
64 bytes from 192.168.30.10: icmp_seq=3 ttl=61 time=1.65 ms  
64 bytes from 192.168.30.10: icmp_seq=4 ttl=61 time=1.73 ms  
64 bytes from 192.168.30.10: icmp_seq=5 ttl=61 time=1.56 ms  
64 bytes from 192.168.30.10: icmp_seq=6 ttl=61 time=1.75 ms  
64 bytes from 192.168.30.10: icmp_seq=7 ttl=61 time=1.84 ms  
64 bytes from 192.168.30.10: icmp_seq=8 ttl=61 time=2.08 ms  
64 bytes from 192.168.30.10: icmp_seq=9 ttl=61 time=23.6 ms  
64 bytes from 192.168.30.10: icmp_seq=10 ttl=61 time=1.71 ms  
  
--- 192.168.30.10 ping statistics ---  
10 packets transmitted, 10 received, 0% packet loss, time 9016ms  
... 1 more lines
```

R1 monitor capture HOSTCAP stop

Stopped capture point : HOSTCAP

R1 show monitor capture HOSTCAP buffer brief

```
-----  
#   size  timestamp      source      destination  dscp  protocol  
-----  
0  170    0.000000    10.10.20.1  -> 10.10.20.2    0  BE   ESP  
1  170    0.000000    10.10.20.2  -> 10.10.20.1    0  BE   ESP  
2  138    0.042997    10.10.20.2  -> 10.10.20.1   48 CS6  ESP  
3  122    0.164985    10.10.20.2  -> 10.10.20.1   48 CS6  ESP  
4  122    0.164985    10.10.20.1  -> 10.10.20.2   48 CS6  ESP  
5  122    0.187994    10.10.20.1  -> 10.10.20.2   48 CS6  ESP  
6  122    0.205983    10.10.20.2  -> 10.10.20.1   48 CS6  ESP  
7  138    0.401972    10.10.20.1  -> 10.10.20.2   48 CS6  ESP  
8  122    0.578025    10.10.20.2  -> 10.10.20.1   48 CS6  ESP  
9  122    0.578025    10.10.20.1  -> 10.10.20.2   48 CS6  ESP  
10 122    0.611012    10.10.20.1  -> 10.10.20.2   48 CS6  ESP  
... 116 more lines
```

R1 no monitor capture HOSTCAP

\${output} =

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

11 Throughput

11.01 PASS Hub To Spoke Hosts Sustain The Target Throughput (31.6s, 2 commands)

h1 -> h2 crosses one tunnel.

Setup (suite) Open All Routers, Open All Hosts, Wait For Fabric

Acceptance criteria

- `${m}` is not empty
- `${mbps} >= ${TP_TARGET}` holds

How it was checked

```
H1 nc -l -p 5001 > /dev/null 2>&1 &
${output} =
```

```
H1 S=$(date +%s); dd if=/dev/zero bs=64k count=1024 2>/dev/null | nc -w 1 192.168.20.10 5001; echo SECS=$(( $(date +%s) - S ))
SECS=30
```

Teardown (suite) Close All Connections

11.02 PASS Spoke To Spoke Hosts Sustain The Target Throughput (32.4s, 2 commands)

h2 -> h3 crosses two tunnels, being decrypted and re-encrypted at the hub, so it is the more demanding path.

Setup (suite) Open All Routers, Open All Hosts, Wait For Fabric

Acceptance criteria

- `${m}` is not empty
- `${mbps} >= ${TP_TARGET}` holds

How it was checked

```
H2 nc -l -p 5001 > /dev/null 2>&1 &
${output} =
```

```
H2 S=$(date +%s); dd if=/dev/zero bs=64k count=1024 2>/dev/null | nc -w 1 192.168.30.10 5001; echo SECS=$(( $(date +%s) - S ))
SECS=32
```

Teardown (suite) Close All Connections

11.03 PASS Bulk Throughput Traffic Is Encrypted (17.0s, 6 commands)

Rate alone says nothing about protection. A transfer of this size must move thousands of packets through ESP, so assert the counters rose in proportion to the bytes actually sent.

Setup (suite) Open All Routers, Open All Hosts, Wait For Fabric

Acceptance criteria

- `${m}` is not empty
- `${data_out} >= ${expected}` holds
- `${data_in} >= ${expected}` holds
- `${ack_in} > 0` holds
- `${ack_out} > 0` holds

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 72858, #pkts encrypt: 72858, #pkts digest: 72858
#pkts decaps: 73367, #pkts decrypt: 73367, #pkts verify: 73367
#pkts compressed: 0, #pkts decompressed: 0
```

```

#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines

```

R2 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 74859, #pkts encrypt: 74859, #pkts digest: 74859
#pkts decaps: 74203, #pkts decrypt: 74203, #pkts verify: 74203
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines

```

H1 nc -l -p 5001 > /dev/null 2>&1 &

```
[1]+  Stopped (tty input) nc -l -p 5001 1>/dev/null 2>&1
```

H1 S=\$(date +%s); dd if=/dev/zero bs=64k count=512 2>/dev/null | nc -w 1 192.168.20.10 5001; echo SECS=\$((\$(date +%s) - S))

```
SECS=16
```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 97191, #pkts encrypt: 97191, #pkts digest: 97191
#pkts decaps: 84882, #pkts decrypt: 84882, #pkts verify: 84882
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 86388, #pkts encrypt: 86388, #pkts digest: 86388
#pkts decaps: 98473, #pkts decrypt: 98473, #pkts verify: 98473
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines

```

Teardown (suite) Close All Connections

12 Nat

12.01 PASS Every Router Has A Static Source NAT Mapping For Its Host (0.3s, 3 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains `${HOST_IP}${h}}` (checked 3×)
- output contains `${HOST_NAT}${h}}` (checked 3×)
- output contains `route-map NAT-TO-DOMAIN` (checked 3×)

How it was checked

H1 show running-config | include ip nat inside source

```
ip nat inside source static 192.168.10.10 10.30.1.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.11 10.30.1.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.12 10.30.1.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.13 10.30.1.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.14 10.30.1.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.15 10.30.1.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.16 10.30.1.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.17 10.30.1.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.18 10.30.1.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.19 10.30.1.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
```

H2 show running-config | include ip nat inside source

```
ip nat inside source static 192.168.20.10 10.30.2.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.11 10.30.2.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.12 10.30.2.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.13 10.30.2.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.14 10.30.2.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.15 10.30.2.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.16 10.30.2.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.17 10.30.2.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.18 10.30.2.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.19 10.30.2.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
```

H3 show running-config | include ip nat inside source

```
ip nat inside source static 192.168.30.10 10.30.3.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.11 10.30.3.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.12 10.30.3.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.13 10.30.3.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.14 10.30.3.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.15 10.30.3.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.16 10.30.3.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.17 10.30.3.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.18 10.30.3.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.19 10.30.3.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
```

Teardown (suite) Close All Connections

12.02 PASS NAT Inside And Outside Interfaces Are Marked Correctly (0.3s, 7 commands)

The LAN faces inside, the tunnels face outside. Getting this backwards silently disables translation rather than erroring.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains `ip nat inside` (checked 3×)
- output contains `ip nat outside` (checked 4×)

How it was checked

R1 show running-config interface GigabitEthernet3

Building configuration...

```
Current configuration : 133 bytes
!
interface GigabitEthernet3
 description LAN toward h1
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
 negotiation auto
```

```

end

R1 show running-config interface Tunnel0
Building configuration...

Current configuration : 281 bytes
!
interface Tunnel0
description to c8000v-r2
ip address 192.168.100.1 255.255.255.252
ip nat outside
bfd interval 500 min_rx 500 multiplier 3
tunnel source GigabitEthernet2
tunnel mode ipsec ipv4
tunnel destination 10.10.10.2
tunnel protection ipsec profile LAB-IPSEC
end

R2 show running-config interface GigabitEthernet3
Building configuration...

Current configuration : 145 bytes
!
interface GigabitEthernet3
description LAN toward the local host
ip address 192.168.20.1 255.255.255.0
ip nat inside
negotiation auto
end

R2 show running-config interface Tunnel0
Building configuration...

Current configuration : 291 bytes
!
interface Tunnel0
description to the hub (c8000v-r1)
ip address 192.168.100.2 255.255.255.252
ip nat outside
bfd interval 500 min_rx 500 multiplier 3
tunnel source GigabitEthernet2
tunnel mode ipsec ipv4
tunnel destination 10.10.10.1
tunnel protection ipsec profile LAB-IPSEC
end

R3 show running-config interface GigabitEthernet3
Building configuration...

Current configuration : 145 bytes
!
interface GigabitEthernet3
description LAN toward the local host
ip address 192.168.30.1 255.255.255.0
ip nat inside
negotiation auto
end

R3 show running-config interface Tunnel0
Building configuration...

Current configuration : 291 bytes
!
interface Tunnel0
description to the hub (c8000v-r1)
ip address 192.168.101.2 255.255.255.252
ip nat outside
bfd interval 500 min_rx 500 multiplier 3
tunnel source GigabitEthernet2
tunnel mode ipsec ipv4
tunnel destination 10.10.20.1
tunnel protection ipsec profile LAB-IPSEC
end

R1 show running-config interface Tunnel1
Building configuration...

Current configuration : 281 bytes
!
interface Tunnel1
description to c8000v-r3
ip address 192.168.101.1 255.255.255.252
ip nat outside
bfd interval 500 min_rx 500 multiplier 3

```

```

tunnel source GigabitEthernet4
tunnel mode ipsec ipv4
tunnel destination 10.10.20.2
tunnel protection ipsec profile LAB-IPSEC
end

```

Teardown (suite) Close All Connections

12.03 PASS Mapped Prefixes Are Advertised Across The Fabric (0.1s, 3 commands)

Return traffic can only find its way home if each site's NAT range is reachable, so the mapped prefixes ride BGP like the real ones.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **Known via "bgp \${R2_ASN}"** (checked 2×)
- output contains **Known via "bgp \${R3_ASN}"**

How it was checked

R2 show ip route 10.30.1.0

```

Routing entry for 10.30.1.0/24
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 192.168.100.1 00:03:44 ago
  Routing Descriptor Blocks:
    * 192.168.100.1, from 192.168.100.1, 00:03:44 ago
      opaque_ptr 0x716DDEC6F658
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65001
      MPLS label: none

```

R2 show ip route 10.30.3.0

```

Routing entry for 10.30.3.0/24
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 192.168.100.1 00:03:44 ago
  Routing Descriptor Blocks:
    * 192.168.100.1, from 192.168.100.1, 00:03:44 ago
      opaque_ptr 0x716DDEC6FA18
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none

```

R3 show ip route 10.30.1.0

```

Routing entry for 10.30.1.0/24
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 192.168.101.1 00:03:44 ago
  Routing Descriptor Blocks:
    * 192.168.101.1, from 192.168.101.1, 00:03:44 ago
      opaque_ptr 0x73EA5EFE07E8
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65001
      MPLS label: none

```

Teardown (suite) Close All Connections

12.04 PASS Every Host Reaches Every Other Host At Its Mapped Address (24.1s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **0% packet loss** (checked 6×)
- ping from \${alias} to \${target} lost packets: \${out}

How it was checked

H2 ping -c 5 10.30.2.10

```

PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.17 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.06 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.25 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.19 ms

```

```
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.24 ms

--- 10.30.2.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 1.060/1.181/1.248/0.067 ms
```

H3 ping -c 5 10.30.3.10

```
PING 10.30.3.10 (10.30.3.10) 56(84) bytes of data.
64 bytes from 10.30.3.10: icmp_seq=1 ttl=62 time=1.35 ms
64 bytes from 10.30.3.10: icmp_seq=2 ttl=62 time=1.20 ms
64 bytes from 10.30.3.10: icmp_seq=3 ttl=62 time=1.13 ms
64 bytes from 10.30.3.10: icmp_seq=4 ttl=62 time=1.24 ms
64 bytes from 10.30.3.10: icmp_seq=5 ttl=62 time=1.18 ms

--- 10.30.3.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.131/1.219/1.349/0.074 ms
```

H1 ping -c 5 10.30.1.10

```
PING 10.30.1.10 (10.30.1.10) 56(84) bytes of data.
64 bytes from 10.30.1.10: icmp_seq=1 ttl=62 time=1.23 ms
64 bytes from 10.30.1.10: icmp_seq=2 ttl=62 time=1.25 ms
64 bytes from 10.30.1.10: icmp_seq=3 ttl=62 time=1.42 ms
64 bytes from 10.30.1.10: icmp_seq=4 ttl=62 time=1.38 ms
64 bytes from 10.30.1.10: icmp_seq=5 ttl=62 time=1.26 ms

--- 10.30.1.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.233/1.306/1.416/0.075 ms
```

H3 ping -c 5 10.30.3.10

```
PING 10.30.3.10 (10.30.3.10) 56(84) bytes of data.
64 bytes from 10.30.3.10: icmp_seq=1 ttl=61 time=1.73 ms
64 bytes from 10.30.3.10: icmp_seq=2 ttl=61 time=2.16 ms
64 bytes from 10.30.3.10: icmp_seq=3 ttl=61 time=1.53 ms
64 bytes from 10.30.3.10: icmp_seq=4 ttl=61 time=1.75 ms
64 bytes from 10.30.3.10: icmp_seq=5 ttl=61 time=1.65 ms

--- 10.30.3.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 1.526/1.763/2.155/0.211 ms
```

H1 ping -c 5 10.30.1.10

```
PING 10.30.1.10 (10.30.1.10) 56(84) bytes of data.
64 bytes from 10.30.1.10: icmp_seq=1 ttl=62 time=1.47 ms
64 bytes from 10.30.1.10: icmp_seq=2 ttl=62 time=1.19 ms
64 bytes from 10.30.1.10: icmp_seq=3 ttl=62 time=1.26 ms
64 bytes from 10.30.1.10: icmp_seq=4 ttl=62 time=1.07 ms
64 bytes from 10.30.1.10: icmp_seq=5 ttl=62 time=1.28 ms

--- 10.30.1.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.073/1.254/1.470/0.130 ms
```

H2 ping -c 5 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=61 time=1.69 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=61 time=1.63 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=61 time=1.68 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=61 time=1.85 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=61 time=1.91 ms

--- 10.30.2.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 1.626/1.749/1.909/0.110 ms
```

Teardown (suite) Close All Connections

12.05 PASS Translations Show The Host Rewritten To Its Mapped Address (4.2s, 2 commands)

The inside-local / inside-global pair is the direct evidence that the source address was rewritten.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **0% packet loss**
ping from \${alias} to \${target} lost packets: \${out}
- output matches `icmpseq=${H1_NAT_IP}:ld+ls+${H1_IP}:ld+ls+${H2_NAT_IP}`

How it was checked

H1 ping -c 5 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=40.8 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=6.75 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.25 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.22 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.28 ms

--- 10.30.2.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 1.215/10.254/40.779/15.410 ms
```

R1 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.1.128	192.168.10.88	---	---
---	10.30.1.129	192.168.10.90	---	---
---	10.30.1.113	192.168.10.91	---	---
---	10.30.1.114	192.168.10.87	---	---
---	10.30.1.116	192.168.10.89	---	---
---	10.30.1.119	192.168.10.71	---	---
---	10.30.1.123	192.168.10.81	---	---
---	10.30.1.120	192.168.10.75	---	---
---	10.30.1.126	192.168.10.72	---	---
---	10.30.1.125	192.168.10.85	---	---
---	10.30.1.101	192.168.10.92	---	---
---	10.30.1.102	192.168.10.83	---	---
---	10.30.1.111	192.168.10.73	---	---
...	35 more lines			

Teardown (suite) Close All Connections

12.06 PASS The Far Host Sees The Mapped Source Not The Real One (4.2s, 2 commands)

On the receiving router the peer appears as the sender's mapped address, which is the whole point of source NAT.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **0% packet loss**
ping from \${alias} to \${target} lost packets: \${out}
- output contains **\${H1_NAT_IP}**
- output does not contain **\${H1_IP}**

How it was checked

H1 ping -c 5 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.77 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.17 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=23.8 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.26 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.18 ms

--- 10.30.2.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 1.172/5.834/23.795/8.983 ms
```

R2 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.2.128	192.168.20.86	---	---
---	10.30.2.129	192.168.20.75	---	---
---	10.30.2.106	192.168.20.80	---	---
---	10.30.2.105	192.168.20.82	---	---
---	10.30.2.111	192.168.20.89	---	---
---	10.30.2.108	192.168.20.79	---	---
---	10.30.2.101	192.168.20.76	---	---
---	10.30.2.102	192.168.20.90	---	---
---	10.30.2.126	192.168.20.74	---	---
---	10.30.2.125	192.168.20.67	---	---
---	10.30.2.123	192.168.20.77	---	---
---	10.30.2.120	192.168.20.66	---	---
---	10.30.2.116	192.168.20.91	---	---
...	37 more lines			

Teardown (suite) Close All Connections

12.07 PASS Spoke To Spoke NAT Works Through The Hub (4.4s, 3 commands)

h2 -> h3 by mapped address crosses two tunnels and two translations, with no direct spoke-to-spoke path.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **0% packet loss**
ping from \${alias} to \${target} lost packets: \${out}
- output contains **\${H3_NAT_IP}**
- output contains **\${H2_NAT_IP}**

How it was checked

H2 ping -c 5 10.30.3.10

```
PING 10.30.3.10 (10.30.3.10) 56(84) bytes of data.  
64 bytes from 10.30.3.10: icmp_seq=1 ttl=61 time=1.75 ms  
64 bytes from 10.30.3.10: icmp_seq=2 ttl=61 time=1.88 ms  
64 bytes from 10.30.3.10: icmp_seq=3 ttl=61 time=1.50 ms  
64 bytes from 10.30.3.10: icmp_seq=4 ttl=61 time=28.8 ms  
64 bytes from 10.30.3.10: icmp_seq=5 ttl=61 time=1.70 ms  
  
--- 10.30.3.10 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4006ms  
rtt min/avg/max/mdev = 1.503/7.135/28.839/10.852 ms
```

R2 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.2.128	192.168.20.86	---	---
---	10.30.2.129	192.168.20.75	---	---
---	10.30.2.106	192.168.20.80	---	---
---	10.30.2.105	192.168.20.82	---	---
---	10.30.2.111	192.168.20.89	---	---
---	10.30.2.108	192.168.20.79	---	---
---	10.30.2.101	192.168.20.76	---	---
---	10.30.2.102	192.168.20.90	---	---
---	10.30.2.126	192.168.20.74	---	---
---	10.30.2.125	192.168.20.67	---	---
---	10.30.2.123	192.168.20.77	---	---
---	10.30.2.120	192.168.20.66	---	---
---	10.30.2.116	192.168.20.91	---	---

... 38 more lines

R3 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.3.104	192.168.30.85	---	---
---	10.30.3.107	192.168.30.92	---	---
---	10.30.3.109	192.168.30.91	---	---
---	10.30.3.110	192.168.30.93	---	---
---	10.30.3.103	192.168.30.90	---	---
---	10.30.3.100	192.168.30.84	---	---
---	10.30.3.124	192.168.30.82	---	---
---	10.30.3.127	192.168.30.88	---	---
---	10.30.3.121	192.168.30.86	---	---
---	10.30.3.122	192.168.30.83	---	---
---	10.30.3.118	192.168.30.89	---	---
---	10.30.3.117	192.168.30.79	---	---
---	10.30.3.115	192.168.30.76	---	---

... 37 more lines

Teardown (suite) Close All Connections

12.08 PASS NATted Traffic Is Still Encrypted (19.4s, 5 commands)

Translation happens before encryption, so NATted traffic must still raise the ESP counters.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **0% packet loss**
ping from \${alias} to \${target} lost packets: \${out}
- **\${r1_enc} >= \${expected}** holds
- **\${r1_dec} >= \${expected}** holds
- **\${r2_enc} >= \${expected}** holds
- **\${r2_dec} >= \${expected}** holds

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 97539, #pkts encrypt: 97539, #pkts digest: 97539
  #pkts decaps: 85231, #pkts decrypt: 85231, #pkts verify: 85231
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 81 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 86732, #pkts encrypt: 86732, #pkts digest: 86732
  #pkts decaps: 98822, #pkts decrypt: 98822, #pkts verify: 98822
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 33 more lines
```

H1 ping -c 20 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.27 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.28 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.24 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.38 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.13 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.27 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.23 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.14 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.20 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=3.93 ms
64 bytes from 10.30.2.10: icmp_seq=11 ttl=62 time=1.34 ms
64 bytes from 10.30.2.10: icmp_seq=12 ttl=62 time=1.18 ms
64 bytes from 10.30.2.10: icmp_seq=13 ttl=62 time=1.25 ms
... 11 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 97672, #pkts encrypt: 97672, #pkts digest: 97672
  #pkts decaps: 85364, #pkts decrypt: 85364, #pkts verify: 85364
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 81 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 86864, #pkts encrypt: 86864, #pkts digest: 86864
  #pkts decaps: 98954, #pkts decrypt: 98954, #pkts verify: 98954
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
```

```

#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines

```

Teardown (suite) Close All Connections

12.09 PASS Mapped Addresses Never Appear In Cleartext On The Link (12.3s, 18 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **0% packet loss**
ping from \${alias} to \${target} lost packets: \${out}
- output contains **ESP**
- output does not contain **\${H1_NAT_IP}**
- output does not contain **\${H2_NAT_IP}**
- output does not contain **\${H1_IP}**

How it was checked

R1 no monitor capture NATCAP

Capture does not exist

R1 monitor capture NATCAP interface GigabitEthernet2 both

\${output} =

R1 monitor capture NATCAP match ipv4 any any

\${output} =

R1 monitor capture NATCAP buffer size 5

\${output} =

R1 monitor capture NATCAP start

Started capture point : NATCAP

R1 ping 10.10.10.2 repeat 2

Type escape sequence to abort.

Sending 2, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:

!!

Success rate is 100 percent (2/2), round-trip min/avg/max = 1/1/2 ms

R1 monitor capture NATCAP stop

Stopped capture point : NATCAP

R1 show monitor capture NATCAP buffer brief

```

-----
#   size   timestamp      source          destination    dscp   protocol
-----
0  114     0.000000    10.10.10.1      -> 10.10.10.2    0 BE   ICMP
1  114     0.000000    10.10.10.2      -> 10.10.10.1    0 BE   ICMP
2  114     0.001007    10.10.10.1      -> 10.10.10.2    0 BE   ICMP
3  114     0.001999    10.10.10.2      -> 10.10.10.1    0 BE   ICMP
4  122     0.029005    10.10.10.2      -> 10.10.10.1    48 CS6  ESP
5  122     0.029005    10.10.10.1      -> 10.10.10.2    48 CS6  ESP
6  122     0.177984    10.10.10.1      -> 10.10.10.2    48 CS6  ESP
7  122     0.178992    10.10.10.2      -> 10.10.10.1    48 CS6  ESP
8  138     0.248980    10.10.10.2      -> 10.10.10.1    48 CS6  ESP
9  138     0.341977    10.10.10.1      -> 10.10.10.2    48 CS6  ESP
10 122     0.508976    10.10.10.2      -> 10.10.10.1    48 CS6  ESP
... 16 more lines

```

R1 no monitor capture NATCAP

\${output} =

R1 no monitor capture NATCAP

Capture does not exist

R1 monitor capture NATCAP interface GigabitEthernet2 both

\${output} =

R1 monitor capture NATCAP match ipv4 any any

\${output} =

R1 monitor capture NATCAP buffer size 5

\${output} =

R1 monitor capture NATCAP start

Started capture point : NATCAP

H1 ping -c 10 10.30.2.10

PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.

64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.35 ms

64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.23 ms

64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.31 ms

```

64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.27 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=34.0 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.12 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.23 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.15 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.33 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=29.8 ms

--- 10.30.2.10 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9013ms
... 1 more lines

```

R1 monitor capture NATCAP stop

Stopped capture point : NATCAP

R1 show monitor capture NATCAP buffer brief

```

-----
#   size  timestamp      source          destination    dscp  protocol
-----
0   170    0.000000    10.10.10.1     -> 10.10.10.2    0 BE   ESP
1   170    0.001007    10.10.10.2     -> 10.10.10.1    0 BE   ESP
2   122    0.107996    10.10.10.2     -> 10.10.10.1    48 CS6  ESP
3   122    0.107996    10.10.10.1     -> 10.10.10.2    48 CS6  ESP
4   122    0.262986    10.10.10.1     -> 10.10.10.2    48 CS6  ESP
5   122    0.263993    10.10.10.2     -> 10.10.10.1    48 CS6  ESP
6   170    0.272996    10.10.10.1     -> 10.10.10.2    48 CS6  ESP
7   138    0.393992    10.10.10.2     -> 10.10.10.1    48 CS6  ESP
8   122    0.538973    10.10.10.2     -> 10.10.10.1    48 CS6  ESP
9   122    0.538973    10.10.10.1     -> 10.10.10.2    48 CS6  ESP
10  138    0.559983    10.10.10.2     -> 10.10.10.1    48 CS6  ESP
... 117 more lines

```

R1 no monitor capture NATCAP

\${output} =

Teardown (suite) Close All Connections

12.10 PASS The Untranslated LAN Path Still Works (8.0s, 2 commands)

Guards the route-map scoping. If translation ever widened to all traffic, replies on this path would arrive from a mapped address the sender never contacted, and this would fail.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **0% packet loss** (checked 2×)
- ping from \${alias} to \${target} lost packets: \${out}

How it was checked

H1 ping -c 5 192.168.20.10

```

PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.
64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=1.58 ms
64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.27 ms
64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=1.34 ms
64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.26 ms
64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.27 ms

--- 192.168.20.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.258/1.345/1.577/0.119 ms

```

H2 ping -c 5 192.168.30.10

```

PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.
64 bytes from 192.168.30.10: icmp_seq=1 ttl=61 time=1.57 ms
64 bytes from 192.168.30.10: icmp_seq=2 ttl=61 time=1.67 ms
64 bytes from 192.168.30.10: icmp_seq=3 ttl=61 time=2.16 ms
64 bytes from 192.168.30.10: icmp_seq=4 ttl=61 time=28.3 ms
64 bytes from 192.168.30.10: icmp_seq=5 ttl=61 time=1.68 ms

--- 192.168.30.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.574/7.077/28.299/10.612 ms

```

Teardown (suite) Close All Connections

13 Nat Pool

13.01 **PASS** Every Host Carries Its Full Block Of Addresses (0.1s, 6 commands)

10 statically mapped plus 30 pool-eligible, on one interface.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **#{lan}.#{o}/24** (checked 12×)
- **#{n}** equals **40** (checked 3×)

How it was checked

H1 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:01:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.229.55/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    ... 70 more lines
```

```
H1 ip addr show eth1 | grep -c 'inet 192.168.10.'
40
```

H2 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:02:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.199.41/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    ... 70 more lines
```

```
H2 ip addr show eth1 | grep -c 'inet 192.168.20.'
40
```

H3 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:03:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.30.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.207.242/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    ... 70 more lines
```

```
H3 ip addr show eth1 | grep -c 'inet 192.168.30.'
40
```

Teardown (suite) Close All Connections

13.02 **PASS** Every Router Configures Ten Static Mappings And A Thirty Address Pool (0.4s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains = **10** (checked 3×)
- output contains **total addresses 30** (checked 3×)

How it was checked

H1 `show running-config | count ^ip nat inside source static`

Number of lines which match regexp = 10

H1 `show ip nat statistics`

Total active translations: 10 (10 static, 0 dynamic; 0 extended)

Outside interfaces:

Tunnel0, Tunnel1

Inside interfaces:

GigabitEthernet3

Hits: 0 Misses: 0

Reserved port setting disabled provisioned no

Dynamic overload mapping configured: 0

Expired translations: 0

Dynamic mappings:

-- Inside Source

[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 0

pool SITE-POOL: id 1, netmask 255.255.255.0

start 10.30.1.100 end 10.30.1.129

... 9 more lines

H2 `show running-config | count ^ip nat inside source static`

Number of lines which match regexp = 10

H2 `show ip nat statistics`

Total active translations: 10 (10 static, 0 dynamic; 0 extended)

Outside interfaces:

Tunnel0

Inside interfaces:

GigabitEthernet3

Hits: 0 Misses: 0

Reserved port setting disabled provisioned no

Dynamic overload mapping configured: 0

Expired translations: 0

Dynamic mappings:

-- Inside Source

[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 0

pool SITE-POOL: id 1, netmask 255.255.255.0

start 10.30.2.100 end 10.30.2.129

... 9 more lines

H3 `show running-config | count ^ip nat inside source static`

Number of lines which match regexp = 10

H3 `show ip nat statistics`

Total active translations: 10 (10 static, 0 dynamic; 0 extended)

Outside interfaces:

Tunnel0

Inside interfaces:

GigabitEthernet3

Hits: 0 Misses: 0

Reserved port setting disabled provisioned no

Dynamic overload mapping configured: 0

Expired translations: 0

Dynamic mappings:

-- Inside Source

[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 0

pool SITE-POOL: id 1, netmask 255.255.255.0

start 10.30.3.100 end 10.30.3.129

... 9 more lines

Teardown (suite) Close All Connections

13.03 **PASS** The Pool Access List Is One Range And Excludes The Static Block (0.1s, 3 commands)

A single /27 range rather than thirty host entries, and it must not overlap the statically mapped addresses -- otherwise which rule wins depends on evaluation order.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- **#{permits}** has exactly **1** entries (checked 3×)
NAT-POOL-SRC should hold one range entry, found #{permits}
- output contains **#{lan}.#{DYN_OCTET_FIRST} 0.0.0.31** (checked 3×)
- output does not contain **host #{lan}.#{o}** (checked 6×)

How it was checked

H1 show ip access-lists NAT-POOL-SRC

```
Extended IP access list NAT-POOL-SRC
 10 permit ip 192.168.10.64 0.0.0.31 10.30.0.0 0.0.255.255
```

H2 show ip access-lists NAT-POOL-SRC

```
Extended IP access list NAT-POOL-SRC
 10 permit ip 192.168.20.64 0.0.0.31 10.30.0.0 0.0.255.255
```

H3 show ip access-lists NAT-POOL-SRC

```
Extended IP access list NAT-POOL-SRC
 10 permit ip 192.168.30.64 0.0.0.31 10.30.0.0 0.0.255.255
```

Teardown (suite) Close All Connections

13.04 PASS All Ten Static Addresses Translate To Their Own Fixed Mapping (9.5s, 2 commands)

Static NAT is one-to-one and deterministic: inside .10-.19 must appear outside as .10-.19, each to its own.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- **#{static}** has exactly **10** entries
only #{static} of the 10 static addresses were translated
- **#{d}** equals **10**
the 10 static addresses shared outside addresses; they must be one-to-one
- **#{aligned}** holds

How it was checked

H1 for i in \$(seq 10 19) \$(seq 64 93); do ping -c 2 -W 2 -I 192.168.10.\$i 10.30.2.10 >/dev/null 2>&1 & done; wait; echo SENT

```
SENT
[40]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[39]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[38]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[37]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[36]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[35]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[34]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[33]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[32]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[31]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[30]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[29]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[28]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
... 27 more lines
```

R1 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.1.128	192.168.10.89	---	---
---	10.30.1.129	192.168.10.85	---	---
---	10.30.1.113	192.168.10.78	---	---
---	10.30.1.114	192.168.10.74	---	---
---	10.30.1.116	192.168.10.83	---	---
---	10.30.1.119	192.168.10.87	---	---
---	10.30.1.123	192.168.10.75	---	---
---	10.30.1.120	192.168.10.70	---	---
---	10.30.1.126	192.168.10.84	---	---
---	10.30.1.125	192.168.10.82	---	---
---	10.30.1.101	192.168.10.86	---	---
---	10.30.1.102	192.168.10.79	---	---
---	10.30.1.111	192.168.10.69	---	---
---	...	68 more lines	---	---

Teardown (suite) Close All Connections

13.05 **PASS** All Thirty Pool Addresses Translate Into The Pool And Are Distinct (0.5s, 1

commands)

Without overload each client takes its own pool address, so 30 clients must yield 30 different outside addresses.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- **\${dyn}** has exactly 30 entries
only \${dyn} of the 30 pool clients were translated
- **\${d}** equals 30
pool clients shared outside addresses; that is overload, not a 1:1 pool
- **\${within}** holds

How it was checked

R1 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.1.128	192.168.10.89	---	---
---	10.30.1.129	192.168.10.85	---	---
---	10.30.1.113	192.168.10.78	---	---
---	10.30.1.114	192.168.10.74	---	---
---	10.30.1.116	192.168.10.83	---	---
---	10.30.1.119	192.168.10.87	---	---
---	10.30.1.123	192.168.10.75	---	---
---	10.30.1.120	192.168.10.70	---	---
---	10.30.1.126	192.168.10.84	---	---
---	10.30.1.125	192.168.10.82	---	---
---	10.30.1.101	192.168.10.86	---	---
---	10.30.1.102	192.168.10.79	---	---
---	10.30.1.111	192.168.10.69	---	---
... 68 more lines				

Teardown (suite) Close All Connections

13.06 **PASS** The Pool Is Exactly Consumed With No Misses (0.0s, 1 commands)

30 clients against a 30 address pool: full allocation and no misses is the evidence that none of them shared or failed.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **allocated 30 (100%)**
- output matches **misses 0**

How it was checked

R1 show ip nat statistics

```
Total active translations: 80 (10 static, 70 dynamic; 40 extended)
Outside interfaces:
  Tunnel0, Tunnel1
Inside interfaces:
  GigabitEthernet3
Hits: 120 Misses: 40
Reserved port setting disabled provisioned no
Dynamic overload mapping configured: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 60
  pool SITE-POOL: id 1, netmask 255.255.255.0
    start 10.30.1.100 end 10.30.1.129
... 9 more lines
```

Teardown (suite) Close All Connections

13.07 PASS A Thirty-first Client Gets No Translation Once The Pool Is Exhausted (12.3s, 9

commands)

The boundary the pool sizing never otherwise reaches. The pool holds exactly as many addresses as there are clients, which means the exhaustion path is never exercised by a normal run -- and an off-by-one in pool bounds would look identical to a correct configuration. So bring up one more client inside the pool ACL and require the router to refuse it: no translation, no borrowing of another client's address, and the miss counted. Without "overload" there is nothing to share, so silently reusing an address would be a correctness bug, not graceful degradation. Run against R1/H1, because it is R1's pool that the tests above fill: the suite drives traffic from H1 first and only reaches the other routers later. Asserting exhaustion on a router whose pool is still empty would test nothing.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **allocated 30 (100%)**
- **`\${n}`** is not empty (checked 4×)
- output contains **100% packet loss**
the thirty-first client was translated even though the pool was exhausted
- **`\${xlate}`** is empty
- **`\${after}`** > **`\${before}`** holds (checked 3×)

How it was checked

R1 show ip nat statistics | include allocated

type generic, total addresses 30, allocated 30 (100%), misses 0

R1 show ip nat statistics

Total active translations: 80 (10 static, 70 dynamic; 40 extended)

Outside interfaces:

Tunnel0, Tunnel1

Inside interfaces:

GigabitEthernet3

Hits: 120 Misses: 40

Reserved port setting disabled provisioned no

Dynamic overload mapping configured: 0

Expired translations: 0

Dynamic mappings:

-- Inside Source

[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 60

pool SITE-POOL: id 1, netmask 255.255.255.0

start 10.30.1.100 end 10.30.1.129

... 9 more lines

H1 sudo ip addr add 192.168.10.94/24 dev eth1

`\${output}` =

H1 ping -c 3 -W 2 -I 192.168.10.94 10.30.2.10

PING 10.30.2.10 (10.30.2.10) from 192.168.10.94 : 56(84) bytes of data.

From 192.168.10.1 icmp_seq=1 Destination Host Unreachable

From 192.168.10.1 icmp_seq=2 Destination Host Unreachable

From 192.168.10.1 icmp_seq=3 Destination Host Unreachable

--- 10.30.2.10 ping statistics ---

3 packets transmitted, 0 received, +3 errors, 100% packet loss, time 2037ms

R1 show ip nat translations | include 192.168.10.94

`\${output}` =

R1 show ip nat statistics

Total active translations: 80 (10 static, 70 dynamic; 40 extended)

Outside interfaces:

Tunnel0, Tunnel1

Inside interfaces:

GigabitEthernet3

Hits: 120 Misses: 40

Reserved port setting disabled provisioned no

Dynamic overload mapping configured: 0

Expired translations: 0

Dynamic mappings:

-- Inside Source

[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 60

pool SITE-POOL: id 1, netmask 255.255.255.0

start 10.30.1.100 end 10.30.1.129

... 9 more lines

R1 show ip nat statistics

Total active translations: 80 (10 static, 70 dynamic; 40 extended)

Outside interfaces:


```

Tunnel0, Tunnel1
Inside interfaces:
GigabitEthernet3
Hits: 120 Misses: 40
Reserved port setting disabled provisioned no
Dynamic overload mapping configured: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 60
pool SITE-POOL: id 1, netmask 255.255.255.0
start 10.30.1.100 end 10.30.1.129
... 9 more lines

```

R1 show ip nat statistics

```

Total active translations: 80 (10 static, 70 dynamic; 40 extended)
Outside interfaces:
Tunnel0, Tunnel1
Inside interfaces:
GigabitEthernet3
Hits: 120 Misses: 40
Reserved port setting disabled provisioned no
Dynamic overload mapping configured: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 60
pool SITE-POOL: id 1, netmask 255.255.255.0
start 10.30.1.100 end 10.30.1.129
... 9 more lines

```

H1 sudo ip addr del 192.168.10.94/24 dev eth1
 \${output} =

Teardown (this test) Remove The Spare Client From H1

13.08 PASS The Pool Is Still Intact After The Exhaustion Attempt (0.5s, 2 commands)

The refusal must cost nothing: the thirty legitimate clients keep the addresses they held, and none was evicted to make room for the one that was turned away.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **allocated 30 (100%)**
- **\${dyn}** has exactly **30** entries
 only \${dyn} of the original thirty pool clients still hold a translation
- **\${d}** equals **30**
 the refused client caused two survivors to share an address

How it was checked

R1 show ip nat statistics | include allocated

type generic, total addresses 30, allocated 30 (100%), misses 3

R1 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.1.128	192.168.10.89	---	---
---	10.30.1.129	192.168.10.85	---	---
---	10.30.1.113	192.168.10.78	---	---
---	10.30.1.114	192.168.10.74	---	---
---	10.30.1.116	192.168.10.83	---	---
---	10.30.1.119	192.168.10.87	---	---
---	10.30.1.123	192.168.10.75	---	---
---	10.30.1.120	192.168.10.70	---	---
---	10.30.1.126	192.168.10.84	---	---
---	10.30.1.125	192.168.10.82	---	---
---	10.30.1.101	192.168.10.86	---	---
---	10.30.1.102	192.168.10.79	---	---
---	10.30.1.111	192.168.10.69	---	---
...	68 more lines			

Teardown (suite) Close All Connections

13.09 PASS Static And Pool Translations Coexist Across The Whole Block (0.5s, 1 commands)

All 40 of a host's addresses in flight, translated by two different mechanisms, with no outside address reused.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- **#{all}** has exactly 40 entries
expected 40 translated addresses, found #{all}
- **#{d}** equals 40
the 40 inside addresses did not receive 40 distinct outside addresses

How it was checked

R1 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.1.128	192.168.10.89	---	---
---	10.30.1.129	192.168.10.85	---	---
---	10.30.1.113	192.168.10.78	---	---
---	10.30.1.114	192.168.10.74	---	---
---	10.30.1.116	192.168.10.83	---	---
---	10.30.1.119	192.168.10.87	---	---
---	10.30.1.123	192.168.10.75	---	---
---	10.30.1.120	192.168.10.70	---	---
---	10.30.1.126	192.168.10.84	---	---
---	10.30.1.125	192.168.10.82	---	---
---	10.30.1.101	192.168.10.86	---	---
---	10.30.1.102	192.168.10.79	---	---
---	10.30.1.111	192.168.10.69	---	---
...	68 more lines			

Teardown (suite) Close All Connections

13.10 PASS Both Mechanisms Work On Every Router (19.6s, 4 commands)

The spokes carry the same scheme, not just the hub.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- **#{static}** has exactly 10 entries (checked 2×)
- **#{dyn}** has exactly 30 entries (checked 2×)
- **#{aligned}** holds (checked 2×)

How it was checked

H2 for i in \$(seq 10 19) \$(seq 64 93); do ping -c 2 -W 2 -I 192.168.20.\$i 10.30.1.10 >/dev/null 2>&1 & done; wait;
echo SENT

```
SENT
[40]+ Done      ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[39]+ Done      ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[38]+ Done      ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[37]+ Done      ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[36]+ Done      ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[35]+ Done      ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[34]+ Done      ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[33]+ Done      ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[32]+ Done      ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[31]+ Done      ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[30]+ Done      ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[29]+ Done      ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[28]+ Done      ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
... 27 more lines
```

H2 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.2.128	192.168.20.87	---	---
---	10.30.2.129	192.168.20.75	---	---
---	10.30.2.106	192.168.20.73	---	---
---	10.30.2.105	192.168.20.79	---	---
---	10.30.2.111	192.168.20.78	---	---
---	10.30.2.108	192.168.20.93	---	---
---	10.30.2.101	192.168.20.92	---	---
---	10.30.2.102	192.168.20.77	---	---
---	10.30.2.126	192.168.20.84	---	---
---	10.30.2.125	192.168.20.90	---	---
---	10.30.2.123	192.168.20.70	---	---

```

--- 10.30.2.120          192.168.20.65      ---          ---
--- 10.30.2.116          192.168.20.85      ---          ---
... 108 more lines

H3 for i in $(seq 10 19) $(seq 64 93); do ping -c 2 -W 2 -I 192.168.30.$i 10.30.1.10 >/dev/null 2>&1 & done; wait;
echo SENT
SENT
[40]+ Done                ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[39]+ Done                ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[38]+ Done                ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[37]+ Done                ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[36]+ Done                ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[35]+ Done                ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[34]+ Done                ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[33]+ Done                ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[32]+ Done                ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[31]+ Done                ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[30]+ Done                ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[29]+ Done                ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[28]+ Done                ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
... 27 more lines

H3 show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
--- 10.30.3.104          192.168.30.88      ---                ---
--- 10.30.3.107          192.168.30.75      ---                ---
--- 10.30.3.109          192.168.30.93      ---                ---
--- 10.30.3.110          192.168.30.90      ---                ---
--- 10.30.3.103          192.168.30.91      ---                ---
--- 10.30.3.100          192.168.30.86      ---                ---
--- 10.30.3.124          192.168.30.84      ---                ---
--- 10.30.3.127          192.168.30.81      ---                ---
--- 10.30.3.121          192.168.30.89      ---                ---
--- 10.30.3.122          192.168.30.87      ---                ---
--- 10.30.3.118          192.168.30.92      ---                ---
--- 10.30.3.117          192.168.30.77      ---                ---
--- 10.30.3.115          192.168.30.73      ---                ---
... 68 more lines

```

Teardown (suite) Close All Connections

13.11 PASS Only The Static Mappings Live In The Configuration (0.1s, 1 commands)

Static mappings are configuration and survive a reload; pool allocations are runtime state and do not. Confusing the two is how NAT surprises people after a reboot.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **static** \${H1_IP} \${H1_NAT_IP}
- output does not contain \${NAT24}[R1].\${POOL_OCTET_FIRST}
- output contains **ip nat inside source list NAT-POOL-SRC pool SITE-POOL**

How it was checked

```

R1 show running-config | include ip nat inside source
ip nat inside source static 192.168.10.10 10.30.1.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.11 10.30.1.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.12 10.30.1.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.13 10.30.1.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.14 10.30.1.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.15 10.30.1.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.16 10.30.1.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.17 10.30.1.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.18 10.30.1.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.19 10.30.1.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL

```

Teardown (suite) Close All Connections

13.12 PASS Traffic From Both Mechanisms Is Encrypted (38.5s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **0% packet loss** (checked 2x)
- ping from \${source} on \${alias} to \${target} lost packets
- **\${r1_enc} >= \${expected}** holds
 - **\${r1_dec} >= \${expected}** holds
 - **\${r2_enc} >= \${expected}** holds
 - **\${r2_dec} >= \${expected}** holds

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 98237, #pkts encrypt: 98237, #pkts digest: 98237
  #pkts decaps: 85926, #pkts decrypt: 85926, #pkts verify: 85926
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 81 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 87429, #pkts encrypt: 87429, #pkts digest: 87429
  #pkts decaps: 99519, #pkts decrypt: 99519, #pkts verify: 99519
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 33 more lines
```

H1 ping -c 20 -I 192.168.10.10 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) from 192.168.10.10 : 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.50 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.26 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.23 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.32 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.24 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.10 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.16 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.37 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.14 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=1.48 ms
64 bytes from 10.30.2.10: icmp_seq=11 ttl=62 time=1.27 ms
64 bytes from 10.30.2.10: icmp_seq=12 ttl=62 time=1.16 ms
64 bytes from 10.30.2.10: icmp_seq=13 ttl=62 time=39.0 ms
... 11 more lines
```

H1 ping -c 20 -I 192.168.10.64 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) from 192.168.10.64 : 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.27 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.28 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.09 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.27 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.36 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.23 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.21 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=23.2 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.27 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=1.23 ms
64 bytes from 10.30.2.10: icmp_seq=11 ttl=62 time=1.19 ms
64 bytes from 10.30.2.10: icmp_seq=12 ttl=62 time=20.7 ms
64 bytes from 10.30.2.10: icmp_seq=13 ttl=62 time=1.29 ms
... 11 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1
protected vrf: (none)
```

```

local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 98502, #pkts encrypt: 98502, #pkts digest: 98502
#pkts decaps: 86190, #pkts decrypt: 86190, #pkts verify: 86190
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 81 more lines

```

R2 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 87695, #pkts encrypt: 87695, #pkts digest: 87695
#pkts decaps: 99784, #pkts decrypt: 99784, #pkts verify: 99784
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines

```

Teardown (suite) Close All Connections

13.13 PASS Neither Real Nor Translated Addresses Appear In Cleartext (10.0s, 9 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **0% packet loss**
ping from \${source} on \${alias} to \${target} lost packets
- output contains **ESP**
- output does not contain \${LAN24}[H1].\${DYN_OCTET_FIRST}
- output does not contain \${NAT24}[R1].\${POOL_OCTET_FIRST}

How it was checked

R1 no monitor capture POOLCAP

Capture does not exist

R1 monitor capture POOLCAP interface GigabitEthernet2 both

\${output} =

R1 monitor capture POOLCAP match ipv4 any any

\${output} =

R1 monitor capture POOLCAP buffer size 5

\${output} =

R1 monitor capture POOLCAP start

Started capture point : POOLCAP

H1 ping -c 10 -I 192.168.10.64 10.30.2.10

```

PING 10.30.2.10 (10.30.2.10) from 192.168.10.64 : 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.34 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.27 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=20.7 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.17 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.15 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.36 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.46 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.25 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.22 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=1.20 ms

```

--- 10.30.2.10 ping statistics ---

10 packets transmitted, 10 received, 0% packet loss, time 9014ms

... 1 more lines

R1 monitor capture POOLCAP stop

Stopped capture point : POOLCAP

R1 show monitor capture POOLCAP buffer brief

#	size	timestamp	source		destination	dscp	protocol
0	170	0.000000	10.10.10.1	->	10.10.10.2	0 BE	ESP
1	170	0.001007	10.10.10.2	->	10.10.10.1	0 BE	ESP
2	122	0.121988	10.10.10.1	->	10.10.10.2	48 CS6	ESP
3	122	0.122995	10.10.10.2	->	10.10.10.1	48 CS6	ESP
4	122	0.222996	10.10.10.2	->	10.10.10.1	48 CS6	ESP
5	122	0.222996	10.10.10.1	->	10.10.10.2	48 CS6	ESP
6	138	0.272981	10.10.10.1	->	10.10.10.2	48 CS6	ESP
7	122	0.551973	10.10.10.1	->	10.10.10.2	48 CS6	ESP
8	122	0.552965	10.10.10.2	->	10.10.10.1	48 CS6	ESP
9	122	0.713968	10.10.10.2	->	10.10.10.1	48 CS6	ESP
10	122	0.713968	10.10.10.1	->	10.10.10.2	48 CS6	ESP

... 118 more lines

R1 no monitor capture P00LCAP

`${output} =`

Teardown (suite) Close All Connections

14 Ntp

14.01 PASS Every Router Is An NTP Client Of The NMS Over The Management VRF (0.3s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **ntp source \${OOB_INTF}[\${r}]** (checked 3×)
- output contains **ntp server vrf \${MGMT_VRF} \${NMS_OOB_IP}** (checked 3×)
- output does not contain **pool.ntp.org** (checked 3×)

How it was checked

R1 show running-config | include ^ntp

```
ntp source GigabitEthernet5
ntp server vrf MGMT 192.168.99.10
```

R2 show running-config | include ^ntp

```
ntp source GigabitEthernet4
ntp server vrf MGMT 192.168.99.10
```

R3 show running-config | include ^ntp

```
ntp source GigabitEthernet4
ntp server vrf MGMT 192.168.99.10
```

Teardown (suite) Close Nms Session, Close All Connections

14.02 PASS The NMS Itself Synchronises To The Public Pool (2.3s, 2 commands)

The precondition the rest of this suite rests on. chrony is configured "local stratum 10", so it serves time whether or not it has an upstream -- meaning the routers can agree perfectly on time that came from nowhere. This requires a real upstream: a reference that is not the local clock, and a stratum that proves it came from somewhere else.

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain **7F7F0101**
the NMS is serving its own local clock, so lab time is fabricated
- **\${m}** is not empty
- **0 < \${m}[0] < \${MAX_STRATUM}** holds
- output matches **^*!s+!d+!l.d+!l.d+!l.d+**
the NMS has selected no upstream source: \${sources}

How it was checked

NMS chronyc tracking

```
Reference ID      : 603CF71D (h96-60-247-29.mdsnwi.broadband.dynamic.tds.net)
Stratum          : 3
Ref time (UTC)   : Mon Sep 07 19:51:50 2026
System time      : 0.000126047 seconds fast of NTP time
Last offset      : -0.000042516 seconds
RMS offset       : 0.000809346 seconds
Frequency        : 8.547 ppm fast
Residual freq    : +0.002 ppm
Skew             : 0.199 ppm
Root delay       : 0.065194733 seconds
Root dispersion  : 0.005409982 seconds
Update interval  : 521.4 seconds
Leap status      : Normal
```

NMS chronyc -n sources

MS Name/IP address	Stratum	Poll	Reach	LastRx	Last sample	
^+ 23.142.248.8	2	10	375	270	+1430us[+1430us]	+/- 50ms
^+ 65.182.224.39	3	10	377	856	+546us[+546us]	+/- 56ms
^+ 73.185.182.209	2	9	377	474	-1896us[-1896us]	+/- 59ms
^+ 45.79.82.45	2	9	377	223	-3000us[-3000us]	+/- 42ms
^* 96.60.247.29	2	10	377	859	-1731us[-1774us]	+/- 37ms
^+ 44.190.5.123	2	9	377	470	-993us[-993us]	+/- 39ms

Teardown (suite) Close Nms Session, Close All Connections

14.03 PASS Every Router Appears As A Client Of The NMS (0.1s, 1 commands)

Seen from the server, which is where a missing client shows up as an absence rather than as a router quietly using something else.

Setup (suite) Open All Routers

Acceptance criteria

- output contains `$$$r}_OOB_IP}` (checked 3×)

How it was checked

NMS `sudo chronyc clients`

Hostname	NTP	Drop	Int	IntL	Last	Cmd	Drop	Int	Last
=====	=====	=====	=====	=====	=====	=====	=====	=====	=====
192.168.99.3	78	0	5	-	104	0	0	-	-
192.168.99.1	76	0	5	-	201	0	0	-	-
192.168.99.2	81	0	6	-	8	0	0	-	-
localhost	0	0	-	-	-	32	0	6	0

Teardown (suite) Close Nms Session, Close All Connections

14.04 PASS Every Router Forms Associations With Pool Servers (0.1s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- `$$$rows}` is not empty (checked 3×)

How it was checked

R1 `show ntp associations`

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 96.60.247.29  3   201   256  377  0.924 -6.495 4.209
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 `show ntp associations`

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 96.60.247.29  3    8   128   77  0.913 -3.482 2.375
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 `show ntp associations`

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 96.60.247.29  3   104   256  377  0.063 -5.824 3.190
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

Teardown (suite) Close Nms Session, Close All Connections

14.05 PASS Associations Become Reachable (0.1s, 3 commands)

A non-zero reach register means packets are actually coming back, which distinguishes a working client from one that has merely been configured.

Setup (suite) Open All Routers

Acceptance criteria

- `$$$up}` is not empty (checked 3×)

How it was checked

R1 `show ntp associations`

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 96.60.247.29  3   201   256  377  0.924 -6.495 4.209
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 `show ntp associations`

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 96.60.247.29  3    8   128   77  0.913 -3.482 2.375
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 `show ntp associations`

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 96.60.247.29  3   104   256  377  0.063 -5.824 3.190
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

Teardown (suite) Close Nms Session, Close All Connections

14.06 PASS Every Router Selects A System Peer (0.1s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- `${peer}` differs from `$(None)` (checked 3×)

How it was checked

R1 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 96.60.247.29  3   201   256  377  0.924 -6.495 4.209
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 96.60.247.29  3    8   128   77  0.913 -3.482 2.375
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 96.60.247.29  3   104   256  377  0.063 -5.824 3.190
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

Teardown (suite) Close Nms Session, Close All Connections

14.07 PASS Every Router Clock Is Synchronised (0.1s, 3 commands)

The end state: IOS declares the clock synchronised. This lags real accuracy by a long way. IOS holds the flag back until its loopfilter leaves the 'FREQ' drift-measurement phase, measured on this platform at about 17 minutes from a cold NTP start -- while the offset is already within milliseconds well before that. The wait below is bounded to cover a freshly provisioned lab; on a lab that has been up it returns on the first poll and costs nothing.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **Clock is synchronized** (checked 3×)

How it was checked

R1 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0017 Hz, precision is 2**10
ntp uptime is 918500 (1/100 of seconds), resolution is 4000
reference time is EE4994D7.74FDF4F8 (19:53:59.457 UTC Mon Sep 7 2026)
clock offset is -6.4950 msec, root delay is 66.12 msec
root dispersion is 24.31 msec, peer dispersion is 4.20 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000006809 s/s
system poll interval is 256, last update was 731 sec ago.
```

R2 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0018 Hz, precision is 2**10
ntp uptime is 918500 (1/100 of seconds), resolution is 4000
reference time is EE49972D.71A9FD20 (20:03:57.444 UTC Mon Sep 7 2026)
clock offset is -3.4826 msec, root delay is 66.12 msec
root dispersion is 14.77 msec, peer dispersion is 2.37 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007266 s/s
system poll interval is 128, last update was 133 sec ago.
```

R3 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0017 Hz, precision is 2**10
ntp uptime is 918400 (1/100 of seconds), resolution is 4000
reference time is EE499001.7CAC0988 (19:33:21.487 UTC Mon Sep 7 2026)
clock offset is -5.8247 msec, root delay is 66.11 msec
root dispersion is 43.39 msec, peer dispersion is 3.19 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000006973 s/s
system poll interval is 256, last update was 1969 sec ago.
```

Teardown (suite) Close Nms Session, Close All Connections

14.08 PASS Stratum Is Sane On Every Router (0.1s, 3 commands)

Stratum 16 means unsynchronised. A client of a stratum 1-2 pool server should land well below the limit.

Setup (suite) Open All Routers

Acceptance criteria

- `${m}` is not empty (checked 3×)
- `${m}[0] < ${MAX_STRATUM}` holds (checked 3×)

How it was checked

R1 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0017 Hz, precision is 2**10
ntp uptime is 918500 (1/100 of seconds), resolution is 4000
reference time is EE4994D7.74FDF4F8 (19:53:59.457 UTC Mon Sep 7 2026)
clock offset is -6.4950 msec, root delay is 66.12 msec
root dispersion is 24.31 msec, peer dispersion is 4.20 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000006809 s/s
system poll interval is 256, last update was 731 sec ago.
```

R2 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0018 Hz, precision is 2**10
ntp uptime is 918500 (1/100 of seconds), resolution is 4000
reference time is EE49972D.71A9FD20 (20:03:57.444 UTC Mon Sep 7 2026)
clock offset is -3.4826 msec, root delay is 66.12 msec
root dispersion is 14.77 msec, peer dispersion is 2.37 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007266 s/s
system poll interval is 128, last update was 133 sec ago.
```

R3 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0017 Hz, precision is 2**10
ntp uptime is 918500 (1/100 of seconds), resolution is 4000
reference time is EE499001.7CAC0988 (19:33:21.487 UTC Mon Sep 7 2026)
clock offset is -5.8247 msec, root delay is 66.11 msec
root dispersion is 43.39 msec, peer dispersion is 3.19 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000006973 s/s
system poll interval is 256, last update was 1969 sec ago.
```

Teardown (suite) Close Nms Session, Close All Connections

14.09 PASS The Reference Is One Of The Peers The Router Actually Polls (0.1s, 6 commands)

Guards against a router calling itself synchronised to something that is not in its association list.

Setup (suite) Open All Routers

Acceptance criteria

- `${ref}` is not empty (checked 3×)
- output contains `${ref}[0]` (checked 3×)

How it was checked

R1 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0017 Hz, precision is 2**10
ntp uptime is 918500 (1/100 of seconds), resolution is 4000
reference time is EE4994D7.74FDF4F8 (19:53:59.457 UTC Mon Sep 7 2026)
clock offset is -6.4950 msec, root delay is 66.12 msec
root dispersion is 24.31 msec, peer dispersion is 4.20 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000006809 s/s
system poll interval is 256, last update was 731 sec ago.
```

R1 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 96.60.247.29  3   201   256   377  0.924  -6.495  4.209
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0018 Hz, precision is 2**10
ntp uptime is 918500 (1/100 of seconds), resolution is 4000
reference time is EE49972D.71A9FD20 (20:03:57.444 UTC Mon Sep 7 2026)
clock offset is -3.4826 msec, root delay is 66.12 msec
root dispersion is 14.77 msec, peer dispersion is 2.37 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007266 s/s
system poll interval is 128, last update was 133 sec ago.
```

R2 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 96.60.247.29  3    8   128    77  0.913  -3.482  2.375
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 show ntp status

```

Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0017 Hz, precision is 2**10
ntp uptime is 918500 (1/100 of seconds), resolution is 4000
reference time is EE499001.7CAC0988 (19:33:21.487 UTC Mon Sep 7 2026)
clock offset is -5.8247 msec, root delay is 66.11 msec
root dispersion is 43.39 msec, peer dispersion is 3.19 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000006973 s/s
system poll interval is 256, last update was 1969 sec ago.

```

R3 show ntp associations

```

address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10  96.60.247.29    3   104   256   377  0.063  -5.824  3.190
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured

```

Teardown (suite) Close Nms Session, Close All Connections

14.10 PASS Clock Offset Is Small On Every Router (0.1s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- $\{\text{peer}\}$ differs from $\{\text{None}\}$ (checked 3×)
- $\{\text{offset}\} < \{\text{MAX_OFFSET_MS}\}$ holds (checked 3×)

How it was checked

R1 show ntp associations

```

address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10  96.60.247.29    3   202   256   377  0.924  -6.495  4.209
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured

```

R2 show ntp associations

```

address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10  96.60.247.29    3     9   128    77  0.913  -3.482  2.375
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured

```

R3 show ntp associations

```

address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10  96.60.247.29    3   105   256   377  0.063  -5.824  3.190
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured

```

Teardown (suite) Close Nms Session, Close All Connections

14.11 PASS All Routers Agree On The Time (0.0s, 3 commands)

The practical payoff: three independently synchronised clocks should read the same, which is what makes logs across the fabric correlatable.

Setup (suite) Open All Routers

Acceptance criteria

- $\{\text{s12}\} < \{\text{MAX_SKEW_SECONDS}\}$ holds
- $\{\text{s13}\} < \{\text{MAX_SKEW_SECONDS}\}$ holds

How it was checked

R1 show clock

```
20:06:10.521 UTC Mon Sep 7 2026
```

R2 show clock

```
20:06:10.525 UTC Mon Sep 7 2026
```

R3 show clock

```
20:06:10.541 UTC Mon Sep 7 2026
```

Teardown (suite) Close Nms Session, Close All Connections

14.12 PASS The Clock Is No Longer At The Boot Epoch (0.1s, 6 commands)

An unsynchronised IOS device reports a 1993 or 1900 clock; a synchronised one reports now. This is the crude check that catches a clock that never moved at all.

Setup (suite) Open All Routers

Acceptance criteria

- `${year} >= 2026` holds (checked 3×)
- output does not contain `*` (checked 3×)

How it was checked

R1 show clock

20:06:10.553 UTC Mon Sep 7 2026

R1 show clock

20:06:10.564 UTC Mon Sep 7 2026

R2 show clock

20:06:10.572 UTC Mon Sep 7 2026

R2 show clock

20:06:10.586 UTC Mon Sep 7 2026

R3 show clock

20:06:10.600 UTC Mon Sep 7 2026

R3 show clock

20:06:10.611 UTC Mon Sep 7 2026

Teardown (suite) Close Nms Session, Close All Connections

15 Syslog

15.01 **PASS** The Collector Is A Service On The NMS Serving Every Router On One Port (0.1s, 2 commands)

What the move bought: one rsyslog listener for all three routers instead of one busybox listener per router, and a system service rather than something a suite starts -- so no run can leave a stale listener swallowing another's messages.

Setup (suite) Start Collector And Routers

Acceptance criteria

- `${state}` equals **active**
- output contains `:${SYSLOG_PORT}`
- output contains **rsyslogd**

How it was checked

NMS `systemctl is-active rsyslog`
active

NMS `sudo ss -ltnp | grep ':514 ' || echo NONE`

UNCONN	0	0	0.0.0.0:514	0.0.0.0:*	users:("rsyslogd",pid=7013,fd=6))
UNCONN	0	0	:::514	:::*	users:("rsyslogd",pid=7013,fd=7))

Teardown (suite) Close Collector, Close All Connections

15.02 **PASS** Every Router Is Configured To Log To The Collector (0.3s, 3 commands)

Setup (suite) Start Collector And Routers

Acceptance criteria

- output contains **logging host** `${SYSLOG_HOST}` (checked 3×)
- output contains **logging source-interface** `${OOB_INTF}${r}` vrf `${MGMT_VRF}` (checked 3×)
- output contains **service sequence-numbers** (checked 3×)

How it was checked

R1 `show running-config | include ^logging|^service (timestamps|sequence)`

```
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
logging source-interface GigabitEthernet5 vrf MGMT
logging host 192.168.99.10 vrf MGMT
```

R2 `show running-config | include ^logging|^service (timestamps|sequence)`

```
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
```

R3 `show running-config | include ^logging|^service (timestamps|sequence)`

```
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
```

Teardown (suite) Close Collector, Close All Connections

15.03 **PASS** Every Router Reports The Collector As An Active Logging Host (0.2s, 3 commands)

The router's own view: it should name the collector and show a non-zero count of messages sent to it.

Setup (suite) Start Collector And Routers

Acceptance criteria

- output contains `${SYSLOG_HOST}` (checked 3×)
- output contains **level** `${TRAP_LEVEL}` (checked 3×)

How it was checked

```
R1 show logging | include Trap logging|Logging to 192.168.99.10
  Trap logging: level informational, 1931 message lines logged
    Logging to 192.168.99.10 (MGMT) (udp port 514, audit disabled,
R2 show logging | include Trap logging|Logging to 192.168.99.10
  Trap logging: level informational, 2427 message lines logged
    Logging to 192.168.99.10 (MGMT) (udp port 514, audit disabled,
R3 show logging | include Trap logging|Logging to 192.168.99.10
  Trap logging: level informational, 1325 message lines logged
    Logging to 192.168.99.10 (MGMT) (udp port 514, audit disabled,
```

Teardown (suite) Close Collector, Close All Connections

15.04 PASS The Hub Message Reaches The Collector (4.0s, 1 commands)

End to end over the LAN: emit a uniquely marked message and find it at the collector.

Setup (suite) Start Collector And Routers

Acceptance criteria

- `${n} > 0` holds

How it was checked

```
NMS sudo grep -c -- 'LABSYSLOG-HUB-160610' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
1
```

Teardown (suite) Close Collector, Close All Connections

15.05 PASS Each Spoke Message Reaches The Collector Across The Tunnel (8.1s, 2 commands)

The spokes have no direct path to h1's LAN; their messages are carried by the tunnels and the hub's routing.

Setup (suite) Start Collector And Routers

Acceptance criteria

- `${n} > 0` holds (checked 2×)

How it was checked

```
NMS sudo grep -c -- 'LABSYSLOG-R2-160610' /var/log/lab/192.168.99.2.log 2>/dev/null || echo 0
1
```

```
NMS sudo grep -c -- 'LABSYSLOG-R3-160610' /var/log/lab/192.168.99.3.log 2>/dev/null || echo 0
1
```

Teardown (suite) Close Collector, Close All Connections

15.06 PASS Received Messages Carry Facility Severity And Mnemonic (4.1s, 2 commands)

Setup (suite) Start Collector And Routers

Acceptance criteria

- `${n} > 0` holds
- `${m}[facility]` equals **FACCHECK**
- `${m}[mnemonic]` equals **MNEMCHECK**
- `${m}[sev]` equals **5**

How it was checked

```
NMS sudo grep -c -- 'LABSYSLOG-FIELDS-160610' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
1
```

```
NMS sudo cat /var/log/lab/192.168.99.1.log 2>/dev/null || true
<189>175: 000171: *Sep  7 17:45:23.653 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>176: 000172: *Sep  7 17:45:23.707 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 17:45:23 UTC Mon Sep 7 2026
<189>177: 000173: *Sep  7 17:45:23.707 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>178: 000174: *Sep  7 17:45:23.915 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0
(10.0.2.2)
<189>179: 000175: *Sep  7 17:45:24.967 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0
(10.0.2.2)
```

```

<190>180: 000176: *Sep 7 17:45:25.069 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>181: 000177: *Sep 7 17:45:25.069 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>182: 000178: *Sep 7 17:45:51.602 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>183: 000179: *Sep 7 17:45:51.656 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 17:45:51 UTC Mon Sep 7 2026
<189>184: 000180: *Sep 7 17:45:51.656 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<190>185: 000181: *Sep 7 17:45:51.788 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>186: 000182: *Sep 7 17:45:51.788 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>187: 000183: *Sep 7 17:48:46.557 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>188: 000184: *Sep 7 17:48:46.611 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 17:48:46 UTC Mon Sep 7 2026
... 1745 more lines

```

Teardown (suite) Close Collector, Close All Connections

15.07 PASS Received Messages Carry Sequence Numbers (0.1s, 1 commands)

'service sequence-numbers' is configured, so every message should be numbered -- which is how a collector detects loss.

Setup (suite) Start Collector And Routers

Acceptance criteria

- `${msgs}` is not empty
- `${uniq} > 1` holds
- `min($seqs) >= 0` holds

How it was checked

NMS `sudo cat /var/log/lab/192.168.99.1.log 2>/dev/null || true`

```

<189>175: 000171: *Sep 7 17:45:23.653 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>176: 000172: *Sep 7 17:45:23.707 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 17:45:23 UTC Mon Sep 7 2026
<189>177: 000173: *Sep 7 17:45:23.707 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>178: 000174: *Sep 7 17:45:23.915 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0
(10.0.2.2)
<189>179: 000175: *Sep 7 17:45:24.967 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0
(10.0.2.2)
<190>180: 000176: *Sep 7 17:45:25.069 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>181: 000177: *Sep 7 17:45:25.069 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>182: 000178: *Sep 7 17:45:51.602 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>183: 000179: *Sep 7 17:45:51.656 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 17:45:51 UTC Mon Sep 7 2026
<189>184: 000180: *Sep 7 17:45:51.656 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<190>185: 000181: *Sep 7 17:45:51.788 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>186: 000182: *Sep 7 17:45:51.788 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>187: 000183: *Sep 7 17:48:46.557 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>188: 000184: *Sep 7 17:48:46.611 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 17:48:46 UTC Mon Sep 7 2026
... 1745 more lines

```

Teardown (suite) Close Collector, Close All Connections

15.08 PASS Message Timestamps Agree With The Router Clock (4.1s, 3 commands)

Ties syslog to NTP: a message's timestamp should match the sending router's clock. Skewed timestamps make a collector's correlation across devices worthless.

Setup (suite) Start Collector And Routers

Acceptance criteria

- `${n} > 0` holds
- `${skew} < ${MAX_TS_SKEW}` holds

How it was checked

```
NMS sudo grep -c -- 'LABSYSLOG-TIME-160610' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
1
```

R1 show clock

20:06:33.885 UTC Mon Sep 7 2026

```
NMS sudo cat /var/log/lab/192.168.99.1.log 2>/dev/null || true
```

```
<189>175: 000171: *Sep 7 17:45:23.653 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>176: 000172: *Sep 7 17:45:23.707 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source: 10.0.2.2] [localport: 22] at 17:45:23 UTC Mon Sep 7 2026
<189>177: 000173: *Sep 7 17:45:23.707 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>178: 000174: *Sep 7 17:45:23.915 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0 (10.0.2.2)
<189>179: 000175: *Sep 7 17:45:24.967 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0 (10.0.2.2)
<190>180: 000176: *Sep 7 17:45:25.069 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>181: 000177: *Sep 7 17:45:25.069 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user '' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>182: 000178: *Sep 7 17:45:51.602 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>183: 000179: *Sep 7 17:45:51.656 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source: 10.0.2.2] [localport: 22] at 17:45:51 UTC Mon Sep 7 2026
<189>184: 000180: *Sep 7 17:45:51.656 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<190>185: 000181: *Sep 7 17:45:51.788 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>186: 000182: *Sep 7 17:45:51.788 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user '' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>187: 000183: *Sep 7 17:48:46.557 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>188: 000184: *Sep 7 17:48:46.611 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source: 10.0.2.2] [localport: 22] at 17:48:46 UTC Mon Sep 7 2026
... 1746 more lines
```

Teardown (suite) Close Collector, Close All Connections

15.09 PASS Messages Below The Trap Level Are Not Forwarded (14.0s, 1 commands)

Negative control. Trap level is informational (6), so a debug-severity (7) message must not reach the collector. If this ever passes traffic, the level filter is not working and the positive tests above prove much less than they appear to.

Setup (suite) Start Collector And Routers

Acceptance criteria

- $\{n\}$ equals 0
a debug-severity message reached the collector despite trap level $\{TRAP_LEVEL\}$

How it was checked

```
NMS sudo grep -c -- 'LABSYSLOG-DEBUG-160610' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
0
0
```

Teardown (suite) Close Collector, Close All Connections

15.10 PASS All Three Routers Are Represented At The Collector (12.1s, 3 commands)

Messages from every router, distinguishable at the collector, which is the point of a central log host.

Setup (suite) Start Collector And Routers

Acceptance criteria

- $\{n\} > 0$ holds (checked 3×)

How it was checked

```
NMS sudo grep -c -- 'LABSYSLOG-ALL-R1-160610' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
1
NMS sudo grep -c -- 'LABSYSLOG-ALL-R2-160610' /var/log/lab/192.168.99.2.log 2>/dev/null || echo 0
1
NMS sudo grep -c -- 'LABSYSLOG-ALL-R3-160610' /var/log/lab/192.168.99.3.log 2>/dev/null || echo 0
1
```


Teardown (suite) Close Collector, Close All Connections

16 Snmpv3

16.01 PASS Both Trap Receivers Are Services On The NMS (0.1s, 2 commands)

snmptrapd on the standard port and the raw capture on a second one, both system services rather than listeners a suite starts -- so no run can leave one behind, which is how a negative test once passed while nothing was arriving.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- `${state}` equals **active active**
- output contains `:${TRAP_PORT}`
- output contains `:${TRAP_RAW_PORT}`

How it was checked

```
NMS systemctl is-active snmptrapd lab-trapcap | tr '\n' ' '
active active

NMS sudo ss -ltnp | grep -E ':(162|1162) ' || echo NONE
UNCONN 0      0      0.0.0.0:1162    0.0.0.0:*      users:(("python3",pid=7175,fd=3))
UNCONN 0      0      0.0.0.0:162    0.0.0.0:*
users:((("snmptrapd",pid=7134,fd=3),("systemd",pid=1,fd=113))
UNCONN 0      0      [::]:162      [::]:*
users:((("snmptrapd",pid=7134,fd=4),("systemd",pid=1,fd=114))
```

Teardown (suite) Close Trap Collector, Close All Connections

16.02 PASS Every Router Has An SNMPv3 User With Authentication And Privacy (0.0s, 3 commands)

v3 without auth and priv is barely better than v2c, so the protocols in use are asserted, not just the user's existence.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- output contains **User name: \${SNMP_USER}** (checked 3×)
- output contains **Authentication Protocol: SHA** (checked 3×)
- output contains **Privacy Protocol: AES128** (checked 3×)
- output contains **Group-name: \${SNMP_GROUP}** (checked 3×)

How it was checked

```
R1 show snmp user
User name: labmon
Engine ID: 800000090300525400C80101
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: LABGRP
```

```
R2 show snmp user
User name: labmon
Engine ID: 800000090300525400C80201
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: LABGRP
```

```
R3 show snmp user
User name: labmon
Engine ID: 800000090300525400C80301
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: LABGRP
```

Teardown (suite) Close Trap Collector, Close All Connections

16.03 PASS Every Router Has A v3 Group Requiring Privacy (0.1s, 3 commands)

Setup (suite) Start Snmp Collectors

Acceptance criteria

- output contains **groupname: \${SNMP_GROUP}** (checked 3×)
- output contains **security model:v3 priv** (checked 3×)

How it was checked

R1 show snmp group

```
groupname: ILMI                      security model:v1
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                     writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: ILMI                      security model:v2c
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                     writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: LABGRP                    security model:v3 priv
contextname: <no context specified>  storage-type: nonvolatile
... 3 more lines
```

R2 show snmp group

```
groupname: ILMI                      security model:v1
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                     writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: ILMI                      security model:v2c
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                     writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: LABGRP                    security model:v3 priv
contextname: <no context specified>  storage-type: nonvolatile
... 3 more lines
```

R3 show snmp group

```
groupname: ILMI                      security model:v1
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                     writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: ILMI                      security model:v2c
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                     writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: LABGRP                    security model:v3 priv
contextname: <no context specified>  storage-type: nonvolatile
... 3 more lines
```

Teardown (suite) Close Trap Collector, Close All Connections

16.04 PASS No Router Accepts SNMPv1 Or v2c Communities (0.3s, 3 commands)

Security posture: configuring v3 is pointless if a community string still gets you in.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- **\${out}** is empty (checked 3×)

How it was checked

R1 show running-config | include ^snmp-server community
\${output} =

R2 show running-config | include ^snmp-server community
\${output} =

R3 show running-config | include ^snmp-server community
\${output} =

Teardown (suite) Close Trap Collector, Close All Connections

16.05 PASS Every Router Targets The Collector With v3 Privacy (0.1s, 3 commands)

Setup (suite) Start Snmp Collectors

Acceptance criteria

- output contains **\${COLLECTOR}** (checked 3×)
- output contains **security model: v3 priv** (checked 3×)
- output contains **user: \${SNMP_USER}** (checked 3×)

How it was checked

R1 show snmp host

```
Notification host: 192.168.99.10      udp-port: 1162  VRFName: MGMT  type: trap
user: labmon      security model: v3 priv
```

```
Notification host: 192.168.99.10      udp-port: 162   VRFName: MGMT  type: trap
user: labmon      security model: v3 priv
```

R2 show snmp host

```
Notification host: 192.168.99.10      udp-port: 1162  VRFName: MGMT  type: trap
user: labmon      security model: v3 priv
```

```
Notification host: 192.168.99.10      udp-port: 162   VRFName: MGMT  type: trap
user: labmon      security model: v3 priv
```

R3 show snmp host

```
Notification host: 192.168.99.10      udp-port: 1162  VRFName: MGMT  type: trap
user: labmon      security model: v3 priv
```

```
Notification host: 192.168.99.10      udp-port: 162   VRFName: MGMT  type: trap
user: labmon      security model: v3 priv
```

Teardown (suite) Close Trap Collector, Close All Connections

16.06 PASS Global Trap Sending Is Enabled (0.1s, 3 commands)

Enabling individual trap types leaves the global switch off: the router builds trap PDUs and never sends them, which looks exactly like a broken collector.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- output contains **SNMP global trap: enabled** (checked 3×)

How it was checked

R1 show snmp | include global trap

```
SNMP global trap: enabled
```

R2 show snmp | include global trap

```
SNMP global trap: enabled
```

R3 show snmp | include global trap

```
SNMP global trap: enabled
```

Teardown (suite) Close Trap Collector, Close All Connections

16.07 PASS Traps From Every Router Reach The Collector (13.2s, 51 commands)

Setup (suite) Start Snmp Collectors

Acceptance criteria

- **\${traps}** is not empty (checked 3×)

How it was checked

R1 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R1 interface Loopback270

```
${output} =
```

R1 ip address 10.99.97.1 255.255.255.255

```
${output} =
```

R1 snmp trap link-status

```

    ${output} =
R1 end
    ${output} =
R1 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R1 interface Loopback270
    ${output} =
R1 shutdown
    ${output} =
R1 end
    ${output} =
R1 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R1 interface Loopback270
    ${output} =
R1 no shutdown
    ${output} =
R1 end
    ${output} =
R1 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R1 no interface Loopback270
    ${output} =
R1 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 interface Loopback270
    ${output} =
R2 ip address 10.99.97.1 255.255.255.255
    ${output} =
R2 snmp trap link-status
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 interface Loopback270
    ${output} =
R2 shutdown
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 interface Loopback270
    ${output} =
R2 no shutdown
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no interface Loopback270
    ${output} =
R2 end
    ${output} =
R3 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R3 interface Loopback270
    ${output} =
R3 ip address 10.99.97.1 255.255.255.255
    ${output} =
R3 snmp trap link-status
    ${output} =

```

```
R3 end
${output} =

R3 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R3 interface Loopback270
${output} =

R3 shutdown
${output} =

R3 end
${output} =

R3 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R3 interface Loopback270
${output} =

R3 no shutdown
${output} =

R3 end
${output} =

R3 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R3 no interface Loopback270
${output} =

R3 end
${output} =

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true
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1788803096.391 192.168.99.2 3081f2020103300d02012c020205dc04010302010304373035040c800000090300525400c8020102
0102020202c204066c61626d6f6e040cb99e87223f51e570ae17f77d04088a5dbc633a502d650481a4a2fd1e95f78522963fae000fc1
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1788803096.645 192.168.99.2 3082013c020103300d02012e020205dc04010302010304373035040c800000090300525400c80201
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1788803124.123 192.168.99.1 3081f2020103300d020142020205dc04010302010304373035040c800000090300525400c8010102
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```

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... 5352 more lines
```

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

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1788803124.123 192.168.99.1 3081f2020103300d020142020205dc04010302010304373035040c800000090300525400c8010102
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... 5352 more lines

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

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1788803125.954 192.168.99.3 3081db020103300d02012e020205dc04010302010304373035040c800000090300525400c8030102


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1788803126.554 192.168.99.3 3081f2020103300d020130020205dc04010302010304373035040c800000090300525400c8030102
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1788803126.805 192.168.99.3 3081db020103300d020132020205dc04010302010304373035040c800000090300525400c8030102
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0c0ea187ed17a61fa340fac549253feb7c0bd3ea
... 5353 more lines
```

Teardown (suite) Close Trap Collector, Close All Connections

16.08 PASS Received Traps Are SNMP Version 3 (0.3s, 3 commands)

Reads the version field out of the datagram itself rather than trusting the configuration.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- `${traps}` is not empty (checked 3x)
- `${versions}` equals `${[[3]]}` (checked 3x)

How it was checked

NMS `sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true`

```
1788803096.140 192.168.99.2 3081f2020103300d02012a020205dc04010302010304373035040c800000090300525400c8020102
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1788803096.391 192.168.99.2 3081f2020103300d02012c020205dc04010302010304373035040c800000090300525400c8020102
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1788803125.704 192.168.99.3 3081f2020103300d02012c020205dc04010302010304373035040c800000090300525400c8030102
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NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

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```

```
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0c0ea187ed17a61fa340fac549253feb7c0bd3ea
... 5354 more lines
```

Teardown (suite) Close Trap Collector, Close All Connections

16.09 PASS Received Traps Carry The Configured User And Privacy Flags (0.3s, 3 commands)

Setup (suite) Start Snmp Collectors

Acceptance criteria

- `#{users}` equals `#{['${SNMP_USER}']}` (checked 3×)
- `#{all_priv}` holds (checked 3×)

How it was checked

NMS `sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true`

```
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```



```
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... 5354 more lines
```

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

```
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```

```
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... 5355 more lines
```

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

```
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1788803125.728 192.168.99.1 3082013d020103300d02014c020205dc04010302010304373035040c800000090300525400c80101
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```

```
1d5e9b106c4e72f7a33a75546face52513fea6af352968fc1f68d0d5ddfc9d69ff6c8abb5e412aa948b1c45a4260939f3b1bcb20791b7
11a2f1d277266a0f726133
1788803125.954 192.168.99.3 3081db020103300d02012e020205dc04010302010304373035040c800000090300525400c8030102
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1788803126.554 192.168.99.3 3081f2020103300d020130020205dc04010302010304373035040c800000090300525400c8030102
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1788803126.805 192.168.99.3 3081db020103300d0201302020205dc04010302010304373035040c800000090300525400c8030102
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0c0ea187ed17a61fa340fac549253feb7c0bd3ea
... 5355 more lines
```

Teardown (suite) Close Trap Collector, Close All Connections

16.10 PASS Trap Payloads Are Encrypted (0.3s, 3 commands)

With privacy in force the scoped PDU arrives as an opaque OCTET STRING. If it were a readable SEQUENCE the varbinds would be on the wire in clear.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- `#{encrypted}` holds (checked 3×)

How it was checked

NMS `sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true`

```
1788803096.140 192.168.99.2 3081f2020103300d02012a020205dc04010302010304373035040c800000090300525400c8020102
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1788803096.391 192.168.99.2 3081f2020103300d02012c020205dc04010302010304373035040c800000090300525400c8020102
0102020202c204066c61626d6f6e040cb99e87223f51e570ae17f77d04088a5dbcb633a502d650481a4a2fd1e95f78522963fae000fc1
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1788803124.123 192.168.99.1 3081f2020103300d020142020205dc04010302010304373035040c800000090300525400c8010102
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1788803124.623 192.168.99.1 3081db020103300d020146020205dc04010302010304373035040c800000090300525400c8010102
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NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

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NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

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ba5cbc3180231980b150ef8dac82d8f49a7e444b2ce8630f6fc5e436b866b17f64dd9e5d07d584308158363d67521861a56232e4abc2c
a5874a28367c44f44f35614b2923277986aa2912eadc0c82b648e3c81d293143c8ca3db5e034f7f36c8e02
1788803124.623 192.168.99.1 3081db020103300d020146020205dc04010302010304373035040c800000090300525400c8010102
0102020202de04066c61626d6f6e040c780611fbd402b12c77c9810c040830d81086869a6e604081d4dfc1ec6d05f298fe18babfda6
868f3261d34859e18e5b182ec8641cd1bcc4853d641f6963b6426bed6abdc497716e582beb3275c9b0f845c855dd395503640340f900
a90271558551f972f878313d4c25ff92432839eed061ed0a81a9afc326b60a6202fdb128a3293ea063e2fbf06204f217a496e9941e
081a5a86a627ffe80ef228fba447629338bee9b
1788803125.229 192.168.99.1 3081f2020103300d020148020205dc04010302010304373035040c800000090300525400c8010102
0102020202df04066c61626d6f6e040c0173b62339bc54a4f11b1f38040830d81086869a6e600481a4bb295c762ce8319549b9f0ee00
ccf4c70e083598f7d0b387e0d4e647515bd2c3e1ac6ec6d3815337c0ff0f47e8e3e13fd81a7b8b977be0b460ead26ccb5e49e2ff653
9223de21757b6651658f767205b58ac1cd21651a38d5d5c1af85920ab2a09ebb8ca96e8674fc82624b970994afbf5fae318623514a45e
5f69012fac10535d257e0d5328b5e73ddee54f8499753c9b8898dceb1001b5b2d4d0d3afad35c725c8c4b60
1788803125.455 192.168.99.3 3081f2020103300d02012a020205dc04010302010304373035040c800000090300525400c8030102
0102020202df04066c61626d6f6e040c294a36477b10366227c93705040835b22fd9d14a5f200481a4959e9f626fdacc78f394d82297
18f155ba4ed4d1d5dc9997246261bee2f460da7bebfdb6f81b1228768bd77b73d7a41c1d2c2c802b210c3240e0d1eb98ca08982dd0e
8a09e72d1ec6db7ba811bbf3002abbc8ffe1e866fa1e7c0f778d1e73b1235631c013e5dec4b2191932a5f0ec89a55d0be466dc1f25e
69631a5152b037dd21f1035a8f58cd496b54d73a03f8630bedb290d66d7dcd9f9cb57754257b418cea6a9
1788803125.478 192.168.99.1 3081db020103300d0201a020205dc04010302010304373035040c800000090300525400c8010102
0102020202df04066c61626d6f6e040cfc149655332b61ccbe054e08040830d81086869a6e6004818d9c16590f4b15372ba5103ba62e
3f3166569d7ee5dd7681d6b70b56a309f52badc89241e1e88027aa91c37b211eaedb71a82eff88e26826da9445fa0d11b3be79e1420a
b6827421c1cf1cf0e49306122bccf34415ce45f4a5bd23827e03c9df7c59443648813d7ac9fc184cdaf7b1927f9c1e19bed98c7abdea3
07157269de7a28288c8d2b66a904054c44795a2c
1788803125.704 192.168.99.3 3081f2020103300d02012c020205dc04010302010304373035040c800000090300525400c8030102
0102020202df04066c61626d6f6e040c2278f01240147c6beb32946040835b22fd9d14a5f220481a48f5b40cd3ea2c3ac5d87048953
8e6dd2b0e507669543ee2851b762db3f6b0ce10c1958391674f303a6e6d9664f077f8980de2f22af5be4997786cf9978cc9b5ab88c76
7e031e3b2e03b4581569a0de282c47d020fcca13d641861c3bd92f561d573db0b88e65e731bd25798ebe8c1fff341952c45ab84621e
4098a04519a6ea69dd8212352c4bbb3010d7e57f00b32ad31ed45d63b9a6902e0a637c83eead7a85edcd7
1788803125.728 192.168.99.1 3082013d020103300d02014c020205dc04010302010304373035040c800000090300525400c80101
020102020202df04066c61626d6f6e040cdaea973e5e017c0c6d2166b5040830d81086869a6e6a0481efabec86b13535efce488aae59
84d8d39ec3eeacff487d7b4f6fe30bc793316145586fe859331159707b47b5aa6d1356d71ced033b92b9ec9d87f9b1b7ba11fab5b09
```

```
483487400b5760b173cdb7d3442d1d4a07e3f6726937bb3d8fa7ed718c94fc27a82805efc82533cfbec03e8b110e7f6077229e74c4b6
7e8ba4566d9b9114c41a0285b500194f68e493bb8b035cf33f064685c3794896a80e2836ed7d22716dc4496b49fd716863fc0b68e653
1d5e9b106c4e72f7a33a75546face52513fea6af352968fc1f68d0d5ddf9c9d69ff6c8abb5e412aa948b1c45a4260939f3b1bc20791b7
11a2f1d277266a0f726133
1788803125.954 192.168.99.3 3081db020103300d02012e020205dc04010302010304373035040c800000090300525400c8030102
0102020202df04066c61626d6f6e040c4c6cd43e7963ccb9b374bcb1040835b22fd9d14a5f2404818d63257883cac272f48279fad535
03f507d2bc453d69cefddbc2447ffc911b255e50f47669e510bc563187271cfd1eccd9b4deb6eefb267ece9a8a39e2a79dc17875b062
109c7c372a183431a20593f59f8c8ecfa827d46c4f700d9d85500c26926ff7efd94451eea875ac256cea34b6d23444c97cbaedbfb622
c7661c2ce777f355306e3a0320d24259f1b3852
1788803126.554 192.168.99.3 3081f2020103300d020130020205dc04010302010304373035040c800000090300525400c8030102
0102020202e004066c61626d6f6e040c47f9363931d78e229e4cd1c6040835b22fd9d14a5f260481a404822a478ef80a4680f6c86623
02fc5ae684697ad85b4ba6edf77c7c6c4d3e4768fa589abfc15dcca9d2df87dc16d114ed99e15d27e3ac42cbfc166cdf8493f26d1c07
715f9bfe7032125e76fdc56ac6c0d751846cf4c9ac72b30008377946f96e5c4e2929163b390db08010b589ae0bfa65e45552bb4c4542
61c34d24ff8c1471063eb73ab6a13d47b0a6746c33241bcd083e2d612306cead450bd29b684e4304dc90e1
1788803126.805 192.168.99.3 3081db020103300d020132020205dc04010302010304373035040c800000090300525400c8030102
0102020202e104066c61626d6f6e040c15b72a4734b81f8f285b4be4040835b22fd9d14a5f2804818df82cff847696fda65bf7ccc44
2a1386b355e5a66aaa7992d2839eb2d4efa5dc08f2e6e834520d2980cb91f3353fd346adeb614b598654a9fc882dc0de23ed5bd611e
993c52c91753dd7a7200bfcf7fe17ab66113ba02fd4cccbef6faade260b687bab23f8d0ed68abe905b37dc59011a0d440b7a70325c2d
0c0ea187ed17a61fa340fac549253feb7c0bd3ea
... 5356 more lines
```

Teardown (suite) Close Trap Collector, Close All Connections

16.11 PASS snmptrapd Decrypts Traps From Every Router (0.2s, 3 commands)

The strongest single statement this suite makes: a real SNMP manager, holding only the configured user and each router's engine ID, authenticated and decrypted the traps. That cannot happen unless the keys, the engine IDs and the privacy protocol all match on both sides.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- **{entries}** is not empty (checked 3×)
- **{oids}** is not empty (checked 3×)

How it was checked

NMS `sudo cat /var/log/lab/snmptrapd.log 2>/dev/null || true`

```
NET-SNMP version 5.9.4.pre2
2026-09-07 18:41:39 <UNKNOWN> [UDP: [192.168.99.2]:52037->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:23.80 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.0.1
iso.3.6.1.4.1.9.2.9.3.1.1.434.1 6 iso.3.6.1.2.1.6.13.1.1.10.0.2.15.22.10.0.2.2.40618 8
iso.3.6.1.4.1.9.2.6.1.1.5.10.0.2.15.22.10.0.2.2.40618 15
iso.3.6.1.4.1.9.2.6.1.1.1.10.0.2.15.22.10.0.2.2.40618 1856
iso.3.6.1.4.1.9.2.6.1.1.2.10.0.2.15.22.10.0.2.2.40618 5587 iso.3.6.1.4.1.9.2.9.2.1.18.434 "lab"
2026-09-07 18:41:50 <UNKNOWN> [UDP: [192.168.99.3]:52396->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:34.21 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.65 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.65 2
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.65 3 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.65 "lab"
2026-09-07 18:41:50 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:40.29 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.85 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.85 2
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.85 3 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.85 "lab"
2026-09-07 18:41:50 <UNKNOWN> [UDP: [192.168.99.3]:52396->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:34.28 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.41.2.0.1
iso.3.6.1.4.1.9.9.41.1.2.3.1.2.27 "AAAA" iso.3.6.1.4.1.9.9.41.1.2.3.1.3.27 5
iso.3.6.1.4.1.9.9.41.1.2.3.1.4.27 "CLI_DEPRECATED" iso.3.6.1.4.1.9.9.41.1.2.3.1.5.27 "WARNING: Command
has been added to the configuration using a type 0 password. However, recommended to migrate to strong
type-6 encryption" iso.3.6.1.4.1.9.9.41.1.2.3.1.6.27 0:1:09:34.28
2026-09-07 18:41:51 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:40.37 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.41.2.0.1
iso.3.6.1.4.1.9.9.41.1.2.3.1.2.34 "AAAA" iso.3.6.1.4.1.9.9.41.1.2.3.1.3.34 5
iso.3.6.1.4.1.9.9.41.1.2.3.1.4.34 "CLI_DEPRECATED" iso.3.6.1.4.1.9.9.41.1.2.3.1.5.34 "WARNING: Command
has been added to the configuration using a type 0 password. However, recommended to migrate to strong
type-6 encryption" iso.3.6.1.4.1.9.9.41.1.2.3.1.6.34 0:1:09:40.37
2026-09-07 18:41:51 <UNKNOWN> [UDP: [192.168.99.3]:52396->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:34.28 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.41.2.0.1
iso.3.6.1.4.1.9.9.41.1.2.3.1.2.28 "AAAA" iso.3.6.1.4.1.9.9.41.1.2.3.1.3.28 5
iso.3.6.1.4.1.9.9.41.1.2.3.1.4.28 "CLI_DEPRECATED" iso.3.6.1.4.1.9.9.41.1.2.3.1.5.28 "WARNING: Command
has been added to the configuration using a type 0 password. However, recommended to migrate to strong
type-6 encryption" iso.3.6.1.4.1.9.9.41.1.2.3.1.6.28 0:1:09:34.28
2026-09-07 18:41:51 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
... 8329 more lines
```

NMS `sudo cat /var/log/lab/snmptrapd.log 2>/dev/null || true`

```
NET-SNMP version 5.9.4.pre2
2026-09-07 18:41:39 <UNKNOWN> [UDP: [192.168.99.2]:52037->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:23.80 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.0.1
```

```

iso.3.6.1.4.1.9.2.9.3.1.1.434.1 6      iso.3.6.1.2.1.6.13.1.1.10.0.2.15.22.10.0.2.2.40618 8
iso.3.6.1.4.1.9.2.6.1.1.5.10.0.2.15.22.10.0.2.2.40618 15
iso.3.6.1.4.1.9.2.6.1.1.1.10.0.2.15.22.10.0.2.2.40618 1856
iso.3.6.1.4.1.9.2.6.1.1.2.10.0.2.15.22.10.0.2.2.40618 5587      iso.3.6.1.4.1.9.2.9.2.1.18.434 "lab"
2026-09-07 18:41:50 <UNKNOWN> [UDP: [192.168.99.3]:52396->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:34.21      iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.65 1      iso.3.6.1.4.1.9.9.43.1.1.6.1.4.65 2
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.65 3      iso.3.6.1.4.1.9.9.43.1.1.6.1.8.65 "lab"
2026-09-07 18:41:50 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:40.29      iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.85 1      iso.3.6.1.4.1.9.9.43.1.1.6.1.4.85 2
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.85 3      iso.3.6.1.4.1.9.9.43.1.1.6.1.8.85 "lab"
2026-09-07 18:41:50 <UNKNOWN> [UDP: [192.168.99.3]:52396->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:34.28      iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.41.2.0.1
iso.3.6.1.4.1.9.9.41.1.2.3.1.2.27 "AAAA"      iso.3.6.1.4.1.9.9.41.1.2.3.1.3.27 5
iso.3.6.1.4.1.9.9.41.1.2.3.1.4.27 "CLI_DEPRECATED"      iso.3.6.1.4.1.9.9.41.1.2.3.1.5.27 "WARNING: Command
has been added to the configuration using a type 0 password. However, recommended to migrate to strong
type-6 encryption"      iso.3.6.1.4.1.9.9.41.1.2.3.1.6.27 0:1:09:34.28
2026-09-07 18:41:51 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:40.37      iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.41.2.0.1
iso.3.6.1.4.1.9.9.41.1.2.3.1.2.34 "AAAA"      iso.3.6.1.4.1.9.9.41.1.2.3.1.3.34 5
iso.3.6.1.4.1.9.9.41.1.2.3.1.4.34 "CLI_DEPRECATED"      iso.3.6.1.4.1.9.9.41.1.2.3.1.5.34 "WARNING: Command
has been added to the configuration using a type 0 password. However, recommended to migrate to strong
type-6 encryption"      iso.3.6.1.4.1.9.9.41.1.2.3.1.6.34 0:1:09:40.37
2026-09-07 18:41:51 <UNKNOWN> [UDP: [192.168.99.3]:52396->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:34.28      iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.41.2.0.1
iso.3.6.1.4.1.9.9.41.1.2.3.1.2.28 "AAAA"      iso.3.6.1.4.1.9.9.41.1.2.3.1.3.28 5
iso.3.6.1.4.1.9.9.41.1.2.3.1.4.28 "CLI_DEPRECATED"      iso.3.6.1.4.1.9.9.41.1.2.3.1.5.28 "WARNING: Command
has been added to the configuration using a type 0 password. However, recommended to migrate to strong
type-6 encryption"      iso.3.6.1.4.1.9.9.41.1.2.3.1.6.28 0:1:09:34.28
2026-09-07 18:41:51 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
... 8329 more lines

```

NMS `sudo cat /var/log/lab/snmptrapd.log 2>/dev/null || true`

```

NET-SNMP version 5.9.4.pre2
2026-09-07 18:41:39 <UNKNOWN> [UDP: [192.168.99.2]:52037->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:23.80      iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.0.1
iso.3.6.1.4.1.9.2.9.3.1.1.434.1 6      iso.3.6.1.2.1.6.13.1.1.10.0.2.15.22.10.0.2.2.40618 8
iso.3.6.1.4.1.9.2.6.1.1.5.10.0.2.15.22.10.0.2.2.40618 15
iso.3.6.1.4.1.9.2.6.1.1.1.10.0.2.15.22.10.0.2.2.40618 1856
iso.3.6.1.4.1.9.2.6.1.1.2.10.0.2.15.22.10.0.2.2.40618 5587      iso.3.6.1.4.1.9.2.9.2.1.18.434 "lab"
2026-09-07 18:41:50 <UNKNOWN> [UDP: [192.168.99.3]:52396->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:34.21      iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.65 1      iso.3.6.1.4.1.9.9.43.1.1.6.1.4.65 2
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.65 3      iso.3.6.1.4.1.9.9.43.1.1.6.1.8.65 "lab"
2026-09-07 18:41:50 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:40.29      iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.85 1      iso.3.6.1.4.1.9.9.43.1.1.6.1.4.85 2
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.85 3      iso.3.6.1.4.1.9.9.43.1.1.6.1.8.85 "lab"
2026-09-07 18:41:50 <UNKNOWN> [UDP: [192.168.99.3]:52396->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:34.28      iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.41.2.0.1
iso.3.6.1.4.1.9.9.41.1.2.3.1.2.27 "AAAA"      iso.3.6.1.4.1.9.9.41.1.2.3.1.3.27 5
iso.3.6.1.4.1.9.9.41.1.2.3.1.4.27 "CLI_DEPRECATED"      iso.3.6.1.4.1.9.9.41.1.2.3.1.5.27 "WARNING: Command
has been added to the configuration using a type 0 password. However, recommended to migrate to strong
type-6 encryption"      iso.3.6.1.4.1.9.9.41.1.2.3.1.6.27 0:1:09:34.28
2026-09-07 18:41:51 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:40.37      iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.41.2.0.1
iso.3.6.1.4.1.9.9.41.1.2.3.1.2.34 "AAAA"      iso.3.6.1.4.1.9.9.41.1.2.3.1.3.34 5
iso.3.6.1.4.1.9.9.41.1.2.3.1.4.34 "CLI_DEPRECATED"      iso.3.6.1.4.1.9.9.41.1.2.3.1.5.34 "WARNING: Command
has been added to the configuration using a type 0 password. However, recommended to migrate to strong
type-6 encryption"      iso.3.6.1.4.1.9.9.41.1.2.3.1.6.34 0:1:09:40.37
2026-09-07 18:41:51 <UNKNOWN> [UDP: [192.168.99.3]:52396->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:34.28      iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.41.2.0.1
iso.3.6.1.4.1.9.9.41.1.2.3.1.2.28 "AAAA"      iso.3.6.1.4.1.9.9.41.1.2.3.1.3.28 5
iso.3.6.1.4.1.9.9.41.1.2.3.1.4.28 "CLI_DEPRECATED"      iso.3.6.1.4.1.9.9.41.1.2.3.1.5.28 "WARNING: Command
has been added to the configuration using a type 0 password. However, recommended to migrate to strong
type-6 encryption"      iso.3.6.1.4.1.9.9.41.1.2.3.1.6.28 0:1:09:34.28
2026-09-07 18:41:51 <UNKNOWN> [UDP: [192.168.99.1]:53747->[192.168.99.10]:162]:
... 8329 more lines

```

Teardown (suite) Close Trap Collector, Close All Connections

16.12 PASS One Raw Listener Receives Every Router (0.1s, 1 commands)

The property the per-router-port workaround existed to dodge: a single socket now serves all three senders.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- output contains **\${R1_OOB_IP}**
- output contains **\${R2_OOB_IP}**
- output contains **\${R3_OOB_IP}**

How it was checked

NMS `sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true`

```
1788803096.140 192.168.99.2 3081f2020103300d02012a020205dc04010302010304373035040c800000090300525400c8020102
010202020202c204066c61626d6f6e040ca2ec33c9c6772141b00df0a604088a5dbc633a502d630481a4a7b547a9cda13405e57b74c4c3
f5b0d542b40991eb028f9746d1b34fdececf4728b8bdf8579bfbb06a99fcdd4a2e7b60cdd4a2cae80bf2664e6a01cbe77167514ad206
51b66bab21b368b295c091ec465ff1f82e3016a03669987ca4c02d220c578e63fefff90d04254ee23c08215f9d0c2323c86a01833dc7a
1d8d4368a84d0968c39552e2d0100cb1f8f8a0e87ebbf8a78b268e67ca27e8d524d78a5de46122216429c1
1788803096.391 192.168.99.2 3081f2020103300d02012c020205dc04010302010304373035040c800000090300525400c8020102
0102020202c204066c61626d6f6e040cb99e87223f51e570ae17f77d04088a5dbc633a502d650481a4a2fd1e95f78522963fae000fc1
942b103469827cc759a47d74971900f4d21fed2e254f818d4ba6a5f6a96cdd76a251f4bdaff8e01a153ef62a6bf01da474edad341a86
7f5a3c677c797ee74a255062e517b8799495ee828b348db481bd1348d64bab05077c93ace035517a7358fbea2377be4b997ddaf07b0f
0e144eca952034340749487a6e4de41f790fd6995cea84a39f6257054fdbf5a59af4659c7697e1d673aa86
1788803096.645 192.168.99.2 3082013c020103300d02012e020205dc04010302010304373035040c800000090300525400c80201
020102020202c204066c61626d6f6e040ce7691a063029064a45f1222b04088a5dbc633a502d670481eed5ee3ab1546abb2c8ef388e3
58318302c2a05d3a164546e52046947cd5dc39ca909d6e6369242ff9a427f52b26cd302b6f8eda36bf90dfabe655b2a415a0485cd092
39b1c7d7bb35214f7dbfbb8f5e9751d8f17290de12ac1c3506f2ff52bc9f852dd7b2d765c4845332e46f8f0e4994d7a9f98e63fb461c
958ff6a1f10e9da7d9e82da525c16c80201c9a9b6c2508c25fad9e0c83aeef2ba78f19e4ec68755545a1a44fea7e0a82634ae428e05
9231d1c93b3b9a938af1933e2b046ca24658d07c9f0ef6d1b0d494e7f70575330d48f23222b3b02cae9551573f91ad2786c5e677a378b0
e3947c42614abee82a37
1788803124.123 192.168.99.1 3081f2020103300d02014a020205dc04010302010304373035040c800000090300525400c8010102
0102020202de04066c61626d6f6e040cc7c17353321d99d084ecf4ac0040830d81086869a6e600481a4e32979013285e1fd25d5247eb0
e6778ef3b68e2d56d6cf1a7d6289fab038960aadbc259c2fe7f4ec3e74a9f2c9f4b5d52e1feb4c33aab463bc14bc283df39ee59b9ef0
9e838f79928eeffb1d42f3eaf793999d80a7d5eea43b646cd9f66cc798abadff3c8361be466bc850129a3fc4baa1e229a5e501430716
6e26151b701b33c1ea6a77d197f93fbfbfcc5f21fe31c6bf89e58fa959a179506befbba3191e6655b4e13
1788803124.373 192.168.99.1 3081f2020103300d02014a020205dc04010302010304373035040c800000090300525400c8010102
0102020202de04066c61626d6f6e040cbdc2bdc67bb32c40835ce6e040830d81086869a6e620481a494bfa2eabf9c0489afeb96a1b4
feea02e4f2482ab16724cc16017d431bd12819bd76b4948f38ef999c2a8a586f21c937f92bb44d6d01373788d73db01fd8c2fb94a3d3
ba5cbc3180231980b150ef8dac82d8f49a7e444b2ce8630f6fc5e436b866b17f64dd9e5d07d584308158363d67521861a56232e4abc2c
a5874a28367c44f44f35614b2923277986aa2912eadc0c82b648e3c81d293143c8ca3db5e034f7f36c8e02
1788803124.623 192.168.99.1 3081db020103300d020146020205dc04010302010304373035040c800000090300525400c8010102
0102020202df04066c61626d6f6e040cc780611fddb402b12c77c9810c040830d81086869a6e6404818d4dfc1ec6d05f298f7e18babfda6
868f3261d34859e18e5b182ec8641cd1bcc4853d641f6963b6426bed6abdc497716e582beb3275c9b0f845c855dd395503640340f900
a90271558551f972f878313d4c25ff92432839eed061ed0a81a9afc326b60a6202fdb128a3293ea063e2ffb06204f217a496e9941e
081a5a86a627ffe80ef228fba447629338bee99b
1788803125.229 192.168.99.1 3081f2020103300d020148020205dc04010302010304373035040c800000090300525400c8010102
0102020202df04066c61626d6f6e040cc0173b62339bc54aaf411b1f38040830d81086869a6e600481a4bb295c762ce8319549b9f0ee00
ccf4c70e083598ff70db387e60d4e647515bd2c3e1ac6ec6d3815337c0f0f047e8e3e13fd81a7b8b977be0b460ead26ccb5e49e2f653
9223de21757b6651658f767205b58ac1cd21651a38d5d5c1af85920ab2a09ebbb8ca96e8674fc82624b970994afbf5fae3186235144a5e
5f69012fac10535d257e0d5328b5e73dde54f8499753c9b8898dceb1001b5b2d40d3afad35c725c8c4b60
1788803125.455 192.168.99.3 3081f2020103300d02012a020205dc04010302010304373035040c800000090300525400c8030102
0102020202df04066c61626d6f6e040cc294a36477b10366227c93705040835b22fd9d14a5f200481a4959e9f626fdacc78f394d82297
18f155ba4ed4d1d5dc9997246261bee2f460da7bebfdb6f81b1228768bd77b73d7a41c1d2c2c802b210c3240e0d1eb98ca08982dd0e
8a09e72d1ec6db7ba811bbf6e3002abbc8ffe1e866fa1e7c0778d1e73b1235631c013e5dec4b2191932a5f0ac89a55d0e466dcd125e
69631a5152b037ddd21f1035a8f58cd496b54d73a03f8630bedb290d60ed7dcd9fcb57754257b418cea6a9
1788803125.478 192.168.99.1 3081db020103300d02014a020205dc04010302010304373035040c800000090300525400c8010102
0102020202df04066c61626d6f6e040ccf149655332b61ccbe054e08040830d81086869a6e6004818d9c16590f44b15372ba5103ba62e
3f3106569d7ee5dd7681d6b70b56a309f52badc89241e1e888027aa91c37b211eaedb71a82eff88e26826da9445fa0d1b3be79e1479a120a
b6827421c1fc1cf0e49306122bccf34415ce45f4a5bd23827e03c9df7c59443648813d7ac9fc184cdafbf1927f9c1e193ed98c7abdea3
07157269de7a28288c8d2b66a904054c44795a2c
1788803125.704 192.168.99.3 3081f2020103300d02012c020205dc04010302010304373035040c800000090300525400c8030102
0102020202df04066c61626d6f6e040ccf2278f01240147c6beb32946040835b22fd9d14a5f200481a48f5b40cd3ea2c34c5d87048953
8e6dd2b0e507669543ee2851b762db3f6b0ce10c1958391674f303a66d9664f077f8980de2f22af5be4997786cf9978cc9b5ab88c76
7e031e3b2e03b4581569a0de282c47d020fcca13d641861c3bd92f561d573db0b88e65e731bd25798ebe8c1fff341952c45ab84621e
4098a04519a6ea69dd8212352c4bb3010d7e57f00b32ad31ed45d63b9a6902e0a637c83ead7a85edcd7
1788803125.728 192.168.99.1 3082013d020103300d02014c020205dc04010302010304373035040c800000090300525400c80101
020102020202df04066c61626d6f6e040cdaea973e5e017c0c6d2166b5040830d81086869a6e6a0481efabec86b13535efce488aae59
84d8d39ec3eeacf487db7b4f6fe6c793316145586fe859331159707b747b5aa6d1356d71ced033b92b9ec9d87f9bd1b7ba11fab5b09
483487400b5760b173cdb7d3442d1d4a07e3f6726937bb3d8fa7ed718c94fc27a82805efc82533cfbec03e8b110e7f6077229e74c4b6
7e8ba4566d9b9114c41a0285f500194f68e493bb8b035cf33f064685c3794896a80e2836ed7d22716dc4496b49fd716863fcb0b68e653
1d5e9b106c4e72f7a33a75546faced52513fea6af352968f1c1f68d0d5ddfc9d69ff6c8abb5e412aa948b1c45a4260939f3b1bc20791b7
11a2f1d277266a0f726133
1788803125.954 192.168.99.3 3081db020103300d02012e020205dc04010302010304373035040c800000090300525400c8030102
0102020202df04066c61626d6f6e040cc6cd43e7963ccb9b374bcb1040835b22fd9d14a5f2404818d63257883cac272f48279fad535
03f507d2bc453d69cfe9fddbc2447ffc911b255e0f47669e510bc563187271cfd1eccd9b4deb6eeffb27e9ce9a8a39e2a79dc17875b062
109c7c372a183431a20593f59f8c8ecfa827d46c4f700d9d85500c26926ff7efd94451eea875ac256cea34b6d23444c97cbaedbfb622
c7661c2ce777f355306e3a0320d24259f1b3852
1788803126.554 192.168.99.3 3081f2020103300d020130020205dc04010302010304373035040c800000090300525400c8030102
0102020202e004066c61626d6f6e040cc47f936393d178e229e4cd1c6040835b22fd9d14a5f260481a404822a478ef80a4680f6c86623
02fc5ae684697ad85b4ba6edf77c7c6c4d3e4768fa589abfc15dcca9d2df87dc16d114ed99e15d27e3ac2c4bfc166cdf8493f26d1c07
715f9bfe7032125e76fdc56ac6c0d751846cf4c9ac72b30008377946f96e5c4e2929163b390db08010b589ae0bfa65e45552bb4c4542
61c34d24ff8c1471063eb73ab6a13d470ba6746c33241bcd083e290d60ced450bd29b684e4304dc90e1
1788803126.805 192.168.99.3 3081db020103300d020132020205dc04010302010304373035040c800000090300525400c8030102
0102020202e104066c61626d6f6e040c15b72a4734b81f8f285b4be4040835b22fd9d14a5f2804818df82cff847696fda65b7fccccf44
2a1386b355e5a66aaa37992d2839eb2d4efa5dc08f2e6e834520d2980cb91f3353fd346adeb614b598654a9fc882dc0de23ed5bd611e
993c52c91753dd7a7200bfcf7fe17ab66113ba02fd4cccbe6faade260b687bab23f8d0ed68abe905b37cd59011a0d440b7a70325c2d
```



```
0c0ea187ed17a61fa340fac549253feb7c0bd3ea
... 5356 more lines
```

Teardown (suite) Close Trap Collector, Close All Connections

16.13 PASS Each Router Identifies Itself By Its Own Engine ID (0.4s, 6 commands)

The engine ID in the received trap must match the one the router reports, and must differ between routers -- that is what lets a collector attribute a trap to a device.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- **`\${local}`** is not empty (checked 3×)
- **`\${ids}`** has exactly **1** entries (checked 3×)
- **`\${ids}[0]`** equals **`\${expected}`** (checked 3×)
- **`\${distinct}`** equals **3**

the three routers did not present three distinct engine IDs: **`\${seen}`**

How it was checked

R1 show snmp engineID | include Local

```
Local SNMP engineID: 800000090300525400C80101
```

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

```
1788803096.140 192.168.99.2 3081f2020103300d02012a020205dc04010302010304373035040c800000090300525400c8020102
0102020202c204066c61626d6f6e040ca2ec33c9c6772141b00df0a604088a5dbc633a502d630481a4a7b547a9cda13405e57b74c4c3
f5b0d542b40991eb028f9746d1b34fdececf4728b8bdf8579bffb06a99fcdd4a2e7b60cdd4a2cae80bf2664e6a01cbe77167514ad206
51b66bab21b368b295c091ec465ff1f82e3016a03669987ca4c02d220c578e63feff90d04254ee23c08215f9d0c2323c86a01833dc7a
1d8d4368a84d0968c39552e2d0100cb1f8f8a0e87ebbf8a78b268e67ca27e8d524d78a5de46122216429c1
1788803096.391 192.168.99.2 3081f2020103300d02012c020205dc04010302010304373035040c800000090300525400c8020102
0102020202c204066c61626d6f6e040cb99e87223f51e570ae17f77d04088a5dbc633a502d650481a4a2fd1e95f78522963fae000fc1
942b103469827cc759a47d74971900f4d21fed2e254f818d4ba6a5f6a96cdd76a251f4bdaff8e01a153ef62a6bf01da474edad341a86
7f5a3c677c797ee74a255062e517b8799495ee828b348db481bd1348d64bab05077c93ace035517a7358fba2377be4b997ddaf07b0f
0e144eca95203430749487a6e4de41f790fd6995cea84a39f6257054fdbf5a59af4659c7697e1d673aa86
1788803096.645 192.168.99.2 3082013c020103300d02012e020205dc04010302010304373035040c800000090300525400c80201
0201020202c204066c61626d6f6e040ce7691a063029064a45f1222b04088a5dbc633a502d670481eed5ee3ab1546abb2c8ef388e3
58318302c2a05d3a164546e52046947cd5dc39ca909d6e6369242ff52bc9f852dd7b2d765c4845332e46f8f0e4994d7a9f98e63fb461c
39b1c7d7bb35214f7dbfbfb8f5e9715d81f7290e12ac1c3506f2ff52bc9f852dd7b2d765c4845332e46f8f0e4994d7a9f98e63fb461c
958ff6a1f10e9da7d9e82da525c16c80201c9a9b6c2508c25fad9e0c83aeef2ba78f19e4ec68755545a1a44fea7e0a82634ae428e05
9231d1c93b3b9a938af1933e2b046ca24658d07c9f0ef6d1b0d494e770575330d48f23222b3b02ae9551573f91ad276fc5e677a3f8b0
e3947c42614abee82a37
1788803124.123 192.168.99.1 3081f2020103300d020142020205dc04010302010304373035040c800000090300525400c8010102
0102020202de04066c61626d6f6e040c7c17353321d99d084ecf4ac0040830d81086869a6e600481a4e32979013285e1fd25d5247eb0
e6778ef3b68e2d56d6cf1a7d6289fab038960aadbc259c2fe7f4ec3e74a9f2c9f4b5d52e1feb4c33aab463bc14bc283df39ee59b9ef0
9e838f79928eeffb1d42f3eaf793999d80a7d5eea43b646cd9fd66cc798abadff3c8361be466bc850129a3fc4baae229a5e501430716
6e26151b701b33c1ea6a77ed197f93fbfbfcc5f21fe31c6bf89e58fa959a179506beffba3191e6655b4e13
1788803124.373 192.168.99.1 3081f2020103300d020144020205dc04010302010304373035040c800000090300525400c8010102
0102020202de04066c61626d6f6e040cbec2bdc67bb32c40835ce6e040830d81086869a6e620481a494bfa2eabf9c0489afeb96a1b4
feea02e4f2482ab16724cc10617d431bd12819bd76b4948f38ef999c2a8a586f21c937f92bb44d6d01373788d73db01fd8c2fb94a3d3
ba5cbc3180231980b150ef8dac82d8f49a7e444b2ce8630f6fce5e436b866b17f64dd9e5d07d584308158363d67521861a56232e4bc2c
a5874a28367c44f44f35614b2923277986aa2912eadc0c82b648e3c81d293143c8ca3db5e034f7f36c8e02
1788803124.623 192.168.99.1 3081db020103300d02014a020205dc04010302010304373035040c800000090300525400c8010102
0102020202de04066c61626d6f6e040c708611fdb402b12c77c9810c040830d81086869a6e6404818d4dfc1ec6d05f298f18babfda6
868f3261d34859e18e5b182ec8641cd1bcc4853d641f6963b6426bed6abdc497716e582beb3275c9b0f845c855dd395503640340f900
a90271558551f972f878313d4c25ff92432839eedd061ed0a81a9afc326b60a6202fdb128a32939ca063e2ffb06204f217a496e9941e
081a5a86a627ffe80ef228fba447629338bee99b
1788803125.229 192.168.99.1 3081f2020103300d020148020205dc04010302010304373035040c800000090300525400c8010102
0102020202df04066c61626d6f6e040c0173b62339bc54a4f11b1f38040830d81086869a6e660481a4bb295c762ce8319549b9f0ee00
ccf4c70e083598fd70bd387e0d4e647515bd2c3e1ac6ec6d3815337c0f0f047e8e3e13fd81a7b8b977be0b460ead26ccb5e49e2ff653
9223de21757b6651658f767205b58ac1cd21651a38d5d5c1af85920ab2a09ebb8ca96e8674fc82624b970994afb5fae318623514a45e
5f69012fac10535d257e0d5328b5e73dd5e54f8499753c9b8898dceb1001b5b2d40d3afad35c725c8c4b60
1788803125.455 192.168.99.3 3081f2020103300d02012a020205dc04010302010304373035040c800000090300525400c8030102
0102020202df04066c61626d6f6e040c294a36477b10366227c93705040835b22fd9d14a5f200481a4959e9f626fdacc78f394d82297
18f155ba4ed4d1d5dc9997246261bee2f460da7bebfdbf6f81b1228768bd77b73d7a41c1d2c2c802b210c3240e0d1eb98ca808982dd0e
8a09e72d1ec6db7ba811bbfe3002abbc8ffe1e866fa1e7c0778d1e73b1235631c013e5dec4b2191932a5f0ec89a55d0e6466dc1f25e
69631a5152b037ddd21f1035a8f58cd496b54d73a03f8630bedb290d60ed7dcd9fcb57754257b418cea6a9
1788803125.478 192.168.99.1 3081db020103300d02014a020205dc04010302010304373035040c800000090300525400c8010102
0102020202df04066c61626d6f6e040cfc149655332b61ccbe054e08040830d81086869a6e6804818d9c16590f4b15372ba5103ba62e
3f3106569d7ee5dd7681d6b70b56a309f52badc89241e1e888027aa91c37b211eaddb71a82eff88e26826da9445fa0d1b1be79e1420a
b6827421c1f1cf0e49306122bccf34415ce45f4a5bd23827e03c9dfa7c59443648813d7ac9fc184cdfab1927f9c1e193ed98c7abdea3
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```
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... 5356 more lines
```

R2 show snmp engineID | include Local

```
Local snmp engineID: 800000090300525400c80201
```

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

```
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... 5356 more lines
```

R3 show snmp engineID | include Local

```
Local SNMP engineID: 800000090300525400C80301
```

NMS sudo cat /var/log/lab/traps-raw.log 2-/dev/null || true

```
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1788803096.391 192.168.99.2 3081f2020103300d02012c020205dc04010302010304373035040c800000090300525400c8020102
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1788803124.623 192.168.99.1 3081db020103300d020146020205dc04010302010304373035040c800000090300525400c8010102
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1788803125.455 192.168.99.3 3081f2020103300d02012a020205dc04010302010304373035040c800000090300525400c8030102
0102020202df04066c61626d6f6e040c294a36477b10366227c93705040835b22fd9d14a5f220481a4959e9f626fdacc78f394d82297
18f155ba4ed4d1d5dc9997246261bee2f460da7bebfdb6f81b1228768bd77b73d7a41c1d2c2c802b210c3240e0d1eb98ca808982dd0e
8a09e72d1ec6db7ba811bbf3002abbc8fffe1e866fa1e7c0778d1e793b1235631c013e5dec4b2191932a5f0ec89a55d0e6466dc1f25e
69631a5152b037dd21f1035a8f58cd496b54d73a03f8630ebd39d060ed7dcd9fcb57754257b418cea6a9
1788803125.478 192.168.99.1 3081db020103300d02014a020205dc04010302010304373035040c800000090300525400c8010102
0102020202df04066c61626d6f6e040cfc149655332b61cc8e054e08040830d81086869a6e6004818d9c16590f4b15372ba5103ba62e
3f3106569d7ee5dd7681db4f6fe30bc793316145586fe859331159707b47b5aa6d1356d71ced033b92b9ec9d87f9bd1b7ba11fab5b09
b6827421c1f1cf0e49306122bccf34415ce45f4a5bd23827e03c9dfa7c59443648813d7ac9fc184cdfab1927f9c1e193ed98c7abdea3
07157269de7a28288c8d2b66a904054c44795a2c
1788803125.704 192.168.99.3 3081f2020103300d02012c020205dc04010302010304373035040c800000090300525400c8030102
0102020202df04066c61626d6f6e040c2278f01240147c6beb32946040835b22fd9d14a5f220481a48f5b40cd3ea2c34c5d87048953
8e6dd2b0e507669543ee2851b762db3f6b0ce10c1958391674f3034e66d9664f077f8980de2f22af5be4997786cf9978cc9b5ab88c76
7e031e3b2e03b4581569a0de282c47d020fcca13d641861c3bd92f561d573db088e65e731bd25798ebe8c1fff341952c45ab84621e
4098a04519a6ea99dd8212352c4dbb3010d7e57f00b32ad31ed473b639a6902e0a637c83ead7a85edcd7
1788803125.728 192.168.99.1 3082013d020103300d02014c020205dc04010302010304373035040c800000090300525400c80101
0201020202df04066c61626d6f6e040cdaea973e5e017c0c6d2166b5040830d81086869a6e6a0481efabec86b13535efce488aae59
84d8d39ec3eeacff487d7b4f6fe30bc793316145586fe859331159707b47b5aa6d1356d71ced033b92b9ec9d87f9bd1b7ba11fab5b09
483487400b5760b173cdb7d3442d1d4a07e3f6726937bb3d8fa7ed718c94fc27a82805efc82533cfbec03e8b110e7f6077229e74c4b6
```

7e8ba4566d9b9114c41a0285b500194f68e493bb8b035cf33f064685c3794896a80e2836ed7d22716dc4496b49fd716863fc0b68e653
1d5e9b106c4e72f7a33a75546face52513fea6af352968fc1f68d0d5ddfc9d69ff6c8abb5e412aa948b1c45a4260939f3b1bc20791b7
11a2f1d277266a0f726133
1788803125.954 192.168.99.3 3081db020103300d02012e020205dc04010302010304373035040c800000090300525400c8030102
0102020202df04066c61626d6f6e040c4c6cd43e7963ccb9b374bcb1040835b22fd9d14a5f2404818d63257883cac272f48279fad535
03f507d2bc453d69cefd9b2447ffc911b255e50f47669e510bc563187271cfd1eccd9b4deb6eefb267ece9a8a39e2a79dc17875b062
109c7c372a183431a20593f59f8c8ecfa827d46c4f700d9d85500c26926ff7efd94451eea875ac256cea34b6d23444c97cbaedbf622
c7661c2ce7777f355306e3a0320d24259f1b3852
1788803126.554 192.168.99.3 3081f2020103300d020130020205dc04010302010304373035040c800000090300525400c8030102
0102020202e004066c61626d6f6e040c47f9363931d78e229e4cd1c6040835b22fd9d14a5f260481a404822a478ef80a4680f6c86623
02fc5ae684697ad85b4ba6edf77c7c6c4d3e4768fa589abfc15dcca9d2df87dc16d114ed99e15d27e3ac42cbfc166cdf8493f26d1c07
715f9bfe7032125e76fdc56ac6c0d751846cf4c9ac72b30008377946f96e5c4e2929163b390db08010b589ae0bfa65e45552bb4c4542
61c34d24ff8c1471063eb73ab6a13d47b0a6746c33241bcd083e2d612306cead450bd29b684e4304dc90e1
1788803126.805 192.168.99.3 3081db020103300d020132020205dc04010302010304373035040c800000090300525400c8030102
0102020202e104066c61626d6f6e040c15b72a4734b81f8f285b4be4040835b22fd9d14a5f2804818df82cfff847696fda65bf7ccc44
2a1386b355e5a66aaa37992d2839eb2d4efa5dc08f2e6e834520d2980cb91f3353fd346adeb614b598654a9fc882dc0de23ed5bd611e
993c52c91753dd7a7200bfcf7fe17ab66113ba02fd4cccbef6faade260b687bab23f8d0ed68abe905b37dc59011a0d440b7a70325c2d
0c0ea187ed17a61fa340fac549253feb7c0bd3ea
... 5356 more lines

Teardown (suite) Close Trap Collector, Close All Connections

17 Snmp Polling

17.01 **PASS** The NMS Runs Real Net-SNMP Tooling (0.1s, 2 commands)

The point of this VM: unlike the cirros hosts it has a genuine SNMP client, so the polling below is the real thing rather than something hand rolled for the test.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- output contains **NET-SNMP version**
- output contains **snmpget**
- output contains **snmpwalk**

How it was checked

NMS `snmpget --version 2>&1 | head -2; snmpwalk --version 2>&1 | head -1`

```
NET-SNMP version: 5.9.4.pre2
NET-SNMP version: 5.9.4.pre2
```

NMS `command -v snmpget snmpwalk snmptrapd`

```
/usr/bin/snmpget
/usr/bin/snmpwalk
/usr/sbin/snmptrapd
```

Teardown (suite) Close Nms, Close All Connections

17.02 **PASS** Every Router Answers An SNMPv3 authPriv Poll (0.2s, 3 commands)

The agent must be reachable and must accept the credentials the provisioner configured.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.1.5.0`

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
```

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0`

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3 1.3.6.1.2.1.1.5.0`

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
```

Teardown (suite) Close Nms, Close All Connections

17.03 **PASS** Polled sysName Matches The Router's Own Hostname (0.4s, 6 commands)

Cross-check: the value SNMP returns is compared against what the device says on its CLI, so a plausible-looking answer from the wrong box cannot pass.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- **`\${polled}`** starts with **`\${expected}`[0]** (checked 3×)
- `msg=SNMP sysName `${polled}` does not match CLI hostname `${expected}`[0]`

How it was checked

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.1.5.0`

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
```

R1 `show running-config | include ^hostname`

```
hostname c8000v-r1
```

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0`

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

```
R2 show running-config | include ^hostname
hostname c8000v-r2
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
```

```
R3 show running-config | include ^hostname
hostname c8000v-r3
```

Teardown (suite) Close Nms, Close All Connections

17.04 PASS Polled sysDescr Identifies The Platform (0.2s, 6 commands)

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- output matches Cisco IOS[-]XE Software|Cisco IOS Software \[IOSXE\] (checked 3×)

How it was checked

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.1.0
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
```

R1 show version | include Version

```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
licensed under the GNU General Public License ("GPL") Version 2.0. The
software code licensed under GPL Version 2.0 is free software that comes
GPL code under the terms of GPL Version 2.0. For more details, see the
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.1.0
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
```

R2 show version | include Version

```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
licensed under the GNU General Public License ("GPL") Version 2.0. The
software code licensed under GPL Version 2.0 is free software that comes
GPL code under the terms of GPL Version 2.0. For more details, see the
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.1.0
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
```

R3 show version | include Version

```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
licensed under the GNU General Public License ("GPL") Version 2.0. The
software code licensed under GPL Version 2.0 is free software that comes
GPL code under the terms of GPL Version 2.0. For more details, see the
```

Teardown (suite) Close Nms, Close All Connections

17.05 PASS sysUpTime Advances Between Two Polls (5.0s, 2 commands)

Proves the answers come from a live agent rather than a cached or replayed response.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- \${second} > \${first} holds

How it was checked

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.3.0
```

```
.1.3.6.1.2.1.1.3.0 = Timeticks: (932870) 2:35:28.70
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.3.0
```

```
.1.3.6.1.2.1.1.3.0 = Timeticks: (933372) 2:35:33.72
```

Teardown (suite) Close Nms, Close All Connections

17.06 PASS Walking The Interface Table Returns Every Interface The CLI Shows (0.2s, 6 commands)

A walk, not a get: the agent must return the whole subtree. Every physical interface named by the CLI must appear in it.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- `${walked}` contains `${intf}` (checked 13×)
msg=\${r}: ifDescr walk is missing `${intf}`

How it was checked

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "GigabitEthernet5"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.10 = STRING: "Tunnel1"
.1.3.6.1.2.1.2.2.1.2.11 = STRING: "Loopback1"
```

R1 show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1	10.0.2.15	YES	DHCP	up	up
GigabitEthernet2	10.10.10.1	YES	TFTP	up	up
GigabitEthernet3	192.168.10.1	YES	TFTP	up	up
GigabitEthernet4	10.10.20.1	YES	TFTP	up	up
GigabitEthernet5	192.168.99.1	YES	TFTP	up	up
Loopback0	172.16.1.1	YES	TFTP	up	up
Loopback1	10.20.1.1	YES	manual	up	up
Tunnel0	192.168.100.1	YES	manual	up	up
Tunnel1	192.168.101.1	YES	manual	up	up

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
```

R2 show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1	10.0.2.15	YES	DHCP	up	up
GigabitEthernet2	10.10.10.2	YES	TFTP	up	up
GigabitEthernet3	192.168.20.1	YES	TFTP	up	up
GigabitEthernet4	192.168.99.2	YES	TFTP	up	up
Loopback0	172.16.2.1	YES	TFTP	up	up
Loopback1	10.20.2.1	YES	manual	up	up
Tunnel0	192.168.100.2	YES	manual	up	up

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
```

```
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
```

R3 show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1	10.0.2.15	YES	DHCP	up	up
GigabitEthernet2	10.10.20.2	YES	TFTP	up	up
GigabitEthernet3	192.168.30.1	YES	TFTP	up	up
GigabitEthernet4	192.168.99.3	YES	TFTP	up	up
Loopback0	172.16.3.1	YES	TFTP	up	up
Loopback1	10.20.3.1	YES	manual	up	up
Tunnel0	192.168.101.2	YES	manual	up	up

Teardown (suite) Close Nms, Close All Connections

17.07 PASS Polled Interface Count Agrees With The Interface Table (0.3s, 6 commands)

ifNumber is a scalar the agent maintains separately from the table, so the two disagreeing would mean a broken agent.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- **\${number}** equals **\${rows}** (checked 3×)
msg=\${r}: ifNumber says \${number} but the table has \${rows} rows

How it was checked

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.2.1.0

```
.1.3.6.1.2.1.2.1.0 = INTEGER: 11
```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.2.2.1.2

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "GigabitEthernet5"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.10 = STRING: "Tunnel1"
.1.3.6.1.2.1.2.2.1.2.11 = STRING: "Loopback1"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.2.1.0

```
.1.3.6.1.2.1.2.1.0 = INTEGER: 9
```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.2.2.1.2

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3 1.3.6.1.2.1.2.1.0

```
.1.3.6.1.2.1.2.1.0 = INTEGER: 9
```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3 1.3.6.1.2.1.2.2.1.2

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
```

Teardown (suite) Close Nms, Close All Connections

17.08 PASS The LAN Interface Is Reported Up By Both SNMP And The CLI (0.4s, 9 commands)

ifOperStatus 1 is up(1). Checked against the CLI line protocol state for the same interface.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- `${index}` differs from **-1** (checked 3×)
msg=\${r}: no Gi3 in the walk
- output contains **1** (checked 3×)
msg=\${r}: SNMP says Gi3 is not up
- output contains **line protocol is up** (checked 3×)

How it was checked

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1  
1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"  
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"  
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"  
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"  
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "GigabitEthernet5"  
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "VoIP-Null0"  
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Null0"  
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback0"  
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"  
.1.3.6.1.2.1.2.2.1.2.10 = STRING: "Tunnel1"  
.1.3.6.1.2.1.2.2.1.2.11 = STRING: "Loopback1"
```

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1  
1.3.6.1.2.1.2.2.1.8
```

```
.1.3.6.1.2.1.2.2.1.8.1 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.2 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.3 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.4 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.5 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.6 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.7 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.8 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.9 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.10 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.11 = INTEGER: 1
```

```
R1 show interfaces GigabitEthernet3 | include line protocol
```

```
GigabitEthernet3 is up, line protocol is up
```

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2  
1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"  
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"  
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"  
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"  
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"  
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"  
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"  
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"  
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
```

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2  
1.3.6.1.2.1.2.2.1.8
```

```
.1.3.6.1.2.1.2.2.1.8.1 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.2 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.3 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.4 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.5 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.6 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.7 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.8 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.9 = INTEGER: 1
```

```
R2 show interfaces GigabitEthernet3 | include line protocol
```

```
GigabitEthernet3 is up, line protocol is up
```

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3  
1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"  
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"  
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"  
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"  
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"  
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
```

```
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.2.2.1.8
.1.3.6.1.2.1.2.2.1.8.1 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.2 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.3 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.4 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.5 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.6 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.7 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.8 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.9 = INTEGER: 1

R3 show interfaces GigabitEthernet3 | include line protocol
GigabitEthernet3 is up, line protocol is up

Teardown (suite) Close Nms, Close All Connections
```

17.09 PASS Spokes Are Polled Across The Encrypted Tunnel (6.2s, 18 commands)

The NMS shares a LAN with the hub only. Traffic to R2 and R3 crosses the IPsec tunnels, so a burst of polling must show up on the ESP counters at both ends.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- **#{a_enc} >= #{expected}** holds (checked 2×)
- **#{a_dec} >= #{expected}** holds (checked 2×)
- **#{b_enc} >= #{expected}** holds (checked 2×)
- **#{b_dec} >= #{expected}** holds (checked 2×)

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 99152, #pkts encrypt: 99152, #pkts digest: 99152
  #pkts decaps: 86839, #pkts decrypt: 86839, #pkts verify: 86839
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 81 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 88344, #pkts encrypt: 88344, #pkts digest: 88344
  #pkts decaps: 100434, #pkts decrypt: 100434, #pkts verify: 100434
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 33 more lines
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
```

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
```

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
```

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 99157, #pkts encrypt: 99157, #pkts digest: 99157
  #pkts decaps: 86844, #pkts decrypt: 86844, #pkts verify: 86844
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 81 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 88350, #pkts encrypt: 88350, #pkts digest: 88350
  #pkts decaps: 100439, #pkts decrypt: 100439, #pkts verify: 100439
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 33 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 99188, #pkts encrypt: 99188, #pkts digest: 99188
  #pkts decaps: 86876, #pkts decrypt: 86876, #pkts verify: 86876
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
```



```

    #send errors 0, #recv errors 0
    ... 81 more lines
R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 88381, #pkts encrypt: 88381, #pkts digest: 88381
  #pkts decaps: 100470, #pkts decrypt: 100470, #pkts verify: 100470
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
  ... 33 more lines

```

Teardown (suite) Close Nms, Close All Connections

17.10 **PASS** A Wrong Authentication Password Is Refused (0.1s, 2 commands)

Negative control. The agent answers the correct credentials first, so a refusal here cannot be a dead router.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A NotTheRightPassword -x AES -X LabPrivPass123 -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0
snmpget: Authentication failure (incorrect password, community or key)

```

Teardown (suite) Close Nms, Close All Connections

17.11 **PASS** A Wrong Privacy Password Is Refused (2.2s, 3 commands)

Authentication succeeds but the payload will not decrypt, so the router drops the request silently and the client times out. Bracketed by a good poll to prove it is still alive.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X NotTheRightPassword -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0
Timeout: No Response from 192.168.99.1.
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"

```

Teardown (suite) Close Nms, Close All Connections

17.12 **PASS** Polling Without Privacy Is Refused By Every Router (0.3s, 6 commands)

The group is configured "v3 priv". Correct credentials at a lower security level must still be denied, otherwise the privacy requirement is decorative.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

NMS `snmpget -On -v3 -l noAuthNoPriv -u labmon -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0`

Error in packet
Reason: authorizationError (access denied to that object)

NMS `snmpget -On -v3 -l authNoPriv -u labmon -a SHA -A LabAuthPass123 -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0`

Error in packet
Reason: authorizationError (access denied to that object)

NMS `snmpget -On -v3 -l noAuthNoPriv -u labmon -t 2 -r 0 192.168.99.2 1.3.6.1.2.1.1.5.0`

Error in packet
Reason: authorizationError (access denied to that object)

NMS `snmpget -On -v3 -l authNoPriv -u labmon -a SHA -A LabAuthPass123 -t 2 -r 0 192.168.99.2 1.3.6.1.2.1.1.5.0`

Error in packet
Reason: authorizationError (access denied to that object)

NMS `snmpget -On -v3 -l noAuthNoPriv -u labmon -t 2 -r 0 192.168.99.3 1.3.6.1.2.1.1.5.0`

Error in packet
Reason: authorizationError (access denied to that object)

NMS `snmpget -On -v3 -l authNoPriv -u labmon -a SHA -A LabAuthPass123 -t 2 -r 0 192.168.99.3 1.3.6.1.2.1.1.5.0`

Error in packet
Reason: authorizationError (access denied to that object)

Teardown (suite) Close Nms, Close All Connections

17.13 PASS An Unknown User Is Refused (0.1s, 1 commands)

Only the provisioned user exists in the USM table.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

NMS `snmpget -On -v3 -l authPriv -u intruder -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0`

snmpget: Unknown user name

Teardown (suite) Close Nms, Close All Connections

17.14 PASS SNMPv2c Gets No Answer At The Wire (6.3s, 6 commands)

Suite 16 asserts no community is configured. This asserts the consequence on the network: a v2c poll gets nothing back, while a v3 poll to the same address succeeds.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- output contains **Timeout** (checked 3×)

How it was checked

NMS `snmpget -On -v2c -c public -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0`

Timeout: No Response from 192.168.99.1.

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.1.5.0`

.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"

NMS `snmpget -On -v2c -c public -t 2 -r 0 192.168.99.2 1.3.6.1.2.1.1.5.0`

Timeout: No Response from 192.168.99.2.

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0`

.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

NMS `snmpget -On -v2c -c public -t 2 -r 0 192.168.99.3 1.3.6.1.2.1.1.5.0`

Timeout: No Response from 192.168.99.3.

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3 1.3.6.1.2.1.1.5.0`

.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"

Teardown (suite) Close Nms, Close All Connections

18 Mtu

18.01 PASS Every Tunnel Reports The Expected Transport MTU (0.4s, 11 commands)

The tunnel MTU must be below the physical one by the ESP overhead. Equal would mean encryption costs nothing, which would mean it is not happening.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- `${tunnels}` is not empty (checked 3×)
- output contains `${TUNNEL_MTU} bytes` (checked 4×)
- output contains `MTU is ${TUNNEL_MTU} bytes` (checked 4×)
- `${TUNNEL_MTU} < ${PHYSICAL_MTU}` holds

How it was checked

```
R1 show ip interface brief | include ^Tunnel
Tunnel0      192.168.100.1  YES manual up
Tunnel1      192.168.101.1  YES manual up

R1 show interfaces Tunnel0 | include Tunnel transport MTU
Tunnel transport MTU 1438 bytes

R1 show ip interface Tunnel0 | include MTU
MTU is 1438 bytes

R1 show interfaces Tunnel1 | include Tunnel transport MTU
Tunnel transport MTU 1438 bytes

R1 show ip interface Tunnel1 | include MTU
MTU is 1438 bytes

R2 show ip interface brief | include ^Tunnel
Tunnel0      192.168.100.2  YES manual up

R2 show interfaces Tunnel0 | include Tunnel transport MTU
Tunnel transport MTU 1438 bytes

R2 show ip interface Tunnel0 | include MTU
MTU is 1438 bytes

R3 show ip interface brief | include ^Tunnel
Tunnel0      192.168.101.2  YES manual up

R3 show interfaces Tunnel0 | include Tunnel transport MTU
Tunnel transport MTU 1438 bytes

R3 show ip interface Tunnel0 | include MTU
MTU is 1438 bytes
```

Teardown (suite) Close All Connections

18.02 PASS The Overhead Is What ESP-AES-256 With SHA256 Costs (0.0s, 0 commands)

Records the number rather than assuming it: 62 bytes of ESP header, IV, padding and ICV. If the transform set changes, this is the test that should fail first.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- `${overhead}` equals 62

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Close All Connections

18.03 PASS A Packet Exactly At The Tunnel MTU Crosses With DF Set (0.1s, 1 commands)

The permissive half of the boundary. Sourced from Loopback1 so the traffic is inside the protected selector.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains **Success rate is 100 percent**
a packet of exactly the tunnel MTU (`${TUNNEL_MTU}`) did not cross

How it was checked

```
R2 ping 10.20.3.1 source Loopback1 size 1438 df-bit repeat 5
Type escape sequence to abort.
Sending 5, 1438-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
Packet sent with the DF bit set
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
```

Teardown (suite) Close All Connections

18.04 **PASS** One Byte Over The Tunnel MTU Does Not Cross With DF Set (10.0s, 1 commands)

The restrictive half. Without this the test above would pass just as well against a tunnel with no MTU limit at all.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains **Success rate is 0 percent**
a packet one byte over the tunnel MTU crossed with DF set

How it was checked

```
R2 ping 10.20.3.1 source Loopback1 size 1439 df-bit repeat 5
Type escape sequence to abort.
Sending 5, 1439-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
Packet sent with the DF bit set
.....
Success rate is 0 percent (0/5)
```

Teardown (suite) Close All Connections

18.05 **PASS** Oversize Traffic Still Crosses When Fragmentation Is Allowed (0.8s, 2 commands)

Distinguishes an MTU limit from a blackhole: the same sizes that fail with DF set must succeed without it.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains **Success rate is 100 percent** (checked 2×)
a \${size}-byte packet failed even with fragmentation allowed

How it was checked

```
R2 ping 10.20.3.1 source Loopback1 size 1500 repeat 5
Type escape sequence to abort.
Sending 5, 1500-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/69/83 ms

R2 ping 10.20.3.1 source Loopback1 size 2000 repeat 5
Type escape sequence to abort.
Sending 5, 2000-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 61/79/85 ms
```

Teardown (suite) Close All Connections

18.06 **PASS** Hosts Pass Sub-MTU Traffic With DF Set (4.0s, 1 commands)

End to end across two encrypt/decrypt hops, from machines that know nothing about the tunnel.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains **0% packet loss**
a host could not pass \${HOST_PAYLOAD} bytes with DF set

How it was checked

H2 `ping -c 5 -M do -s 1410 192.168.30.10`

```
PING 192.168.30.10 (192.168.30.10) 1410(1438) bytes of data.  
1418 bytes from 192.168.30.10: icmp_seq=1 ttl=61 time=1.67 ms  
1418 bytes from 192.168.30.10: icmp_seq=2 ttl=61 time=1.70 ms  
1418 bytes from 192.168.30.10: icmp_seq=3 ttl=61 time=1.69 ms  
1418 bytes from 192.168.30.10: icmp_seq=4 ttl=61 time=1.64 ms  
1418 bytes from 192.168.30.10: icmp_seq=5 ttl=61 time=1.75 ms  
  
--- 192.168.30.10 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4007ms  
rtt min/avg/max/mdev = 1.636/1.687/1.747/0.036 ms
```

Teardown (suite) Close All Connections

18.07 **PASS** An End Host Learns The Tunnel Path MTU (2.0s, 1 commands)

The test that matters most in practice. The host is two hops and an encryption boundary away from the constraint, yet it must end up knowing the real limit -- otherwise a client retransmits into a black hole instead of backing off. The host reporting the exact tunnel MTU is proof that path MTU discovery survived the tunnel.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains `mtu=${TUNNEL_MTU}`
the host did not learn the `${TUNNEL_MTU}`-byte path MTU: `${out}`

How it was checked

H2 `ping -c 2 -M do -s 1472 192.168.30.10`

```
PING 192.168.30.10 (192.168.30.10) 1472(1500) bytes of data.  
ping: local error: message too long, mtu=1438  
ping: local error: message too long, mtu=1438  
  
--- 192.168.30.10 ping statistics ---  
2 packets transmitted, 0 received, 100% packet loss, time 1019ms
```

Teardown (suite) Close All Connections

18.08 **PASS** Hosts Pass Oversize Traffic Without DF Set (4.1s, 1 commands)

The counterpart: the same payload that was refused with DF set gets through when it may be fragmented.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains **0% packet loss**
oversize host traffic failed even with fragmentation allowed

How it was checked

H2 `ping -c 5 -s 1472 192.168.30.10`

```
PING 192.168.30.10 (192.168.30.10) 1472(1500) bytes of data.  
1480 bytes from 192.168.30.10: icmp_seq=1 ttl=61 time=43.0 ms  
1480 bytes from 192.168.30.10: icmp_seq=2 ttl=61 time=42.9 ms  
1480 bytes from 192.168.30.10: icmp_seq=3 ttl=61 time=77.1 ms  
1480 bytes from 192.168.30.10: icmp_seq=4 ttl=61 time=42.2 ms  
1480 bytes from 192.168.30.10: icmp_seq=5 ttl=61 time=42.8 ms  
  
--- 192.168.30.10 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4004ms  
rtt min/avg/max/mdev = 42.241/49.607/77.059/13.728 ms
```

Teardown (suite) Close All Connections

19 Oob

19.01 PASS Every Router Has A Management VRF (0.0s, 3 commands)

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **MGMT_VRF** (checked 3×)
- output contains **ipv4** (checked 3×)

How it was checked

R1 show vrf

Name	Default RD	Protocols	Interfaces
MGMT	<not set>	ipv4	Gi5

R2 show vrf

Name	Default RD	Protocols	Interfaces
MGMT	<not set>	ipv4	Gi4

R3 show vrf

Name	Default RD	Protocols	Interfaces
MGMT	<not set>	ipv4	Gi4

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.02 PASS Only The OOB Interface Is In The Management VRF (0.1s, 3 commands)

A VRF containing a data interface would defeat the point, so this asserts membership is exactly one interface.

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **short** (checked 3×)
- output does not contain **data** (checked 9×)

How it was checked

R1 show vrf MGMT

Name	Default RD	Protocols	Interfaces
MGMT	<not set>	ipv4	Gi5

R2 show vrf MGMT

Name	Default RD	Protocols	Interfaces
MGMT	<not set>	ipv4	Gi4

R3 show vrf MGMT

Name	Default RD	Protocols	Interfaces
MGMT	<not set>	ipv4	Gi4

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.03 PASS Every Router Holds Its OOB Address On The Flat Network (0.1s, 3 commands)

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **OOB_IP** (checked 3×)
- output contains **up** (checked 3×)

How it was checked

R1 show ip interface brief | include GigabitEthernet5

GigabitEthernet5	192.168.99.1	YES	TFTP	up	up
------------------	--------------	-----	------	----	----

R2 show ip interface brief | include GigabitEthernet4

GigabitEthernet4	192.168.99.2	YES	TFTP	up	up
------------------	--------------	-----	------	----	----

R3 show ip interface brief | include GigabitEthernet4

GigabitEthernet4	192.168.99.3	YES	TFTP	up	up
------------------	--------------	-----	------	----	----

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.04 PASS The Management VRF Routing Table Holds Only The Flat Network (0.1s, 3

commands)

The isolation, stated as routing: the VRF must know how to reach the management network and nothing else. A default route or a data prefix here would be a leak.

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **#{OOB_NET}** (checked 3×)
- output does not contain **#{data}** (checked 12×)

How it was checked

R1 show ip route vrf MGMT

Routing Table: MGMT

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected

... 5 more lines

R2 show ip route vrf MGMT

Routing Table: MGMT

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected

... 5 more lines

R3 show ip route vrf MGMT

Routing Table: MGMT

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected

... 5 more lines

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.05 PASS The Global Table Cannot Reach The Management Network (0.1s, 3 commands)

The other direction. Traffic in the global table must have no path to the OOB network, otherwise the VRF is decorative.

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **not in table** (checked 3×)

How it was checked

```
R1 show ip route 192.168.99.0
% Network not in table

R2 show ip route 192.168.99.0
% Network not in table

R3 show ip route 192.168.99.0
% Network not in table
```

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.06 PASS Every Router Answers SNMP On Its OOB Address (0.3s, 6 commands)

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- `${name}` starts with `${expected}[0]` (checked 3×)
- polling `${OOB_IP}${r}` answered as `'${name}'`, not `${r}`

How it was checked

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"

R1 show running-config | include ^hostname
hostname c8000v-r1

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

R2 show running-config | include ^hostname
hostname c8000v-r2

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"

R3 show running-config | include ^hostname
hostname c8000v-r3
```

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.07 PASS Management Traffic Never Touches The Data Path (3.3s, 28 commands)

The negative property that makes this out-of-band. Polling a spoke used to cross an IPsec tunnel and was asserted to advance the ESP counters; now it must do the opposite. The comparison is against an idle window of the same length, because BGP and BFD tick those counters continuously regardless -- so an absolute delta proves nothing.

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- `${attributable} < ${POLL_BURST}` holds

How it was checked

```
R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  current_peer 10.10.10.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 99442, #pkts encrypt: 99442, #pkts digest: 99442
    #pkts decaps: 87127, #pkts decrypt: 87127, #pkts verify: 87127
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
  ... 81 more lines

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"  
  
NMS snmpset -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"  
  
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"  
  
NMS snmpset -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"  
  
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"  
  
NMS snmpset -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"  
  
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"  
  
NMS snmpset -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"  
  
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"  
  
NMS snmpset -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"  
  
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"  
  
NMS snmpset -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```



```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
```

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 99453, #pkts encrypt: 99453, #pkts digest: 99453
  #pkts decaps: 87138, #pkts decrypt: 87138, #pkts verify: 87138
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 81 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 99453, #pkts encrypt: 99453, #pkts digest: 99453
  #pkts decaps: 87138, #pkts decrypt: 87138, #pkts verify: 87138
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 81 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.10.10.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 10.10.10.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 99459, #pkts encrypt: 99459, #pkts digest: 99459
  #pkts decaps: 87144, #pkts decrypt: 87144, #pkts verify: 87144
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 81 more lines
```

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.08 PASS Syslog Reaches The Collector From The OOB Address (12.2s, 4 commands)

rsyslog files are named for the sender. A file named after a LAN address would mean the router is logging from the data path despite the configuration. Each router is made to emit a marked message rather than waiting on ambient traffic: whether a router happens to have logged recently is not what this test is about, and depending on it makes the result a matter of timing.

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains `${OOB_IP}${r}` (checked 3×)
- `${n} > 0` holds (checked 3×)
- output does not contain `${r}_LAN_IP.log` (checked 3×)

How it was checked

```
NMS sudo grep -c -- 'LAB00B-SRC-R1-160830' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
1
```

```
NMS sudo grep -c -- 'LAB00B-SRC-R2-160830' /var/log/lab/192.168.99.2.log 2>/dev/null || echo 0
1
```

```
NMS sudo grep -c -- 'LAB00B-SRC-R3-160830' /var/log/lab/192.168.99.3.log 2>/dev/null || echo 0
```

1

NMS `sudo ls -l /var/log/lab`

```
total 5020
-rw-r--r-- 1 syslog syslog 260466 Sep  7 20:08 192.168.99.1.log
-rw-r--r-- 1 syslog syslog 311672 Sep  7 20:08 192.168.99.2.log
-rw-r--r-- 1 syslog syslog 182534 Sep  7 20:08 192.168.99.3.log
-rw-r--r-- 1 root  root  1410997 Sep  7 20:08 snmptrapd.log
-rw-r--r-- 1 root  root   2943164 Sep  7 20:08 traps-raw.log
```

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.09 PASS A Spoke Stays Manageable With Its Tunnel Down (14.4s, 15 commands)

The reason out-of-band management exists. Shutting R2's tunnel removes every data path between the hub and the spoke -- BGP drops, the LAN becomes unreachable -- and the spoke must still answer SNMP and keep logging, because none of that ever used the tunnel.

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **not in table**
the hub can still reach R2's LAN, so the data path is not down yet
- **`\${before}` equals `\${after}`**
R2 stopped answering SNMP once its tunnel went down, so management is not out-of-band
- **`\${n}` > 0** holds
- output matches **`\${peer}`ls+4ls+`\${peer\_asn}`(ls+ld+){5}ls+S+ls+ld+** (checked 3×)

How it was checked

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0`

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Tunnel0

```
${output} =
```

R2 shutdown

```
${output} =
```

R2 end

```
${output} =
```

R1 show ip route 192.168.20.0

```
% Network not in table
```

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0`

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS `sudo grep -c -- 'LAB00B-160843' /var/log/lab/192.168.99.2.log 2>/dev/null || echo 0`

```
1
```

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Tunnel0

```
${output} =
```

R2 no shutdown

```
${output} =
```

R2 end

```
${output} =
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 13, main routing table version 13
6 network entries using 1488 bytes of memory
6 path entries using 816 bytes of memory
4/4 BGP path/bestpath attribute entries using 1184 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
3 BGP community entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3584 total bytes of memory
BGP activity 343/337 prefixes, 359/353 paths, scan interval 60 secs
22 networks peaked at 17:59:32 Sep  7 2026 UTC (02:09:15.250 ago)
Neighbor          V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
```

... 2 more lines

R1 show bgp ipv4 unicast summary

BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 13, main routing table version 13
6 network entries using 1488 bytes of memory
6 path entries using 816 bytes of memory
4/4 BGP path/bestpath attribute entries using 1184 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
3 BGP community entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3584 total bytes of memory
BGP activity 343/337 prefixes, 359/353 paths, scan interval 60 secs
22 networks peaked at 17:59:32 Sep 7 2026 UTC (02:09:20.286 ago)

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
... 2 more lines									

R1 show bgp ipv4 unicast summary

BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 16, main routing table version 16
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
3 BGP route-map cache entries using 216 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5936 total bytes of memory
BGP activity 346/337 prefixes, 362/353 paths, scan interval 60 secs
22 networks peaked at 17:59:32 Sep 7 2026 UTC (02:09:25.326 ago)

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
... 2 more lines									

Teardown (this test) Restore Tunnels

20 Tacacs

20.01 **PASS** The TACACS Server Runs On The NMS (0.1s, 2 commands)

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- `${state}` equals **active**
tac_plus is not running on the NMS
- output contains `:${TAC_PORT}`
- output contains **tac_plus**

How it was checked

```
NMS systemctl is-active tacacs-lab
active
```

```
NMS sudo ss -ltnp | grep ':49 ' || echo NONE
LISTEN 0      128          0.0.0.0:49      0.0.0.0:*      users: (("tac_plus",pid=7347,fd=4))
LISTEN 0      128          [::]:49         [::]:*         users: (("tac_plus",pid=7347,fd=5))
```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.02 **PASS** The Server Holds Both Remote Users And The Automation Account (0.1s, 1 commands)

The automation account has to exist on the server as well as locally on each router. Without it the first router to get AAA becomes unmanageable by this very test suite, while the server sits there perfectly healthy.

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- output contains `${ADMIN_USER}`
 - output contains `${OPS_USER}`
 - output contains `${USERNAME}`
- the automation account is missing from the server, which would lock the suite out

How it was checked

```
NMS sudo grep -oE '^user = [A-Za-z0-9_-]+' /etc/tacacs+/tac_plus.conf
user = netadmin
user = netops
user = lab
```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.03 **PASS** Every Router Points At The Server Through The Management VRF (0.4s, 3 commands)

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- output contains **aaa new-model** (checked 3×)
- output contains **address ipv4 \${NMS_OOB_IP}** (checked 3×)
- output contains **ip vrf forwarding \${MGMT_VRF}** (checked 3×)
- output contains **ip tacacs source-interface \${OOB_INTF}[\${r}]** (checked 3×)

How it was checked

```
R1 show running-config | include ^aaa|^tacacs server|^ address ipv4|^ ip vrf forwarding|^ ip tacacs source
aaa new-model
aaa group server tacacs+ LABTACGRP
ip vrf forwarding MGMT
ip tacacs source-interface GigabitEthernet5
aaa authentication login default group LABTACGRP local
aaa authentication enable default group LABTACGRP enable
aaa authorization exec default group LABTACGRP local if-authenticated
aaa accounting exec default start-stop group LABTACGRP
```

```
aaa accounting commands 15 default start-stop group LABTACGRP
aaa session-id common
tacacs server LAB-TAC
address ipv4 192.168.99.10
```

R2 `show running-config | include ^aaa |^tacacs server|^ address ipv4|^ ip vrf forwarding|^ ip tacacs source`

```
aaa new-model
aaa group server tacacs+ LABTACGRP
ip vrf forwarding MGMT
ip tacacs source-interface GigabitEthernet4
aaa authentication login default group LABTACGRP local
aaa authentication enable default group LABTACGRP enable
aaa authorization exec default group LABTACGRP local if-authenticated
aaa accounting exec default start-stop group LABTACGRP
aaa accounting commands 15 default start-stop group LABTACGRP
aaa session-id common
tacacs server LAB-TAC
address ipv4 192.168.99.10
```

R3 `show running-config | include ^aaa |^tacacs server|^ address ipv4|^ ip vrf forwarding|^ ip tacacs source`

```
aaa new-model
aaa group server tacacs+ LABTACGRP
ip vrf forwarding MGMT
ip tacacs source-interface GigabitEthernet4
aaa authentication login default group LABTACGRP local
aaa authentication enable default group LABTACGRP enable
aaa authorization exec default group LABTACGRP local if-authenticated
aaa accounting exec default start-stop group LABTACGRP
aaa accounting commands 15 default start-stop group LABTACGRP
aaa session-id common
tacacs server LAB-TAC
address ipv4 192.168.99.10
```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.04 **PASS** Every Method List Falls Back To The Local Database (0.4s, 3 commands)

A method list with no local fallback locks the lab out the moment the server is unreachable -- including from the console, which is the one way back in.

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- output contains **login default group \${TAC_GROUP} local** (checked 3×)
- output contains **exec default group \${TAC_GROUP} local** (checked 3×)

How it was checked

R1 `show running-config | include ^aaa authentication login|^aaa authorization exec`

```
aaa authentication login default group LABTACGRP local
aaa authorization exec default group LABTACGRP local if-authenticated
```

R2 `show running-config | include ^aaa authentication login|^aaa authorization exec`

```
aaa authentication login default group LABTACGRP local
aaa authorization exec default group LABTACGRP local if-authenticated
```

R3 `show running-config | include ^aaa authentication login|^aaa authorization exec`

```
aaa authentication login default group LABTACGRP local
aaa authorization exec default group LABTACGRP local if-authenticated
```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.05 **PASS** Every Router Reports The Server Alive (0.1s, 3 commands)

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- output contains **Server Status: Alive** (checked 3×)
- output contains **\${NMS_OOB_IP}** (checked 3×)

How it was checked

R1 `show tacacs`

```
Tacacs+ Server - public :
Server name: LAB-TAC
Server address: 192.168.99.10
Server port: 49
```

```

        Socket opens:      1666
        Socket closes:    1666
        Socket aborts:     0
        Socket errors:     0
        Socket Timeouts:   0
    Failed Connect Attempts: 0
    Total Packets Sent:    1801
    Total Packets Recv:    1801
        Server Status: Alive
    Continous Authc fail count: 0
    ... 1 more lines

```

R2 show tacacs

```

Tacacs+ Server - public :
    Server name: LAB-TAC
    Server address: 192.168.99.10
    Server port: 49
        Socket opens:      2555
        Socket closes:    2555
        Socket aborts:     0
        Socket errors:     0
        Socket Timeouts:   0
    Failed Connect Attempts: 0
    Total Packets Sent:    2611
    Total Packets Recv:    2611
        Server Status: Alive
    Continous Authc fail count: 0
    ... 1 more lines

```

R3 show tacacs

```

Tacacs+ Server - public :
    Server name: LAB-TAC
    Server address: 192.168.99.10
    Server port: 49
        Socket opens:      907
        Socket closes:    907
        Socket aborts:     0
        Socket errors:     0
        Socket Timeouts:   0
    Failed Connect Attempts: 0
    Total Packets Sent:    1032
    Total Packets Recv:    1032
        Server Status: Alive
    Continous Authc fail count: 0
    ... 1 more lines

```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.06 PASS A Remote Administrator Authenticates And Is Given Privilege 15 (0.3s, 0 commands)

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- `$_{priv}` equals `$_{ADMIN_PRIV}` (checked 3×)

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.07 PASS A Remote Operator Authenticates And Is Held At Privilege 1 (0.3s, 0 commands)

The pair that proves the level comes from the server. Two users, same router, same method list, different levels -- which cannot happen if the router is applying a default.

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- `$_{priv}` equals `$_{OPS_PRIV}` (checked 3×)

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.08 PASS A Wrong Password Is Refused (2.0s, 0 commands)

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.09 PASS An Unknown User Is Refused (2.1s, 0 commands)

The server rejects the user and the router honours that rather than quietly trying the local database.

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.10 PASS The Server Records Authentication In Its Accounting Log (0.1s, 1 commands)

Accounting is what makes remote authentication auditable; without it the router knows who logged in and nobody else ever finds out.

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

```
NMS sudo tail -200 /var/log/tac_plus.acct 2>/dev/null || true
Sep  7 20:06:07 192.168.99.1 lab      tty434 10.0.2.2      stop    task_id=5245      timezone=UTC
service=shell  start_time=1788811567
priv-lvl=15    cmd=no monitor capture POOLCAP <cr>
Sep  7 20:06:07 192.168.99.2 lab      tty435 10.0.2.2      stop    task_id=6119      timezone=UTC
service=shell  start_time=1788811474
disc-cause=1   disc-cause-ext=9   pre-session-time=0
elapsed_time=93 stop_time=1788811567
Sep  7 20:06:07 192.168.99.1 lab      tty435 10.0.2.2      start   task_id=5247      timezone=UTC
service=shell  start_time=1788811567
Sep  7 20:06:07 192.168.99.3 lab      tty434 10.0.2.2      stop    task_id=4501      timezone=UTC
service=shell  start_time=1788811474
disc-cause=1   disc-cause-ext=9   pre-session-time=0
elapsed_time=93 stop_time=1788811567
Sep  7 20:06:07 192.168.99.1 lab      tty434 10.0.2.2      stop    task_id=5226      timezone=UTC
service=shell  start_time=1788811474
disc-cause=1   disc-cause-ext=9   pre-session-time=0
elapsed_time=93 stop_time=1788811567
Sep  7 20:06:07 192.168.99.2 lab      tty434 10.0.2.2      start   task_id=6127      timezone=UTC
service=shell  start_time=1788811567
Sep  7 20:06:07 192.168.99.3 lab      tty435 10.0.2.2      start   task_id=4507      timezone=UTC
service=shell  start_time=1788811567
Sep  7 20:06:07 192.168.99.1 lab      tty435 10.0.2.2      stop    task_id=5247      timezone=UTC
service=shell  start_time=1788811567
priv-lvl=15    cmd=show running-config <cr>
Sep  7 20:06:07 192.168.99.2 lab      tty434 10.0.2.2      stop    task_id=6127      timezone=UTC
service=shell  start_time=1788811567
priv-lvl=15    cmd=show running-config <cr>
Sep  7 20:06:07 192.168.99.3 lab      tty435 10.0.2.2      stop    task_id=4507      timezone=UTC
service=shell  start_time=1788811567
priv-lvl=15    cmd=show running-config <cr>
Sep  7 20:06:10 192.168.99.1 lab      tty435 10.0.2.2      stop    task_id=5247      timezone=UTC
service=shell  start_time=1788811567
disc-cause=1   disc-cause-ext=9   pre-session-time=0
elapsed_time=3  stop_time=1788811570
Sep  7 20:06:10 192.168.99.1 lab      tty434 10.0.2.2      start   task_id=5249      timezone=UTC
service=shell  start_time=1788811570
Sep  7 20:06:10 192.168.99.2 lab      tty434 10.0.2.2      stop    task_id=6127      timezone=UTC
service=shell  start_time=1788811567
disc-cause=1   disc-cause-ext=9   pre-session-time=0
elapsed_time=3  stop_time=1788811570
Sep  7 20:06:10 192.168.99.3 lab      tty435 10.0.2.2      stop    task_id=4507      timezone=UTC
service=shell  start_time=1788811567
disc-cause=1   disc-cause-ext=9   pre-session-time=0
elapsed_time=3  stop_time=1788811570
... 186 more lines
```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.11 PASS Accounting Records Carry The Privilege Level And Come From The OOB Address (0.1s, 1 commands)

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- output contains **service=shell**
- output contains **\${R3_OOB_IP}**

the record did not come from R3's out-of-band address

How it was checked

NMS `sudo tail -200 /var/log/tac_plus.acct 2>/dev/null || true`

```
Sep  7 20:06:07 192.168.99.1 lab      tty434 10.0.2.2      stop    task_id=5245    timezone=UTC
service=shell  start_time=1788811567
priv-lvl=15    cmd=no monitor capture P00LCAP <cr>
Sep  7 20:06:07 192.168.99.2 lab      tty435 10.0.2.2      stop    task_id=6119    timezone=UTC
service=shell  start_time=1788811474
disc-cause=1   disc-cause-ext=9   pre-session-time=0
elapsed_time=93 stop_time=1788811567
Sep  7 20:06:07 192.168.99.1 lab      tty435 10.0.2.2      start   task_id=5247    timezone=UTC
service=shell  start_time=1788811567
Sep  7 20:06:07 192.168.99.3 lab      tty434 10.0.2.2      stop    task_id=4501    timezone=UTC
service=shell  start_time=1788811474
disc-cause=1   disc-cause-ext=9   pre-session-time=0
elapsed_time=93 stop_time=1788811567
Sep  7 20:06:07 192.168.99.1 lab      tty434 10.0.2.2      stop    task_id=5226    timezone=UTC
service=shell  start_time=1788811474
disc-cause=1   disc-cause-ext=9   pre-session-time=0
elapsed_time=93 stop_time=1788811567
Sep  7 20:06:07 192.168.99.2 lab      tty434 10.0.2.2      start   task_id=6127    timezone=UTC
service=shell  start_time=1788811567
Sep  7 20:06:07 192.168.99.3 lab      tty435 10.0.2.2      start   task_id=4507    timezone=UTC
service=shell  start_time=1788811567
Sep  7 20:06:07 192.168.99.1 lab      tty435 10.0.2.2      stop    task_id=5247    timezone=UTC
service=shell  start_time=1788811567
priv-lvl=15    cmd=show running-config <cr>
Sep  7 20:06:07 192.168.99.2 lab      tty434 10.0.2.2      stop    task_id=6127    timezone=UTC
service=shell  start_time=1788811567
priv-lvl=15    cmd=show running-config <cr>
Sep  7 20:06:07 192.168.99.3 lab      tty435 10.0.2.2      stop    task_id=4507    timezone=UTC
service=shell  start_time=1788811567
priv-lvl=15    cmd=show running-config <cr>
Sep  7 20:06:10 192.168.99.1 lab      tty435 10.0.2.2      stop    task_id=5247    timezone=UTC
service=shell  start_time=1788811567
disc-cause=1   disc-cause-ext=9   pre-session-time=0
elapsed_time=3  stop_time=1788811570
Sep  7 20:06:10 192.168.99.1 lab      tty434 10.0.2.2      start   task_id=5249    timezone=UTC
service=shell  start_time=1788811570
Sep  7 20:06:10 192.168.99.2 lab      tty434 10.0.2.2      stop    task_id=6127    timezone=UTC
service=shell  start_time=1788811567
disc-cause=1   disc-cause-ext=9   pre-session-time=0
elapsed_time=3  stop_time=1788811570
Sep  7 20:06:10 192.168.99.3 lab      tty435 10.0.2.2      stop    task_id=4507    timezone=UTC
service=shell  start_time=1788811567
disc-cause=1   disc-cause-ext=9   pre-session-time=0
elapsed_time=3  stop_time=1788811570
... 186 more lines
```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.12 PASS The Local Account Still Works When The Server Is Unreachable (16.6s, 20 commands)

The resilience property. R2's out-of-band interface is shut, which is the only path to the server, so TACACS becomes unreachable rather than merely unwilling. The automation account must still get in through the local database, and a remote-only user must not -- it exists nowhere else. The precondition is a ping inside the VRF, not "show tacacs". That field reports cached state -- it keeps saying Alive until a transaction actually fails -- so waiting on it would either hang or, worse, let this test run with the server still reachable and prove nothing.

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- output matches **Success rate is 0 percent**{does not have a usable source address
 - **\${priv}** equals **15**
- the local account could not get in with the server unreachable
- output contains **Success rate is 100 percent** (checked 3×)

How it was checked

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface GigabitEthernet4

`${output} =`

R2 shutdown

`${output} =`

R2 end


```

${output} =
R2 ping vrf MGMT 192.168.99.10 repeat 2
% VRF MGMT does not have a usable source address
R1 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1 interface GigabitEthernet5
${output} =
R1 no shutdown
${output} =
R1 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 interface GigabitEthernet4
${output} =
R2 no shutdown
${output} =
R2 end
${output} =
R3 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3 interface GigabitEthernet4
${output} =
R3 no shutdown
${output} =
R3 end
${output} =
R2 ping vrf MGMT 192.168.99.10 repeat 5
% VRF MGMT does not have a usable source address
R2 ping vrf MGMT 192.168.99.10 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.99.10, timeout is 2 seconds:
.....
Success rate is 60 percent (3/5), round-trip min/avg/max = 1/1/1 ms
R2 ping vrf MGMT 192.168.99.10 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.99.10, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

```

Teardown (this test) Restore Oob Interfaces

21 Vty Acl

21.01 **PASS** Every Router Applies The ACL Inbound On All VTY Lines (0.3s, 3 commands)

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- output contains **access-class \${VTY_ACL} in vrf-also** (checked 3×)

How it was checked

R1 **show running-config | section ^line vty**

```
line vty 0 4
access-class VTY-MGMT in vrf-also
exec-timeout 0 0
activation-character 13
transport input ssh
line vty 5 15
access-class VTY-MGMT in vrf-also
exec-timeout 0 0
activation-character 13
transport input ssh
```

R2 **show running-config | section ^line vty**

```
line vty 0 4
access-class VTY-MGMT in vrf-also
exec-timeout 0 0
activation-character 13
transport input ssh
line vty 5 15
access-class VTY-MGMT in vrf-also
exec-timeout 0 0
activation-character 13
transport input ssh
```

R3 **show running-config | section ^line vty**

```
line vty 0 4
access-class VTY-MGMT in vrf-also
exec-timeout 0 0
activation-character 13
transport input ssh
line vty 5 15
access-class VTY-MGMT in vrf-also
exec-timeout 0 0
activation-character 13
transport input ssh
```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.02 **PASS** Only SSH Is Accepted On The VTY Lines (0.3s, 3 commands)

An ACL restricting who may connect is worth less if telnet is still an option for those who may.

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- output contains **transport input ssh** (checked 3×)
- output does not contain **transport input all** (checked 3×)
- output does not contain **transport input telnet** (checked 3×)

How it was checked

R1 **show running-config | section ^line vty**

```
line vty 0 4
access-class VTY-MGMT in vrf-also
exec-timeout 0 0
activation-character 13
transport input ssh
line vty 5 15
access-class VTY-MGMT in vrf-also
exec-timeout 0 0
activation-character 13
transport input ssh
```

R2 **show running-config | section ^line vty**

```
line vty 0 4
access-class VTY-MGMT in vrf-also
```

```

exec-timeout 0 0
activation-character 13
transport input ssh
line vty 5 15
access-class VTY-MGMT in vrf-also
exec-timeout 0 0
activation-character 13
transport input ssh

```

R3 show running-config | section ^line vty

```

line vty 0 4
access-class VTY-MGMT in vrf-also
exec-timeout 0 0
activation-character 13
transport input ssh
line vty 5 15
access-class VTY-MGMT in vrf-also
exec-timeout 0 0
activation-character 13
transport input ssh

```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.03 **PASS** The ACL Permits The Management Network And Denies The Rest (0.1s, 3 commands)

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- output contains **permit** **`\${OOB\_NET}`**, wildcard bits **0.0.0.255** (checked 3×)
 - output contains **deny** (checked 3×)
- any log

How it was checked

R1 show ip access-lists VTY-MGMT

```

Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255 (6 matches)
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (114 matches)
 30 deny any log (7 matches)

```

R2 show ip access-lists VTY-MGMT

```

Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (128 matches)
 30 deny any log (2 matches)

```

R3 show ip access-lists VTY-MGMT

```

Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (112 matches)
 30 deny any log (2 matches)

```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.04 **PASS** The Documented Exception Is Present And Is Only The Hypervisor Path (0.1s, 3 commands)

Pinned deliberately. This permit is the one concession in the policy, so it should be visible in a test rather than only in a comment -- and widening it should break something.

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- output contains **permit** **`\${MGMT\_NAT\_NET}`**, wildcard bits **0.0.0.255** (checked 3×)
- **`\${permits}`** has exactly **2** entries (checked 3×)

How it was checked

R1 show ip access-lists VTY-MGMT

```

Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255 (6 matches)
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (114 matches)
 30 deny any log (7 matches)

```

R2 show ip access-lists VTY-MGMT

```
Standard IP access list VTY-MGMT
10 permit 192.168.99.0, wildcard bits 0.0.0.255
20 permit 10.0.2.0, wildcard bits 0.0.0.255 (128 matches)
30 deny any log (2 matches)
```

R3 show ip access-lists VTY-MGMT

```
Standard IP access list VTY-MGMT
10 permit 192.168.99.0, wildcard bits 0.0.0.255
20 permit 10.0.2.0, wildcard bits 0.0.0.255 (112 matches)
30 deny any log (2 matches)
```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.05 PASS SSH From The Management Network Is Allowed (0.0s, 1 commands)

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- `${banner}` starts with SSH-

How it was checked

NMS python3 -c "import socket,sys

```
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.99.10',0))
try:
    s.connect(('192.168.99.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
```

SSH-2.0-Cisco-1.25

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.06 PASS SSH From Outside The Management Network Is Refused (0.1s, 1 commands)

The same machine, a different source address, and a target it is directly connected to -- so nothing but the ACL can account for the difference. Before the access-class was applied this path returned a banner just like the one above.

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

NMS python3 -c "import socket,sys

```
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.10.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
```

REFUSED ConnectionRefusedError: [Errno 111] Connection refused

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.07 PASS Every Router Refuses A Login From Outside The Management Network (0.2s, 3 commands)

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

NMS python3 -c "import socket,sys

```
s=socket.socket()
s.settimeout(6)
```

```
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.10.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
REFUSED ConnectionRefusedError: [Errno 111] Connection refused
```

NMS python3 -c "import socket,sys

```
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.20.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
REFUSED ConnectionRefusedError: [Errno 111] Connection refused
```

NMS python3 -c "import socket,sys

```
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.30.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
REFUSED ConnectionRefusedError: [Errno 111] Connection refused
```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.08 PASS The Refusal Is Logged And Reaches The Collector (0.1s, 2 commands)

"deny any log" makes a rejected connection auditable rather than merely blocked, and the message travels the OOB network to the collector like every other log.

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- output matches **SEC-6-IPACCESSLOGS***
- no access-list denial logged by \${router} reached the collector

How it was checked

NMS python3 -c "import socket,sys

```
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.10.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
REFUSED ConnectionRefusedError: [Errno 111] Connection refused
```

NMS sudo cat /var/log/lab/192.168.99.1.log 2>/dev/null || true

```
<189>175: 000171: *Sep 7 17:45:23.653 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>176: 000172: *Sep 7 17:45:23.707 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source: 10.0.2.2] [localport: 22] at 17:45:23 UTC Mon Sep 7 2026
<189>177: 000173: *Sep 7 17:45:23.707 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>178: 000174: *Sep 7 17:45:23.915 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0 (10.0.2.2)
<189>179: 000175: *Sep 7 17:45:24.967 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0 (10.0.2.2)
<190>180: 000176: *Sep 7 17:45:25.069 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>181: 000177: *Sep 7 17:45:25.069 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user 'lab' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>182: 000178: *Sep 7 17:45:51.602 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>183: 000179: *Sep 7 17:45:51.656 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source: 10.0.2.2] [localport: 22] at 17:45:51 UTC Mon Sep 7 2026
<189>184: 000180: *Sep 7 17:45:51.656 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
```

```

<190>185: 000181: *Sep  7 17:45:51.788 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>186: 000182: *Sep  7 17:45:51.788 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>187: 000183: *Sep  7 17:48:46.557 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>188: 000184: *Sep  7 17:48:46.611 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 17:48:46 UTC Mon Sep 7 2026
... 1819 more lines

```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.09 PASS The Harness Path Still Works (0.1s, 3 commands)

The reason the second permit exists. If this fails the lab is unreachable by every other suite, so it is asserted here rather than discovered in the next run.

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- output contains `$(MGMT_NAT_HOST)` (checked 3×)

How it was checked

R1 show users

Line	User	Host(s)	Idle	Location
*434 vty 0	lab	idle	00:00:00	10.0.2.2

Interface	User	Mode	Idle	Peer Address
-----------	------	------	------	--------------

R2 show users

Line	User	Host(s)	Idle	Location
*434 vty 0	lab	idle	00:00:00	10.0.2.2

Interface	User	Mode	Idle	Peer Address
-----------	------	------	------	--------------

R3 show users

Line	User	Host(s)	Idle	Location
*434 vty 0	lab	idle	00:00:00	10.0.2.2

Interface	User	Mode	Idle	Peer Address
-----------	------	------	------	--------------

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

22 Banner

22.01 **PASS** Every Router Has A Login Banner Configured (0.3s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **banner login** (checked 3×)

How it was checked

R1 `show running-config | include ^banner`
banner login ^C

R2 `show running-config | include ^banner`
banner login ^C

R3 `show running-config | include ^banner`
banner login ^C

Teardown (suite) Close All Connections

22.02 **PASS** The Banner Is Presented Before Authentication (6.1s, 0 commands)

Captured from a login that is refused, so the banner demonstrably reaches an unauthenticated client.

Setup (suite) Open All Routers

Acceptance criteria

- `${banner}` is not empty (checked 3×)

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Close All Connections

22.03 **PASS** The Presented Banner Matches The Source File Exactly (6.2s, 0 commands)

Line for line, not a substring match: a device carrying a stale or truncated copy should fail rather than pass on one memorable phrase still being present.

Setup (suite) Open All Routers

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Close All Connections

22.04 **PASS** The Banner States The Warning It Exists To Give (2.0s, 0 commands)

A banner that does not actually warn is decoration. These are the elements that make it one: restricted access, a statement of monitoring, and consent following from use.

Setup (suite) Open All Routers

Acceptance criteria

- output contains `${phrase}` (checked 4×)
the banner does not state: `${phrase}`

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Close All Connections

22.05 **PASS** The Banner Discloses Nothing About The Platform (6.2s, 0 commands)

An unauthenticated visitor should learn that they are not welcome, and nothing else. A banner naming the vendor, model, software or hostname helps whoever should not be there more than it helps anyone who should.

Setup (suite) Open All Routers

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Close All Connections

22.06 **PASS** The Banner Survives A Configuration Save (0.1s, 3 commands)

Guards the case where a banner is applied to the running configuration and never written -- present until the next reload, and absent exactly when it would be needed.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **banner login** (checked 3×)

How it was checked

R1 `show startup-config | include ^banner`
banner login ^C

R2 `show startup-config | include ^banner`
banner login ^C

R3 `show startup-config | include ^banner`
banner login ^C

Teardown (suite) Close All Connections

Appendix — captured device state

Written by suite *05 Capture State* during the run. Routing tables and protocol state first, then what the NMS sees for each router -- polled over SNMPv3, and received as syslog and traps -- then the full running configuration of each router.

R1 — ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR
& - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0

```
S* 0.0.0.0/0 [254/0] via 10.0.2.2
10.0.0.0/8 is variably subnetted, 12 subnets, 3 masks
C    10.0.2.0/24 is directly connected, GigabitEthernet1
L    10.0.2.15/32 is directly connected, GigabitEthernet1
C    10.10.10.0/30 is directly connected, GigabitEthernet2
L    10.10.10.1/32 is directly connected, GigabitEthernet2
C    10.10.20.0/30 is directly connected, GigabitEthernet4
L    10.10.20.1/32 is directly connected, GigabitEthernet4
C    10.20.1.1/32 is directly connected, Loopback1
B    10.20.2.1/32 [20/0] via 192.168.100.2, 00:17:31
B    10.20.3.1/32 [20/0] via 192.168.101.2, 00:17:31
S    10.30.1.0/24 is directly connected, Null0
B    10.30.2.0/24 [20/0] via 192.168.100.2, 00:17:31
B    10.30.3.0/24 [20/0] via 192.168.101.2, 00:17:31
172.16.0.0/32 is subnetted, 3 subnets
C    172.16.1.1 is directly connected, Loopback0
S    172.16.2.1 is directly connected, Tunnel0
S    172.16.3.1 is directly connected, Tunnel1
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet3
L    192.168.10.1/32 is directly connected, GigabitEthernet3
B    192.168.20.0/24 [20/0] via 192.168.100.2, 00:17:31
B    192.168.30.0/24 [20/0] via 192.168.101.2, 00:17:31
192.168.100.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.100.0/30 is directly connected, Tunnel0
L    192.168.100.1/32 is directly connected, Tunnel0
192.168.101.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.101.0/30 is directly connected, Tunnel1
L    192.168.101.1/32 is directly connected, Tunnel1
```

R2 — ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR
& - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0

```
S* 0.0.0.0/0 [254/0] via 10.0.2.2
10.0.0.0/8 is variably subnetted, 10 subnets, 3 masks
C    10.0.2.0/24 is directly connected, GigabitEthernet1
L    10.0.2.15/32 is directly connected, GigabitEthernet1
C    10.10.10.0/30 is directly connected, GigabitEthernet2
L    10.10.10.2/32 is directly connected, GigabitEthernet2
B    10.20.1.1/32 [20/0] via 192.168.100.1, 00:17:31
C    10.20.2.1/32 is directly connected, Loopback1
```

```

B      10.20.3.1/32 [20/0] via 192.168.100.1, 00:17:31
B      10.30.1.0/24 [20/0] via 192.168.100.1, 00:17:31
S      10.30.2.0/24 is directly connected, Null0
B      10.30.3.0/24 [20/0] via 192.168.100.1, 00:17:31
      172.16.0.0/32 is subnetted, 2 subnets
S      172.16.1.1 is directly connected, Tunnel0
C      172.16.2.1 is directly connected, Loopback0
B      192.168.10.0/24 [20/0] via 192.168.100.1, 00:17:31
      192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.20.0/24 is directly connected, GigabitEthernet3
L      192.168.20.1/32 is directly connected, GigabitEthernet3
B      192.168.30.0/24 [20/0] via 192.168.100.1, 00:17:31
      192.168.100.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.100.0/30 is directly connected, Tunnel0
L      192.168.100.2/32 is directly connected, Tunnel0

```

R3 — ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR
& - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0

```

S*    0.0.0.0/0 [254/0] via 10.0.2.2
      10.0.0.0/8 is variably subnetted, 10 subnets, 3 masks
C      10.0.2.0/24 is directly connected, GigabitEthernet1
L      10.0.2.15/32 is directly connected, GigabitEthernet1
C      10.10.20.0/30 is directly connected, GigabitEthernet2
L      10.10.20.2/32 is directly connected, GigabitEthernet2
B      10.20.1.1/32 [20/0] via 192.168.101.1, 00:17:31
B      10.20.2.1/32 [20/0] via 192.168.101.1, 00:17:31
C      10.20.3.1/32 is directly connected, Loopback1
B      10.30.1.0/24 [20/0] via 192.168.101.1, 00:17:31
B      10.30.2.0/24 [20/0] via 192.168.101.1, 00:17:31
S      10.30.3.0/24 is directly connected, Null0
      172.16.0.0/32 is subnetted, 2 subnets
S      172.16.1.1 is directly connected, Tunnel0
C      172.16.3.1 is directly connected, Loopback0
B      192.168.10.0/24 [20/0] via 192.168.101.1, 00:17:31
B      192.168.20.0/24 [20/0] via 192.168.101.1, 00:17:31
      192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.30.0/24 is directly connected, GigabitEthernet3
L      192.168.30.1/32 is directly connected, GigabitEthernet3
      192.168.101.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.101.0/30 is directly connected, Tunnel0
L      192.168.101.2/32 is directly connected, Tunnel0

```

R1 — bgp summary

BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 10, main routing table version 10
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5720 total bytes of memory
BGP activity 9/0 prefixes, 9/0 paths, scan interval 60 secs
9 networks peaked at 00:07:06 Sep 8 2026 UTC (00:17:33.655 ago)

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
192.168.100.2	4	65002	117	121	10	0	0	00:17:46	3
192.168.101.2	4	65003	117	121	10	0	0	00:17:43	3

R2 — bgp summary

BGP router identifier 172.16.2.1, local AS number 65002
BGP table version is 10, main routing table version 10
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5736 total bytes of memory
BGP activity 9/0 prefixes, 9/0 paths, scan interval 60 secs
9 networks peaked at 00:07:07 Sep 8 2026 UTC (00:17:32.736 ago)

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
192.168.100.1	4	65001	121	117	10	0	0	00:17:46	6

R3 — bgp summary

BGP router identifier 172.16.3.1, local AS number 65003
BGP table version is 10, main routing table version 10
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5736 total bytes of memory
BGP activity 9/0 prefixes, 9/0 paths, scan interval 60 secs
9 networks peaked at 00:07:07 Sep 8 2026 UTC (00:17:32.808 ago)

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
192.168.101.1	4	65001	121	117	10	0	0	00:17:43	6

R1 — bgp table

BGP table version is 10, local router ID is 172.16.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-state,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.1.1/32	0.0.0.0	0		32768	i
*>	10.20.2.1/32	192.168.100.2	0		0	65002 i
*>	10.20.3.1/32	192.168.101.2	0		0	65003 i
*>	10.30.1.0/24	0.0.0.0	0		32768	i
*>	10.30.2.0/24	192.168.100.2	0		0	65002 i
*>	10.30.3.0/24	192.168.101.2	0		0	65003 i
*>	192.168.10.0	0.0.0.0	0		32768	i
*>	192.168.20.0	192.168.100.2	0		0	65002 i
*>	192.168.30.0	192.168.101.2	0		0	65003 i

R2 — bgp table

BGP table version is 10, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-state,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.1.1/32	192.168.100.1	0		0	65001 i
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.20.3.1/32	192.168.100.1	0		0	65001 65003 i
*>	10.30.1.0/24	192.168.100.1	0		0	65001 i
*>	10.30.2.0/24	0.0.0.0	0		32768	i
*>	10.30.3.0/24	192.168.100.1	0		0	65001 65003 i
*>	192.168.10.0	192.168.100.1	0		0	65001 i
*>	192.168.20.0	0.0.0.0	0		32768	i
*>	192.168.30.0	192.168.100.1	0		0	65001 65003 i

R3 — bgp table

BGP table version is 10, local router ID is 172.16.3.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.1.1/32	192.168.101.1	0		0	65001 i
*>	10.20.2.1/32	192.168.101.1			0	65001 65002 i
*>	10.20.3.1/32	0.0.0.0	0		32768	i
*>	10.30.1.0/24	192.168.101.1	0		0	65001 i
*>	10.30.2.0/24	192.168.101.1			0	65001 65002 i
*>	10.30.3.0/24	0.0.0.0	0		32768	i
*>	192.168.10.0	192.168.101.1	0		0	65001 i
*>	192.168.20.0	192.168.101.1			0	65001 65002 i
*>	192.168.30.0	0.0.0.0	0		32768	i

R1 — crypto ikev2 sa

IPv4 Crypto IKEv2 SA

Tunnel-id	Local	Remote	fvr/ivrf	Status
3	10.10.20.1/500	10.10.20.2/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/2025 sec				
CE id: 1003, Session-id: 2				
Local spi: 3CA74A659346024C		Remote spi: 375A6F1573C47ED7		

Tunnel-id	Local	Remote	fvr/ivrf	Status
4	10.10.10.1/500	10.10.10.2/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/2025 sec				
CE id: 1004, Session-id: 1				
Local spi: 667A50DE686D5330		Remote spi: 795CC02AD1A5D549		

IPv6 Crypto IKEv2 SA

R2 — crypto ikev2 sa

IPv4 Crypto IKEv2 SA

Tunnel-id	Local	Remote	fvr/ivrf	Status
1	10.10.10.2/500	10.10.10.1/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/2026 sec				
CE id: 1002, Session-id: 1				
Local spi: 795CC02AD1A5D549		Remote spi: 667A50DE686D5330		

IPv6 Crypto IKEv2 SA

R3 — crypto ikev2 sa

IPv4 Crypto IKEv2 SA

Tunnel-id	Local	Remote	fvr/ivrf	Status
1	10.10.20.2/500	10.10.20.1/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/2026 sec				
CE id: 1002, Session-id: 1				
Local spi: 375A6F1573C47ED7		Remote spi: 3CA74A659346024C		

IPv6 Crypto IKEv2 SA

H1 — ip addr

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
```

```

inet6 ::1/128 scope host
    valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
   link/ether 52:54:00:aa:01:01 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
      valid_lft 85365sec preferred_lft 74565sec
   inet6 fec0::5054:ff:feaa:101/64 scope site dynamic noprefixroute flags 100
      valid_lft 86223sec preferred_lft 14223sec
   inet6 fe80::5054:ff:feaa:101/64 scope link
      valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
   link/ether 52:54:00:aa:01:02 brd ff:ff:ff:ff:ff:ff
   inet 192.168.10.10/24 scope global eth1
      valid_lft forever preferred_lft forever
   inet 169.254.229.55/16 brd 169.254.255.255 scope global noprefixroute eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.11/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.12/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.13/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.14/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.15/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.16/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.17/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.18/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.19/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.64/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.65/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.66/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.67/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.68/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.69/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.70/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.71/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.72/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.73/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.74/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.75/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.76/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.77/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.78/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.79/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.80/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.81/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.82/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.83/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.84/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.85/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.86/24 scope global secondary eth1
      valid_lft forever preferred_lft forever
   inet 192.168.10.87/24 scope global secondary eth1

```

```

    valid_lft forever preferred_lft forever
inet 192.168.10.88/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.10.89/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.10.90/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.10.91/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.10.92/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.10.93/24 scope global secondary eth1
    valid_lft forever preferred_lft forever

```

H1 — ip route

```

default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.10.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.229.55 metric 1003
192.168.10.0/24 dev eth1 scope link src 192.168.10.10
192.168.20.0/24 via 192.168.10.1 dev eth1
192.168.30.0/24 via 192.168.10.1 dev eth1

```

H2 — ip addr

```

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:02:01 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 85365sec preferred_lft 74565sec
    inet6 fec0::5054:ff:feaa:201/64 scope site dynamic noprefixroute flags 100
        valid_lft 85988sec preferred_lft 13988sec
    inet6 fe80::5054:ff:feaa:201/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:02:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.199.41/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.15/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.16/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.17/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.18/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.19/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.64/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.65/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.66/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.67/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.68/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.69/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.70/24 scope global secondary eth1
        valid_lft forever preferred_lft forever

```

```

inet 192.168.20.71/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.72/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.73/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.74/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.75/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.76/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.77/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.78/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.79/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.80/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.81/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.82/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.83/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.84/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.85/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.86/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.87/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.88/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.89/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.90/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.91/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.92/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.93/24 scope global secondary eth1
    valid_lft forever preferred_lft forever

```

H2 — ip route

```

default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.20.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.199.41 metric 1003
192.168.10.0/24 via 192.168.20.1 dev eth1
192.168.20.0/24 dev eth1 scope link src 192.168.20.10
192.168.30.0/24 via 192.168.20.1 dev eth1

```

H3 — ip addr

```

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:03:01 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 85364sec preferred_lft 74564sec
    inet6 fec0::5054:ff:feaa:301/64 scope site dynamic noprefixroute flags 100
        valid_lft 86293sec preferred_lft 14293sec
    inet6 fe80::5054:ff:feaa:301/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:03:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.30.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.207.242/16 brd 169.254.255.255 scope global noprefixroute eth1

```

[illegible]

H3 — ip route

```
default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.30.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.207.242 metric 1003
192.168.10.0/24 via 192.168.30.1 dev eth1
192.168.20.0/24 via 192.168.30.1 dev eth1
192.168.30.0/24 dev eth1 scope link src 192.168.30.10
```

NMS — network

```
lo UNKNOWN 127.0.0.1/8 ::1/128
ens3 UP 10.0.2.15/24 metric 100 fec0::5054:ff:febb:101/64 fe80::5054:ff:febb:101/64
ens4 UP 192.168.10.20/24 fe80::5054:ff:febb:102/64
ens5 UP 192.168.99.10/24 fe80::5054:ff:febb:103/64
default via 10.0.2.2 dev ens3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev ens3 proto kernel scope link src 10.0.2.15 metric 100
10.0.2.2 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.0.2.3 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.30.0.0/16 via 192.168.10.1 dev ens4
192.168.10.0/24 dev ens4 proto kernel scope link src 192.168.10.20
192.168.20.0/24 via 192.168.10.1 dev ens4
192.168.30.0/24 via 192.168.10.1 dev ens4
192.168.99.0/24 dev ens5 proto kernel scope link src 192.168.99.10
```

NMS — snmp version

```
NET-SNMP version: 5.9.4.pre2
```

NMS — collectors

```
active
active
active
```

NMS — collector files

```
total 48
-rw-r--r-- 1 syslog syslog 4896 Sep  8 00:24 192.168.99.1.log
-rw-r--r-- 1 syslog syslog 3591 Sep  8 00:24 192.168.99.2.log
-rw-r--r-- 1 syslog syslog 3591 Sep  8 00:24 192.168.99.3.log
-rw-r--r-- 1 root root 11308 Sep  8 00:24 snmptrapd.log
-rw-r--r-- 1 root root 17747 Sep  8 00:24 traps-raw.log
```

NMS — R1 syslog received

```
tail: cannot open '/var/log/lab/192.168.10.1.log' for reading: No such file or directory
```

NMS — R2 syslog received

```
tail: cannot open '/var/log/lab/192.168.20.1.log' for reading: No such file or directory
```

NMS — R3 syslog received

```
tail: cannot open '/var/log/lab/192.168.30.1.log' for reading: No such file or directory
```

NMS — snmptrapd

```
2026-09-08 00:23:39 <UNKNOWN> [UDP: [192.168.99.1]:53275->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:33:32.15 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.2
iso.3.6.1.4.1.9.9.43.1.1.1.0 201214 iso.3.6.1.4.1.9.9.43.1.1.6.1.6.5 5
iso.3.6.1.4.1.9.9.43.1.1.6.1.8.5 "lab"
2026-09-08 00:23:39 <UNKNOWN> [UDP: [192.168.99.1]:53275->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:33:32.30 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.36 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.36 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.36 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.36 "lab"
2026-09-08 00:23:40 <UNKNOWN> [UDP: [192.168.99.1]:53275->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:33:32.31 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
```

C8000V IPsec Testbed - test evidence

```
iso.3.6.1.2.1.1.3.0 0:0:34:23.87 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.41 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.41 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.41 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.41 "lab"
```

NMS — R1 polled over SNMPv3: snmp system

```
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (207428) 0:34:34.28
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
```

NMS — R1 polled over SNMPv3: snmp interfaces

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "GigabitEthernet5"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.10 = STRING: "Tunnel1"
.1.3.6.1.2.1.2.2.1.2.11 = STRING: "Loopback1"
```

NMS — R2 polled over SNMPv3: snmp system

```
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (206968) 0:34:29.68
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
```

NMS — R2 polled over SNMPv3: snmp interfaces

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
```

NMS — R3 polled over SNMPv3: snmp system

```
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (206974) 0:34:29.74
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
```

NMS — R3 polled over SNMPv3: snmp interfaces

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
```

R1 — running config

Building configuration...

[illegible]

[illegible]

```

80DDCD16 D6BACECA EEB7CF9 8428787B 35202CDC 60E4616A B623CDBD 230E3AFB
418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F 719BB2F0
D697DF7F 28
quit
!
!
!
!
!
!
!
!
license udi pid C8000V sn 9ZZD3DIMBWB
license boot level network-premier addon dna-premier
memory free low-watermark processor 31774
diagnostic bootstrap level minimal
!
!
spanning-tree extend system-id
!
!
enable secret 9 $9$3EoNBc6Q2eJBE.$3047TTMdhTNzASp1ANWS7fp7xP3q/jy6G5vRE8pSd22
!
username lab privilege 15 secret 9 $9$DfbNgBKfs7BDP.$rsDLRF9AcBs6tKn9UxtGuDYZxiNSij6adtQ06PC8qGM
!
redundancy
!
!
!
crypto ikev2 proposal LAB-PROP
encryption aes-cbc-256
integrity sha256
group 14
!
crypto ikev2 policy LAB-POL
proposal LAB-PROP
!
crypto ikev2 keyring LAB-KR
peer R2
address 10.10.10.2
pre-shared-key LabPreSharedKey123
!
peer R3
address 10.10.20.2
pre-shared-key LabPreSharedKey123
!
!
!
crypto ikev2 profile LAB-PROF
match identity remote address 10.10.10.2 255.255.255.255
match identity remote address 10.10.20.2 255.255.255.255
authentication remote pre-share
authentication local pre-share
keyring local LAB-KR
!
!
!
!
!
!
!
!
!
!
!
!
crypto ipsec transform-set LAB-TS esp-aes 256 esp-sha256-hmac
mode tunnel
!
crypto ipsec profile LAB-IPSEC
set transform-set LAB-TS
set ikev2-profile LAB-PROF
!
!
!
!
!
!
!
```

```

!
!
!
interface Loopback0
description Protected endpoint
ip address 172.16.1.1 255.255.255.255
!
interface Loopback1
description BGP-advertised prefix
ip address 10.20.1.1 255.255.255.255
!
interface Tunnel0
description to c8000v-r2
ip address 192.168.100.1 255.255.255.252
ip nat outside
bfd interval 500 min_rx 500 multiplier 3
tunnel source GigabitEthernet2
tunnel mode ipsec ipv4
tunnel destination 10.10.10.2
tunnel protection ipsec profile LAB-IPSEC
!
interface Tunnel1
description to c8000v-r3
ip address 192.168.101.1 255.255.255.252
ip nat outside
bfd interval 500 min_rx 500 multiplier 3
tunnel source GigabitEthernet4
tunnel mode ipsec ipv4
tunnel destination 10.10.20.2
tunnel protection ipsec profile LAB-IPSEC
!
interface GigabitEthernet1
description MGMT - QEMU user-mode net (DHCP)
ip address dhcp
negotiation auto
!
interface GigabitEthernet2
description P2P link to c8000v-r2
ip address 10.10.10.1 255.255.255.252
negotiation auto
!
interface GigabitEthernet3
description LAN toward h1
ip address 192.168.10.1 255.255.255.0
ip nat inside
negotiation auto
!
interface GigabitEthernet4
description P2P link to c8000v-r3
ip address 10.10.20.1 255.255.255.252
negotiation auto
!
interface GigabitEthernet5
description OOB management - NTP, syslog, SNMP
vrf forwarding MGMT
ip address 192.168.99.1 255.255.255.0
negotiation auto
!
router bgp 65001
bgp router-id 172.16.1.1
bgp log-neighbor-changes
neighbor 192.168.100.2 remote-as 65002
neighbor 192.168.100.2 password LabBgpSecret
neighbor 192.168.100.2 timers 10 30
neighbor 192.168.100.2 fall-over bfd
neighbor 192.168.101.2 remote-as 65003
neighbor 192.168.101.2 password LabBgpSecret
neighbor 192.168.101.2 timers 10 30
neighbor 192.168.101.2 fall-over bfd
!
address-family ipv4
network 10.20.1.1 mask 255.255.255.255
network 10.30.1.0 mask 255.255.255.0
network 192.168.10.0
neighbor 192.168.100.2 activate
neighbor 192.168.100.2 send-community both
neighbor 192.168.100.2 route-map RM-IN-R2 in
neighbor 192.168.100.2 route-map RM-OUT-R2 out
neighbor 192.168.100.2 maximum-prefix 15 80
neighbor 192.168.101.2 activate
neighbor 192.168.101.2 send-community both

```

```

neighbor 192.168.101.2 route-map RM-IN-R3 in
neighbor 192.168.101.2 route-map RM-OUT-R3 out
neighbor 192.168.101.2 maximum-prefix 15 80
exit-address-family
!
ip forward-protocol nd
!
ip bgp-community new-format
ip community-list standard CL-VALID permit 65001:1
ip community-list standard CL-VALID permit 65001:2
ip community-list standard CL-VALID permit 65001:3
ip community-list standard CL-VALID permit 65002:1
ip community-list standard CL-VALID permit 65002:2
ip community-list standard CL-VALID permit 65002:3
ip community-list standard CL-VALID permit 65003:1
ip community-list standard CL-VALID permit 65003:2
ip community-list standard CL-VALID permit 65003:3
ip as-path access-list 10 permit ^65002$
ip as-path access-list 20 permit ^65003$
no ip http server
ip http secure-server
ip route 10.30.1.0 255.255.255.0 Null0
ip route 172.16.2.1 255.255.255.255 Tunnel0
ip route 172.16.3.1 255.255.255.255 Tunnel1
ip nat pool SITE-POOL 10.30.1.100 10.30.1.129 prefix-length 24
ip nat inside source static 192.168.10.10 10.30.1.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.11 10.30.1.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.12 10.30.1.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.13 10.30.1.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.14 10.30.1.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.15 10.30.1.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.16 10.30.1.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.17 10.30.1.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.18 10.30.1.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.19 10.30.1.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
ip ssh bulk-mode 131072
ip ssh server algorithm authentication password
!
ip prefix-list PL-FAR-R2 seq 40 permit 10.20.2.1/32
ip prefix-list PL-FAR-R2 seq 45 permit 10.30.2.0/24
ip prefix-list PL-FAR-R2 seq 50 permit 192.168.20.0/24
!
ip prefix-list PL-FAR-R3 seq 40 permit 10.20.3.1/32
ip prefix-list PL-FAR-R3 seq 45 permit 10.30.3.0/24
ip prefix-list PL-FAR-R3 seq 50 permit 192.168.30.0/24
!
ip prefix-list PL-OWN-LAN seq 5 permit 192.168.10.0/24
!
ip prefix-list PL-OWN-L0 seq 5 permit 10.20.1.1/32
!
ip prefix-list PL-OWN-NAT seq 5 permit 10.30.1.0/24
!
ip access-list standard VTY-MGMT
 10 remark out-of-band management network
 10 permit 192.168.99.0 0.0.0.255
 20 remark hypervisor user-mode NAT: how the harness reaches these devices
 20 permit 10.0.2.0 0.0.0.255
 30 deny any log
!
ip access-list extended NAT-DOMAIN
 10 permit ip host 192.168.10.10 10.30.0.0 0.0.255.255
 20 permit ip host 192.168.10.11 10.30.0.0 0.0.255.255
 30 permit ip host 192.168.10.12 10.30.0.0 0.0.255.255
 40 permit ip host 192.168.10.13 10.30.0.0 0.0.255.255
 50 permit ip host 192.168.10.14 10.30.0.0 0.0.255.255
 60 permit ip host 192.168.10.15 10.30.0.0 0.0.255.255
 70 permit ip host 192.168.10.16 10.30.0.0 0.0.255.255
 80 permit ip host 192.168.10.17 10.30.0.0 0.0.255.255
 90 permit ip host 192.168.10.18 10.30.0.0 0.0.255.255
100 permit ip host 192.168.10.19 10.30.0.0 0.0.255.255
ip access-list extended NAT-POOL-SRC
 10 permit ip 192.168.10.64 0.0.0.31 10.30.0.0 0.0.255.255
logging source-interface GigabitEthernet5 vrf MGMT
logging host 192.168.99.10 vrf MGMT
route-map RM-OUT-R2 permit 10
description R1's own L0, tagged 65001:1
match ip address prefix-list PL-OWN-L0
set community 65001:1
!
route-map RM-OUT-R2 permit 20

```



```

description R1's own NAT, tagged 65001:2
match ip address prefix-list PL-OWN-NAT
set community 65001:2
!
route-map RM-OUT-R2 permit 30
description R1's own LAN, tagged 65001:3
match ip address prefix-list PL-OWN-LAN
set community 65001:3
!
route-map RM-OUT-R2 permit 55
description re-advertise R3, preserving its communities
match ip address prefix-list PL-FAR-R3
!
route-map RM-OUT-R3 permit 10
description R1's own L0, tagged 65001:1
match ip address prefix-list PL-OWN-L0
set community 65001:1
!
route-map RM-OUT-R3 permit 20
description R1's own NAT, tagged 65001:2
match ip address prefix-list PL-OWN-NAT
set community 65001:2
!
route-map RM-OUT-R3 permit 30
description R1's own LAN, tagged 65001:3
match ip address prefix-list PL-OWN-LAN
set community 65001:3
!
route-map RM-OUT-R3 permit 55
description re-advertise R2, preserving its communities
match ip address prefix-list PL-FAR-R2
!
route-map NAT-T0-DOMAIN permit 10
match ip address NAT-DOMAIN
!
route-map RM-IN-R2 permit 10
description accept only known communities on an AS path R2 can produce
match as-path 10
match community CL-VALID
!
route-map RM-IN-R3 permit 10
description accept only known communities on an AS path R3 can produce
match as-path 20
match community CL-VALID
!
snmp-server group LABGRP v3 priv read LABVIEW notify LABVIEW
snmp-server view LABVIEW iso included
snmp-server trap-source GigabitEthernet5
snmp-server enable traps snmp authentication linkdown linkup coldstart warmstart
snmp-server enable traps vrrp
snmp-server enable traps flowmon
snmp-server enable traps pfr
snmp-server enable traps ds1
snmp-server enable traps sdwan security policy system omp bfd
snmp-server enable traps entity-perf throughput-notif
snmp-server enable traps call-home message-send-fail server-fail
snmp-server enable traps tty
snmp-server enable traps ospf state-change
snmp-server enable traps ospf errors
snmp-server enable traps ospf retransmit
snmp-server enable traps ospf lsa
snmp-server enable traps ospf cisco-specific state-change nssa-trans-change
snmp-server enable traps ospf cisco-specific state-change shamlink interface
snmp-server enable traps ospf cisco-specific state-change shamlink neighbor
snmp-server enable traps ospf cisco-specific errors
snmp-server enable traps ospf cisco-specific retransmit
snmp-server enable traps ospf cisco-specific lsa
snmp-server enable traps eigrp
snmp-server enable traps casa
snmp-server enable traps xgcp
snmp-server enable traps cef resource-failure peer-state-change peer-fib-state-change inconsistency
snmp-server enable traps smart-license
snmp-server enable traps diameter
snmp-server enable traps entity-qfp mem-res-thresh throughput-notif
snmp-server enable traps entity-state
snmp-server enable traps ethernet cfm cc mep-up mep-down cross-connect loop config
snmp-server enable traps ethernet cfm crosscheck mep-missing mep-unknown service-up
snmp-server enable traps ether-oam
snmp-server enable traps ethernet evc status create delete
snmp-server enable traps aaa_server
snmp-server enable traps config-copy

```

```

snmp-server enable traps config
snmp-server enable traps config-ctid
snmp-server enable traps memory bufferpeak
snmp-server enable traps fru-ctrl
snmp-server enable traps entity-sensor
snmp-server enable traps resource-policy
snmp-server enable traps flash insertion removal lowspace
snmp-server enable traps hsrp
snmp-server enable traps entity
snmp-server enable traps ip local pool
snmp-server enable traps cpu threshold
snmp-server enable traps auth-framework sec-violation
snmp-server enable traps atm subif
snmp-server enable traps frame-relay
snmp-server enable traps frame-relay subif
snmp-server enable traps bgp state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp threshold prefix
snmp-server enable traps bgp cbgp2 state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp cbgp2 threshold prefix
snmp-server enable traps pim neighbor-change rp-mapping-change invalid-pim-message
snmp-server enable traps ipmulticast
snmp-server enable traps msdp
snmp-server enable traps mvpn
snmp-server enable traps rsvp
snmp-server enable traps isis
snmp-server enable traps dhcp
snmp-server enable traps nhrp nhs
snmp-server enable traps nhrp nhc
snmp-server enable traps nhrp nhp
snmp-server enable traps nhrp quota-exceeded
snmp-server enable traps ipsla
snmp-server enable traps bfd
snmp-server enable traps cnpd
snmp-server enable traps event-manager
snmp-server enable traps dlsw
snmp-server enable traps ospfv3 state-change
snmp-server enable traps ospfv3 errors
snmp-server enable traps pppoe
snmp-server enable traps l2tun session
snmp-server enable traps l2tun pseudowire status
snmp-server enable traps l2tun tunnel
snmp-server enable traps pki
snmp-server enable traps pimstdmib neighbor-loss invalid-register invalid-join-prune rp-mapping-change
interface-election
snmp-server enable traps adslline
snmp-server enable traps entity-diag boot-up-fail hm-test-recover hm-thresh-reached scheduled-test-fail
snmp-server enable traps vds12line
snmp-server enable traps vtp
snmp-server enable traps vlancreate
snmp-server enable traps vlandelete
snmp-server enable traps port-security
snmp-server enable traps isdn call-information
snmp-server enable traps isdn layer2
snmp-server enable traps isdn chan-not-avail
snmp-server enable traps isdn ietf
snmp-server enable traps c3g
snmp-server enable traps LTE
snmp-server enable traps lisp
snmp-server enable traps firewall serverstatus
snmp-server enable traps ike policy add
snmp-server enable traps ike policy delete
snmp-server enable traps ike tunnel start
snmp-server enable traps ike tunnel stop
snmp-server enable traps ipsec cryptomap add
snmp-server enable traps ipsec cryptomap delete
snmp-server enable traps ipsec cryptomap attach
snmp-server enable traps ipsec cryptomap detach
snmp-server enable traps ipsec tunnel start
snmp-server enable traps ipsec tunnel stop
snmp-server enable traps ipsec too-many-sas
snmp-server enable traps gdoi gm-start-registration
snmp-server enable traps gdoi gm-registration-complete
snmp-server enable traps gdoi gm-re-register
snmp-server enable traps gdoi gm-rekey-rcvd
snmp-server enable traps gdoi gm-rekey-fail
snmp-server enable traps gdoi ks-rekey-pushed
snmp-server enable traps gdoi gm-incomplete-cfg
snmp-server enable traps gdoi ks-no-rsa-keys
snmp-server enable traps gdoi ks-new-registration
snmp-server enable traps gdoi ks-reg-complete
snmp-server enable traps gdoi ks-role-change

```

```

snmp-server enable traps gdoi ks-gm-deleted
snmp-server enable traps gdoi ks-peer-reachable
snmp-server enable traps gdoi ks-peer-unreachable
snmp-server enable traps mpls rfc ldp
snmp-server enable traps mpls ldp
snmp-server enable traps mpls rfc traffic-eng
snmp-server enable traps mpls traffic-eng
snmp-server enable traps mpls fast-reroute protected
snmp-server enable traps otn
snmp-server enable traps pw vc
snmp-server enable traps frame-relay multilink bundle-mismatch
snmp-server enable traps dial
snmp-server enable traps dsp card-status
snmp-server enable traps dsp oper-state
snmp-server enable traps dsp video-usage
snmp-server enable traps dsp video-out-of-resource
snmp-server enable traps syslog
snmp-server enable traps sbc adj-status
snmp-server enable traps sbc blacklist
snmp-server enable traps sbc congestion-alarm
snmp-server enable traps sbc h248-ctrlr-status
snmp-server enable traps sbc media-source
snmp-server enable traps sbc radius-conn-status
snmp-server enable traps sbc sla-violation
snmp-server enable traps sbc sla-violation-rev1
snmp-server enable traps sbc svc-state
snmp-server enable traps sbc qos-statistics
snmp-server enable traps bulkstat collection transfer
snmp-server enable traps ethernet cfm alarm
snmp-server enable traps alarms informational
snmp-server enable traps transceiver all
snmp-server enable traps rf
snmp-server enable traps vrfmib vrf-up vrf-down vnet-trunk-up vnet-trunk-down
snmp-server enable traps ccme
snmp-server enable traps srst
snmp-server enable traps mpls vpn
snmp-server enable traps mpls rfc vpn
snmp-server enable traps voice
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon udp-port 1162
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon
tacacs server LAB-TAC
address ipv4 192.168.99.10
key LabTacacsKey123
!
!
!
!
!
control-plane
!
!
mgcp behavior rsip-range tgcp-only
mgcp behavior comedia-role none
mgcp behavior comedia-check-media-src disable
mgcp behavior comedia-sdp-force disable
!
mgcp profile default
!
!
!
!
!
banner login ^C
*****
                AUTHORISED ACCESS ONLY

Access to this device is restricted to authorised personnel. If you are not
an authorised user, disconnect now.

All activity is monitored, logged and audited. Continued use of this system
constitutes consent to that monitoring.

Unauthorised access or use may result in disciplinary action and civil or
criminal penalties.
*****
^C
!
line con 0
exec-timeout 0 0
activation-character 13
logging synchronous

```



```
C7479096 B4CB2D62 EA2F505D C7B062A4 6811D95B E8250FC4 5D5D5FB8 8F27D191
C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B DF5F4368 95135E44
DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 0101FF04 04030201
06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 0E041604 1449DC85
4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 86F70D01 010B0500
03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 49631C78 240DA905
604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1 6C9E3D8B
D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 5575B146 8DFC66A8
467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2 55A9232C
7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDBEB86 C71E3B49 1765308B
5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69 39F08678
80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A B623CDBD 230E3AFB
418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862F0AE 0275156F 719BB2F0
D697DF7F 28
quit

!
!
!
!
!
!
!
!
!
license udi pid C8000V sn 9IM1MG1HPJU
license boot level network-premier addon dna-premier
memory free low-watermark processor 31774
diagnostic bootstrap level minimal
!
!
spanning-tree extend system-id
!
!
enable secret 9 $9$TP0hQLBPymXlbE$/z45v0LkFIyZhmarHI.HzRzmprK53pitlGDCuhLOGI
!
username lab privilege 15 secret 9 $9$2Fs/2GuPI4jt1k$oCwnK81TwinWnZK56lD16dv8XXxKG1A7qni5ZkCf0iFY
!
redundancy
!
!
!
!
crypto ikev2 proposal LAB-PROP
encryption aes-cbc-256
integrity sha256
group 14
!
crypto ikev2 policy LAB-POL
proposal LAB-PROP
!
crypto ikev2 keyring LAB-KR
peer R1
address 10.10.10.1
pre-shared-key LabPreSharedKey123
!
!
!
!
crypto ikev2 profile LAB-PROF
match identity remote address 10.10.10.1 255.255.255.255
authentication remote pre-share
authentication local pre-share
keyring local LAB-KR
!
!
!
!
!
!
!
!
!
!
!
!
crypto ipsec transform-set LAB-TS esp-aes 256 esp-sha256-hmac
mode tunnel
!
crypto ipsec profile LAB-IPSEC
set transform-set LAB-TS
set ikev2-profile LAB-PROF
!
```

```

!
!
!
!
!
!
!
!
!
interface Loopback0
  description Protected endpoint
  ip address 172.16.2.1 255.255.255.255
!
interface Loopback1
  description BGP-advertised prefix
  ip address 10.20.2.1 255.255.255.255
!
interface Tunnel0
  description to the hub (c8000v-r1)
  ip address 192.168.100.2 255.255.255.252
  ip nat outside
  bfd interval 500 min_rx 500 multiplier 3
  tunnel source GigabitEthernet2
  tunnel mode ipsec ipv4
  tunnel destination 10.10.10.1
  tunnel protection ipsec profile LAB-IPSEC
!
interface GigabitEthernet1
  description MGMT - QEMU user-mode net (DHCP)
  ip address dhcp
  negotiation auto
!
interface GigabitEthernet2
  description P2P link to the hub (c8000v-r1)
  ip address 10.10.10.2 255.255.255.252
  negotiation auto
!
interface GigabitEthernet3
  description LAN toward the local host
  ip address 192.168.20.1 255.255.255.0
  ip nat inside
  negotiation auto
!
interface GigabitEthernet4
  description OOB management - NTP, syslog, SNMP
  vrf forwarding MGMT
  ip address 192.168.99.2 255.255.255.0
  negotiation auto
!
router bgp 65002
  bgp router-id 172.16.2.1
  bgp log-neighbor-changes
  neighbor 192.168.100.1 remote-as 65001
  neighbor 192.168.100.1 password LabBgpSecret
  neighbor 192.168.100.1 timers 10 30
  neighbor 192.168.100.1 fall-over bfd
!
  address-family ipv4
    network 10.20.2.1 mask 255.255.255.255
    network 10.30.2.0 mask 255.255.255.0
    network 192.168.20.0
    neighbor 192.168.100.1 activate
    neighbor 192.168.100.1 send-community both
    neighbor 192.168.100.1 route-map RM-IN-R1 in
    neighbor 192.168.100.1 route-map RM-OUT-R1 out
    neighbor 192.168.100.1 maximum-prefix 15 80
  exit-address-family
!
ip forward-protocol nd
!
ip bgp-community new-format
ip community-list standard CL-VALID permit 65001:1
ip community-list standard CL-VALID permit 65001:2
ip community-list standard CL-VALID permit 65001:3
ip community-list standard CL-VALID permit 65002:1
ip community-list standard CL-VALID permit 65002:2
ip community-list standard CL-VALID permit 65002:3
ip community-list standard CL-VALID permit 65003:1
ip community-list standard CL-VALID permit 65003:2
ip community-list standard CL-VALID permit 65003:3
ip as-path access-list 10 permit ^65001$

```

```

ip as-path access-list 10 permit ^65001_65003$
no ip http server
ip http secure-server
ip route 10.30.2.0 255.255.255.0 Null0
ip route 172.16.1.1 255.255.255.255 Tunnel0
ip nat pool SITE-POOL 10.30.2.100 10.30.2.129 prefix-length 24
ip nat inside source static 192.168.20.10 10.30.2.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.11 10.30.2.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.12 10.30.2.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.13 10.30.2.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.14 10.30.2.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.15 10.30.2.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.16 10.30.2.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.17 10.30.2.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.18 10.30.2.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.19 10.30.2.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
ip ssh bulk-mode 131072
ip ssh server algorithm authentication password
!
ip prefix-list PL-OWN-LAN seq 5 permit 192.168.20.0/24
!
ip prefix-list PL-OWN-LO seq 5 permit 10.20.2.1/32
!
ip prefix-list PL-OWN-NAT seq 5 permit 10.30.2.0/24
!
ip access-list standard VTY-MGMT
 10 remark out-of-band management network
 10 permit 192.168.99.0 0.0.0.255
 20 remark hypervisor user-mode NAT: how the harness reaches these devices
 20 permit 10.0.2.0 0.0.0.255
 30 deny any log
!
ip access-list extended NAT-DOMAIN
 10 permit ip host 192.168.20.10 10.30.0.0 0.0.255.255
 20 permit ip host 192.168.20.11 10.30.0.0 0.0.255.255
 30 permit ip host 192.168.20.12 10.30.0.0 0.0.255.255
 40 permit ip host 192.168.20.13 10.30.0.0 0.0.255.255
 50 permit ip host 192.168.20.14 10.30.0.0 0.0.255.255
 60 permit ip host 192.168.20.15 10.30.0.0 0.0.255.255
 70 permit ip host 192.168.20.16 10.30.0.0 0.0.255.255
 80 permit ip host 192.168.20.17 10.30.0.0 0.0.255.255
 90 permit ip host 192.168.20.18 10.30.0.0 0.0.255.255
100 permit ip host 192.168.20.19 10.30.0.0 0.0.255.255
ip access-list extended NAT-POOL-SRC
 10 permit ip 192.168.20.64 0.0.0.31 10.30.0.0 0.0.255.255
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
route-map RM-OUT-R1 permit 10
 description R2's own LO, tagged 65002:1
 match ip address prefix-list PL-OWN-LO
 set community 65002:1
!
route-map RM-OUT-R1 permit 20
 description R2's own NAT, tagged 65002:2
 match ip address prefix-list PL-OWN-NAT
 set community 65002:2
!
route-map RM-OUT-R1 permit 30
 description R2's own LAN, tagged 65002:3
 match ip address prefix-list PL-OWN-LAN
 set community 65002:3
!
route-map NAT-TO-DOMAIN permit 10
 match ip address NAT-DOMAIN
!
route-map RM-IN-R1 permit 10
 description accept only known communities on an AS path R1 can produce
 match as-path 10
 match community CL-VALID
!
snmp-server group LABGRP v3 priv read LABVIEW notify LABVIEW
snmp-server view LABVIEW iso included
snmp-server trap-source GigabitEthernet4
snmp-server enable traps snmp authentication linkdown linkup coldstart warmstart
snmp-server enable traps vrrp
snmp-server enable traps flowmon
snmp-server enable traps pfr
snmp-server enable traps ds1
snmp-server enable traps sdwan security policy system omp bfd
snmp-server enable traps entity-perf throughput-notif

```



```

snmp-server enable traps call-home message-send-fail server-fail
snmp-server enable traps tty
snmp-server enable traps ospf state-change
snmp-server enable traps ospf errors
snmp-server enable traps ospf retransmit
snmp-server enable traps ospf lsa
snmp-server enable traps ospf cisco-specific state-change nssa-trans-change
snmp-server enable traps ospf cisco-specific state-change shamlink interface
snmp-server enable traps ospf cisco-specific state-change shamlink neighbor
snmp-server enable traps ospf cisco-specific errors
snmp-server enable traps ospf cisco-specific retransmit
snmp-server enable traps ospf cisco-specific lsa
snmp-server enable traps eigrp
snmp-server enable traps casa
snmp-server enable traps xgcp
snmp-server enable traps cef resource-failure peer-state-change peer-fib-state-change inconsistency
snmp-server enable traps smart-license
snmp-server enable traps diameter
snmp-server enable traps entity-qfp mem-res-thresh throughput-notif
snmp-server enable traps entity-state
snmp-server enable traps ethernet cfm cc mep-up mep-down cross-connect loop config
snmp-server enable traps ethernet cfm crosscheck mep-missing mep-unknown service-up
snmp-server enable traps ether-oam
snmp-server enable traps ethernet evc status create delete
snmp-server enable traps aaa_server
snmp-server enable traps config-copy
snmp-server enable traps config
snmp-server enable traps config-ctid
snmp-server enable traps memory bufferpeak
snmp-server enable traps fru-ctrl
snmp-server enable traps entity-sensor
snmp-server enable traps resource-policy
snmp-server enable traps flash insertion removal lowspace
snmp-server enable traps hsrp
snmp-server enable traps entity
snmp-server enable traps ip local pool
snmp-server enable traps cpu threshold
snmp-server enable traps auth-framework sec-violation
snmp-server enable traps atm subif
snmp-server enable traps frame-relay
snmp-server enable traps frame-relay subif
snmp-server enable traps bgp state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp threshold prefix
snmp-server enable traps bgp cbgp2 state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp cbgp2 threshold prefix
snmp-server enable traps pim neighbor-change rp-mapping-change invalid-pim-message
snmp-server enable traps ipmulticast
snmp-server enable traps msdp
snmp-server enable traps mvpn
snmp-server enable traps rsvp
snmp-server enable traps isis
snmp-server enable traps dhcp
snmp-server enable traps nhrp nhs
snmp-server enable traps nhrp nhc
snmp-server enable traps nhrp nhp
snmp-server enable traps nhrp quota-exceeded
snmp-server enable traps ipsla
snmp-server enable traps bfd
snmp-server enable traps cnpd
snmp-server enable traps event-manager
snmp-server enable traps dlsw
snmp-server enable traps ospfv3 state-change
snmp-server enable traps ospfv3 errors
snmp-server enable traps pppoe
snmp-server enable traps l2tun session
snmp-server enable traps l2tun pseudowire status
snmp-server enable traps l2tun tunnel
snmp-server enable traps pki
snmp-server enable traps pimstdmib neighbor-loss invalid-register invalid-join-prune rp-mapping-change
interface-election
snmp-server enable traps adslline
snmp-server enable traps entity-diag boot-up-fail hm-test-recover hm-thresh-reached scheduled-test-fail
snmp-server enable traps vds12line
snmp-server enable traps vtp
snmp-server enable traps vlancreate
snmp-server enable traps vlandelete
snmp-server enable traps port-security
snmp-server enable traps isdn call-information
snmp-server enable traps isdn layer2
snmp-server enable traps isdn chan-not-avail
snmp-server enable traps isdn ietf

```

```

snmp-server enable traps c3g
snmp-server enable traps LTE
snmp-server enable traps lisp
snmp-server enable traps firewall serverstatus
snmp-server enable traps ike policy add
snmp-server enable traps ike policy delete
snmp-server enable traps ike tunnel start
snmp-server enable traps ike tunnel stop
snmp-server enable traps ipsec cryptomap add
snmp-server enable traps ipsec cryptomap delete
snmp-server enable traps ipsec cryptomap attach
snmp-server enable traps ipsec cryptomap detach
snmp-server enable traps ipsec tunnel start
snmp-server enable traps ipsec tunnel stop
snmp-server enable traps ipsec too-many-sas
snmp-server enable traps gdoi gm-start-registration
snmp-server enable traps gdoi gm-registration-complete
snmp-server enable traps gdoi gm-re-register
snmp-server enable traps gdoi gm-rekey-rcvd
snmp-server enable traps gdoi gm-rekey-fail
snmp-server enable traps gdoi ks-rekey-pushed
snmp-server enable traps gdoi gm-incomplete-cfg
snmp-server enable traps gdoi ks-no-rsa-keys
snmp-server enable traps gdoi ks-new-registration
snmp-server enable traps gdoi ks-reg-complete
snmp-server enable traps gdoi ks-role-change
snmp-server enable traps gdoi ks-gm-deleted
snmp-server enable traps gdoi ks-peer-reachable
snmp-server enable traps gdoi ks-peer-unreachable
snmp-server enable traps mpls rfc ldp
snmp-server enable traps mpls ldp
snmp-server enable traps mpls rfc traffic-eng
snmp-server enable traps mpls traffic-eng
snmp-server enable traps mpls fast-reroute protected
snmp-server enable traps otn
snmp-server enable traps pw vc
snmp-server enable traps frame-relay multilink bundle-mismatch
snmp-server enable traps dial
snmp-server enable traps dsp card-status
snmp-server enable traps dsp oper-state
snmp-server enable traps dsp video-usage
snmp-server enable traps dsp video-out-of-resource
snmp-server enable traps syslog
snmp-server enable traps sbc adj-status
snmp-server enable traps sbc blacklist
snmp-server enable traps sbc congestion-alarm
snmp-server enable traps sbc h248-ctrlr-status
snmp-server enable traps sbc media-source
snmp-server enable traps sbc radius-conn-status
snmp-server enable traps sbc sla-violation
snmp-server enable traps sbc sla-violation-rev1
snmp-server enable traps sbc svc-state
snmp-server enable traps sbc qos-statistics
snmp-server enable traps bulkstat collection transfer
snmp-server enable traps ethernet cfm alarm
snmp-server enable traps alarms informational
snmp-server enable traps transceiver all
snmp-server enable traps rf
snmp-server enable traps vrfmib vrf-up vrf-down vnet-trunk-up vnet-trunk-down
snmp-server enable traps ccme
snmp-server enable traps srst
snmp-server enable traps mpls vpn
snmp-server enable traps mpls rfc vpn
snmp-server enable traps voice
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon udp-port 1162
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon
tacacs server LAB-TAC
  address ipv4 192.168.99.10
  key LabTacacsKey123
!
!
!
!
!
control-plane
!
!
mgcp behavior rsip-range tgcp-only
mgcp behavior comedia-role none
mgcp behavior comedia-check-media-src disable
mgcp behavior comedia-sdp-force disable

```

```

!
mgcp profile default
!
!
!
!
!
banner login ^C
*****
                        AUTHORISED ACCESS ONLY

Access to this device is restricted to authorised personnel. If you are not
an authorised user, disconnect now.

All activity is monitored, logged and audited. Continued use of this system
constitutes consent to that monitoring.

Unauthorised access or use may result in disciplinary action and civil or
criminal penalties.
*****
^C
!
line con 0
  exec-timeout 0 0
  activation-character 13
  logging synchronous
  stopbits 1
line aux 0
  activation-character 13
line vty 0 4
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
line vty 5 15
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
!
ntp source GigabitEthernet4
ntp server vrf MGMT 192.168.99.10
!
!
!
!
!
!
netconf-yang
restconf
end

```

R3 — running config

Building configuration...

Current configuration : 19759 bytes

```

!
! Last configuration change at 00:07:14 UTC Tue Sep 8 2026 by lab
!
version 17.15
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
platform qfp utilization monitor load 80
platform sslvpn use-pd
platform console serial
!
hostname c8000v-r3
!
boot-start-marker
boot-end-marker
!
!
vrf definition MGMT
  description out-of-band management
  !
  address-family ipv4
  exit-address-family
!

```

[illegible]


```

!
crypto ikev2 profile LAB-PROF
  match identity remote address 10.10.20.1 255.255.255.255
  authentication remote pre-share
  authentication local pre-share
  keyring local LAB-KR
!
!
!
!
!
!
!
!
!
!
!
crypto ipsec transform-set LAB-TS esp-aes 256 esp-sha256-hmac
  mode tunnel
!
crypto ipsec profile LAB-IPSEC
  set transform-set LAB-TS
  set ikev2-profile LAB-PROF
!
!
!
!
!
!
!
!
!
interface Loopback0
  description Protected endpoint
  ip address 172.16.3.1 255.255.255.255
!
interface Loopback1
  description BGP-advertised prefix
  ip address 10.20.3.1 255.255.255.255
!
interface Tunnel0
  description to the hub (c8000v-r1)
  ip address 192.168.101.2 255.255.255.252
  ip nat outside
  bfd interval 500 min_rx 500 multiplier 3
  tunnel source GigabitEthernet2
  tunnel mode ipsec ipv4
  tunnel destination 10.10.20.1
  tunnel protection ipsec profile LAB-IPSEC
!
interface GigabitEthernet1
  description MGMT - QEMU user-mode net (DHCP)
  ip address dhcp
  negotiation auto
!
interface GigabitEthernet2
  description P2P link to the hub (c8000v-r1)
  ip address 10.10.20.2 255.255.255.252
  negotiation auto
!
interface GigabitEthernet3
  description LAN toward the local host
  ip address 192.168.30.1 255.255.255.0
  ip nat inside
  negotiation auto
!
interface GigabitEthernet4
  description OOB management - NTP, syslog, SNMP
  vrf forwarding MGMT
  ip address 192.168.99.3 255.255.255.0
  negotiation auto
!
router bgp 65003
  bgp router-id 172.16.3.1
  bgp log-neighbor-changes
  neighbor 192.168.101.1 remote-as 65001
  neighbor 192.168.101.1 password LabBgpSecret
  neighbor 192.168.101.1 timers 10 30

```

```

neighbor 192.168.101.1 fall-over bfd
!
address-family ipv4
network 10.20.3.1 mask 255.255.255.255
network 10.30.3.0 mask 255.255.255.0
network 192.168.30.0
neighbor 192.168.101.1 activate
neighbor 192.168.101.1 send-community both
neighbor 192.168.101.1 route-map RM-IN-R1 in
neighbor 192.168.101.1 route-map RM-OUT-R1 out
neighbor 192.168.101.1 maximum-prefix 15 80
exit-address-family
!
ip forward-protocol nd
!
ip bgp-community new-format
ip community-list standard CL-VALID permit 65001:1
ip community-list standard CL-VALID permit 65001:2
ip community-list standard CL-VALID permit 65001:3
ip community-list standard CL-VALID permit 65002:1
ip community-list standard CL-VALID permit 65002:2
ip community-list standard CL-VALID permit 65002:3
ip community-list standard CL-VALID permit 65003:1
ip community-list standard CL-VALID permit 65003:2
ip community-list standard CL-VALID permit 65003:3
ip as-path access-list 10 permit ^65001$
ip as-path access-list 10 permit ^65001_65002$
no ip http server
ip http secure-server
ip route 10.30.3.0 255.255.255.0 Null0
ip route 172.16.1.1 255.255.255.255 Tunnel0
ip nat pool SITE-POOL 10.30.3.100 10.30.3.129 prefix-length 24
ip nat inside source static 192.168.30.10 10.30.3.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.11 10.30.3.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.12 10.30.3.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.13 10.30.3.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.14 10.30.3.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.15 10.30.3.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.16 10.30.3.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.17 10.30.3.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.18 10.30.3.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.19 10.30.3.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
ip ssh bulk-mode 131072
ip ssh server algorithm authentication password
!
ip prefix-list PL-OWN-LAN seq 5 permit 192.168.30.0/24
!
ip prefix-list PL-OWN-L0 seq 5 permit 10.20.3.1/32
!
ip prefix-list PL-OWN-NAT seq 5 permit 10.30.3.0/24
!
ip access-list standard VTY-MGMT
10 remark out-of-band management network
10 permit 192.168.99.0 0.0.0.255
20 remark hypervisor user-mode NAT: how the harness reaches these devices
20 permit 10.0.2.0 0.0.0.255
30 deny any log
!
ip access-list extended NAT-DOMAIN
10 permit ip host 192.168.30.10 10.30.0.0 0.0.255.255
20 permit ip host 192.168.30.11 10.30.0.0 0.0.255.255
30 permit ip host 192.168.30.12 10.30.0.0 0.0.255.255
40 permit ip host 192.168.30.13 10.30.0.0 0.0.255.255
50 permit ip host 192.168.30.14 10.30.0.0 0.0.255.255
60 permit ip host 192.168.30.15 10.30.0.0 0.0.255.255
70 permit ip host 192.168.30.16 10.30.0.0 0.0.255.255
80 permit ip host 192.168.30.17 10.30.0.0 0.0.255.255
90 permit ip host 192.168.30.18 10.30.0.0 0.0.255.255
100 permit ip host 192.168.30.19 10.30.0.0 0.0.255.255
ip access-list extended NAT-POOL-SRC
10 permit ip 192.168.30.64 0.0.0.31 10.30.0.0 0.0.255.255
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
route-map RM-OUT-R1 permit 10
description R3's own L0, tagged 65003:1
match ip address prefix-list PL-OWN-L0
set community 65003:1
!
route-map RM-OUT-R1 permit 20
description R3's own NAT, tagged 65003:2

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match ip address prefix-list PL-OWN-NAT
set community 65003:2
!
route-map RM-OUT-R1 permit 30
description R3's own LAN, tagged 65003:3
match ip address prefix-list PL-OWN-LAN
set community 65003:3
!
route-map NAT-T0-DOMAIN permit 10
match ip address NAT-DOMAIN
!
route-map RM-IN-R1 permit 10
description accept only known communities on an AS path R1 can produce
match as-path 10
match community CL-VALID
!
snmp-server group LABGRP v3 priv read LABVIEW notify LABVIEW
snmp-server view LABVIEW iso included
snmp-server trap-source GigabitEthernet4
snmp-server enable traps snmp authentication linkdown linkup coldstart warmstart
snmp-server enable traps vrrp
snmp-server enable traps flowmon
snmp-server enable traps pfr
snmp-server enable traps ds1
snmp-server enable traps sdwan security policy system omp bfd
snmp-server enable traps entity-perf throughput-notif
snmp-server enable traps call-home message-send-fail server-fail
snmp-server enable traps tty
snmp-server enable traps ospf state-change
snmp-server enable traps ospf errors
snmp-server enable traps ospf retransmit
snmp-server enable traps ospf lsa
snmp-server enable traps ospf cisco-specific state-change nssa-trans-change
snmp-server enable traps ospf cisco-specific state-change shamlink interface
snmp-server enable traps ospf cisco-specific state-change shamlink neighbor
snmp-server enable traps ospf cisco-specific errors
snmp-server enable traps ospf cisco-specific retransmit
snmp-server enable traps ospf cisco-specific lsa
snmp-server enable traps eigrp
snmp-server enable traps casa
snmp-server enable traps xgcp
snmp-server enable traps cef resource-failure peer-state-change peer-fib-state-change inconsistency
snmp-server enable traps smart-license
snmp-server enable traps diameter
snmp-server enable traps entity-qfp mem-res-thresh throughput-notif
snmp-server enable traps entity-state
snmp-server enable traps ethernet cfm cc mep-up mep-down cross-connect loop config
snmp-server enable traps ethernet cfm crosscheck mep-missing mep-unknown service-up
snmp-server enable traps ether-oam
snmp-server enable traps ethernet evc status create delete
snmp-server enable traps aaa_server
snmp-server enable traps config-copy
snmp-server enable traps config
snmp-server enable traps config-ctid
snmp-server enable traps memory bufferpeak
snmp-server enable traps fru-ctrl
snmp-server enable traps entity-sensor
snmp-server enable traps resource-policy
snmp-server enable traps flash insertion removal lowspace
snmp-server enable traps hsrp
snmp-server enable traps entity
snmp-server enable traps ip local pool
snmp-server enable traps cpu threshold
snmp-server enable traps auth-framework sec-violation
snmp-server enable traps atm subif
snmp-server enable traps frame-relay
snmp-server enable traps frame-relay subif
snmp-server enable traps bgp state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp threshold prefix
snmp-server enable traps bgp cbgp2 state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp cbgp2 threshold prefix
snmp-server enable traps pim neighbor-change rp-mapping-change invalid-pim-message
snmp-server enable traps ipmulticast
snmp-server enable traps msdp
snmp-server enable traps mvpn
snmp-server enable traps rsvp
snmp-server enable traps isis
snmp-server enable traps dhcp
snmp-server enable traps nhrp nhs
snmp-server enable traps nhrp nhc
snmp-server enable traps nhrp nhp

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snmp-server enable traps nhrp quota-exceeded
snmp-server enable traps ipsla
snmp-server enable traps bfd
snmp-server enable traps cnpd
snmp-server enable traps event-manager
snmp-server enable traps dlsr
snmp-server enable traps ospfv3 state-change
snmp-server enable traps ospfv3 errors
snmp-server enable traps pppoe
snmp-server enable traps l2tun session
snmp-server enable traps l2tun pseudowire status
snmp-server enable traps l2tun tunnel
snmp-server enable traps pki
snmp-server enable traps pimstdmib neighbor-loss invalid-register invalid-join-prune rp-mapping-change
interface-election
snmp-server enable traps adsl-line
snmp-server enable traps entity-diag boot-up-fail hm-test-recover hm-thresh-reached scheduled-test-fail
snmp-server enable traps vds-l2-line
snmp-server enable traps vtp
snmp-server enable traps vlancreate
snmp-server enable traps vlandelete
snmp-server enable traps port-security
snmp-server enable traps isdn call-information
snmp-server enable traps isdn layer2
snmp-server enable traps isdn chan-not-avail
snmp-server enable traps isdn ietf
snmp-server enable traps c3g
snmp-server enable traps LTE
snmp-server enable traps lisp
snmp-server enable traps firewall serverstatus
snmp-server enable traps ike policy add
snmp-server enable traps ike policy delete
snmp-server enable traps ike tunnel start
snmp-server enable traps ike tunnel stop
snmp-server enable traps ipsec cryptomap add
snmp-server enable traps ipsec cryptomap delete
snmp-server enable traps ipsec cryptomap attach
snmp-server enable traps ipsec cryptomap detach
snmp-server enable traps ipsec tunnel start
snmp-server enable traps ipsec tunnel stop
snmp-server enable traps ipsec too-many-sas
snmp-server enable traps gdoi gm-start-registration
snmp-server enable traps gdoi gm-registration-complete
snmp-server enable traps gdoi gm-re-register
snmp-server enable traps gdoi gm-rekey-rcvd
snmp-server enable traps gdoi gm-rekey-fail
snmp-server enable traps gdoi ks-rekey-pushed
snmp-server enable traps gdoi gm-incomplete-cfg
snmp-server enable traps gdoi ks-no-rsa-keys
snmp-server enable traps gdoi ks-new-registration
snmp-server enable traps gdoi ks-reg-complete
snmp-server enable traps gdoi ks-role-change
snmp-server enable traps gdoi ks-gm-deleted
snmp-server enable traps gdoi ks-peer-reachable
snmp-server enable traps gdoi ks-peer-unreachable
snmp-server enable traps mpls rfc ldp
snmp-server enable traps mpls ldp
snmp-server enable traps mpls rfc traffic-eng
snmp-server enable traps mpls traffic-eng
snmp-server enable traps mpls fast-reroute protected
snmp-server enable traps otn
snmp-server enable traps pw vc
snmp-server enable traps frame-relay multilink bundle-mismatch
snmp-server enable traps dial
snmp-server enable traps dsp card-status
snmp-server enable traps dsp oper-state
snmp-server enable traps dsp video-usage
snmp-server enable traps dsp video-out-of-resource
snmp-server enable traps syslog
snmp-server enable traps sbc adj-status
snmp-server enable traps sbc blacklist
snmp-server enable traps sbc congestion-alarm
snmp-server enable traps sbc h248-ctrlr-status
snmp-server enable traps sbc media-source
snmp-server enable traps sbc radius-conn-status
snmp-server enable traps sbc sla-violation
snmp-server enable traps sbc sla-violation-rev1
snmp-server enable traps sbc svc-state
snmp-server enable traps sbc qos-statistics
snmp-server enable traps bulkstat collection transfer
snmp-server enable traps ethernet cfm alarm

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snmp-server enable traps alarms informational
snmp-server enable traps transceiver all
snmp-server enable traps rf
snmp-server enable traps vrfmib vrf-up vrf-down vnet-trunk-up vnet-trunk-down
snmp-server enable traps ccme
snmp-server enable traps srst
snmp-server enable traps mpls vpn
snmp-server enable traps mpls rfc vpn
snmp-server enable traps voice
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon udp-port 1162
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon
tacacs server LAB-TAC
  address ipv4 192.168.99.10
  key LabTacacsKey123
!
!
!
!
!
control-plane
!
!
mgcp behavior rsip-range tgcp-only
mgcp behavior comedia-role none
mgcp behavior comedia-check-media-src disable
mgcp behavior comedia-sdp-force disable
!
mgcp profile default
!
!
!
!
!
banner login ^C
*****
                        AUTHORISED ACCESS ONLY

Access to this device is restricted to authorised personnel. If you are not
an authorised user, disconnect now.

All activity is monitored, logged and audited. Continued use of this system
constitutes consent to that monitoring.

Unauthorised access or use may result in disciplinary action and civil or
criminal penalties.
*****
^C
!
line con 0
  exec-timeout 0 0
  activation-character 13
  logging synchronous
  stopbits 1
line aux 0
  activation-character 13
line vty 0 4
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
line vty 5 15
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
!
ntp source GigabitEthernet4
ntp server vrf MGMT 192.168.99.10
!
!
!
!
!
netconf-yang
restconf
end

```